Open Dumps and the Global Trade in Garbage

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October 2026

Abstract

A large fraction of the world's solid waste is disposed of in open dumps, generating serious environmental and health externalities. These effects are difficult to study as data on waste are scarce where mismanagement is worst. We show how to efficiently collect data on dumps when starting with a set of non-representative examples. Our methods combine crowdsourced data and satellite imagery with machine learning and active sampling methods to create the first globally representative human-verified time-series data on open dumps. The data show that open dumps are concentrated in densely populated areas of low- and middle-income countries, they are frequently on fire, and they expand substantially in response to increases in waste imports. Analysis of Java, Indonesia, where we observe nearly the universe of open dumps, shows that higher waste imports following China's 2018 import ban generated just $13,000 in consumer surplus per premature death caused by the increase in garbage-fire pollution.

JEL: C45, C81, Q53, O17

Keywords: Solid Waste, Satellite Data, Machine Learning, Trade

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1 Introduction

What are the impacts of the global trade in waste? Since Larry Summers argued, "the economic logic behind dumping a load of toxic waste in the lowest wage country is impeccable" in a controversial 1991 World Bank memo, cross-country flows of post-consumer plastic and paper have more than doubled.[1] Despite Summers' assertion, economic theory does not give an unambiguous prediction as to the welfare impacts of this trade. This will depend on the magnitude of the externalities generated by waste imports relative to the private benefits (Chichilnisky, 1994; Copeland et al., 2022). Yet studying these tradeoffs has been difficult, as we lack the most basic descriptive facts about whether waste imports are recycled or discarded in environmentally harmful ways.

An estimated 90% of domestic waste is disposed of in open dumps or burned in low-income countries (Kaza et al., 2018). Open dumps cause substantial environmental and social harm. The open burning of waste releases pollutants dangerous to human health and may significantly contribute to poor air quality in developing countries.[2] Dumped waste is also a well-documented source of plastic leakage to marine ecosystems, with possible adverse effects on the productivity and sustainability of ecosystem services such as fishing, recreation, and pest control (Rochman et al., 2016). However, the scale and incidence of these externalities remain understudied due to the lack of data on open dumping. Existing research attempts to study mismanaged waste using rough assumptions about per capita waste generation and improper disposal (Jambeck et al., 2015; Law et al., 2020; Meijer et al., 2021). These modeled estimates make it difficult to study causal relationships between changes in policies, waste management, and environmental and social outcomes.

In this paper, we devise an approach to collect the data required to begin to study the scale and impacts of open dumping. We start by collecting a crowdsourced sample of non-representative open dumps. We use these examples to train a machine learning model that predicts dumps using satellite data. Finally, we use the model's predictions to select and manually verify an approximately optimal subsample of predictions. These verified predictions form the basis of a global time-series dataset of open dump prevalence. This approach allows us to gather globally representative data, even when the machine learning

[1] Calculated from BACI trade data (Gaulier and Zignago, 2010). We sum trade across all country pairs in the two ban categories used in our subsequent analyses —plastic waste and scrap (HS 3915) and recovered paper and paperboard (HS 4707)—in 1995 and 2024. Global trade in these categories grew from 22.3 to 51.9 million metric tons, a 2.3-fold increase.

[2] Examples include black carbon, a major contributor to particulate matter, and polycyclic aromatic hydrocarbons (Pathak et al., 2023).

model's predictions are arbitrarily biased or miscalibrated, though better model accuracy yields more precise estimation. Our data show that open-dumps tend to be located in dense cities in low- and middle-income countries, and that they are frequently on fire, generating substantial amounts of air pollution. We analyze the welfare impacts of a shock to waste imports, exploiting a natural experiment resulting from the 2018 China waste import ban. In Indonesia, we estimate that waste imports generate about $13,000 in consumer surplus for every premature death resulting from the air pollution.

The first step of our pipeline is collecting non-representative 'seed' data. For this, we partnered with an international network of NGOs, researchers, and activists to crowdsource an initial set of open-air waste sites. This initiative, which we called "The Atlas of Plastic Waste", yielded over 270 submissions across 24 countries. We supplement our crowdsourced data collection effort with existing public repositories containing examples of open dumps. We verify the extent and location of all sites using high-resolution satellite imagery.

Next, we use these examples to train a machine learning model that can identify open dumps using satellite imagery. These methods build on recent work using satellite imagery to identify plastic waste in more limited temporal and geographic settings (Kruse et al., 2023). We train a gradient-boosted tree model that differentiates open dumps from other land uses with high accuracy. The area under the receiver-operating characteristic curve (AUC) in our validation sample is 0.96 out of 1, and the balanced accuracy is 90%, demonstrating that our model can find a distinctive spectral signature associated with open dumps.

Since our training data are heavily selected, and dumps are a rare land-use class, even small biases in our model's predictions when deployed out-of-sample could have an outsized relative impact on inferences drawn about the prevalence of open dumps. Thus, rather than use our model's predictions directly, we use them to inform a sampling and verification procedure that aims to minimize the variance of estimates of dump prevalence and changes over time. This procedure relies on optimal sampling methods that devote relatively more sampling effort to observations that our model is most uncertain about (Neyman, 1934; Zrnic and Candès, 2024). For selected observations, we manually verify the existence of dumps using high-resolution satellite images. This results in globally representative, time-series data of open dump prevalence.

We use these data to generate a set of novel descriptive facts about the global prevalence of open dumps. We document an Environmental Kuznets Curve—open dumps are rare in very low-income countries, peak in middle-income countries, and decline slightly from the peak in high-income countries. We also document that dumps are disproportionately found in

areas with high population density in low- and middle-income countries (LMICs), whereas in high-income countries dumps tend to be in slightly less dense areas. We also find that fires in open dumps are common: 16.7% of dumps in LMICs experience a fire at least once a year. The latter two facts–dense nearby populations and frequent fires—highlight some of the public health concerns about open dumping, and raise the question about the welfare consequences of shipping waste to LMICs.

We then investigate whether waste imports have a causal effect on open dumping in receiving countries. Our empirical strategy exploits a shock to the global trade network caused by China's sudden ban on waste imports, announced in 2017 and enforced in 2018. China previously accounted for 50–90% of global waste imports, depending on the category. After the ban, shippers diverted a large fraction of this waste to countries in Southeast Asia and the rest of the world. We use a shift-share instrumental variables strategy that exploits this shock, instrumenting observed changes in each country's waste imports with the diversion it would have received had every exporter reallocated its pre-2018 China export volumes to its other trade partners. We find an increase of 0.054–0.078 m^2 open dump area per kilogram of additional net imports, an elasticity of 1.4–1.8 that implies a substantial share of imported waste either ends up in open dumps or overwhelms domestic waste and recycling facilities. These results are robust to a range of alternative specifications, and placebo tests on commodities and policy timing show precise null effects.

Finally, we look at three smaller geographies—the island of Java in Indonesia, the US state of California, and three states in Southeast Brazil—where more exhaustive search methods become feasible to attempt to catalog the universe of dumps. In all three locations, we find dumps that are not present in administrative or other data sources. This more exhaustive data collection also allows for the more precise identification of the externalities associated with dumps. In Java, we find that the surge in waste imports after the China waste ban increased $PM_{2.5}$ by 6% close to dumps resulting from an increase in garbage fires. We can rule out large increases in pollution close to dumps in California—if anything, we find a modest decrease. A back-of-the-envelope calculation for Indonesia estimates that the ratio of consumer surplus gained from these imports amounts to $13,465 for each premature death caused by the resulting air pollution. This ratio is far lower than most estimates of the value of a statistical life, casting doubt on the efficiency of the international trade in waste.

Our data collection approach builds on the growing literature on using machine learning and satellite data for policy evaluation (see Donaldson and Storeygard (2016) and Jain (2020) for reviews). Previous work has documented the potential biases that can result from using machine-learned proxies for economic or environmental variables in causal inference (Fowlie

et al., 2019; Alix-Garcia and Millimet, 2023; Gordon et al., 2026). Several recent papers propose methods that can correct for non-classical measurement error using a sample of observations with 'ground-truth' data, where true labels are known with certainty (Ratledge et al., 2022; Proctor et al., 2026; Angelopoulos et al., 2023; Rambachan et al., 2025; Sanford et al., 2026; Kluger et al., 2025; Carlson and Dell, 2025).

These approaches take the labeled data as given and assume either that the labeled set is a representative sample of the broader population or that the labeling process can be modeled as a function of observables. Our approach, which is informed by the active statistical learning literature in computer science (Zrnic and Candès, 2024), is fundamentally different in that we study how to choose which data to label when starting with selected non-representative data. While these sampling techniques are common in the active sampling literature in computer science and remote sensing (Tuia et al., 2009; Stehman, 2012; Zrnic and Candès, 2024; Gordon et al., 2026), they have not been widely applied to study policy questions in the social sciences. The method results in unbiased and precise estimates of outcomes of interest. These methods are widely applicable to other hard-to-study land uses, and more generally, they can be used in any setting where data collection is expensive and true positives are rare in the population of interest, but can be predicted with reasonable accuracy.

We derive a simple way to estimate the ex-post efficiency of our method relative to uniform random sampling in the presence of model miscalibration. Efficiency gains depend on the distribution of uncertainty in the data, relative to the distribution of miscalibration. We find that our ex-post realized efficiency gains, accounting for model miscalibration indicate that we reduced the variance of our estimates of dump prevalence by 96% relative to uniform random sampling. This means that our 14,086 pixel verification sample gives us the same statistical power as if we had collected more than 318,000 verified pixels using uniform random sampling.

In addition to the data collection and descriptive statistics, we contribute to long-standing debates in economics about the efficiency and distributional implications of trade in the presence of environmental externalities. Using the quasi-natural experiment of China's import ban on various waste categories, we show clear evidence of the pollution havens hypothesis—the idea that environmental regulations in one country can result in the migration of polluting industries to less-regulated jurisdictions (see Copeland et al. (2022) for a recent review of this hypothesis and associated literature).

Prior work has demonstrated that industry location is influenced by environmental regulations

in the context of lead-acid battery recycling (Tanaka et al., 2022), and the Clean Air Act in the US (Hanna and Oliva, 2015). A few studies have examined the pollution havens hypothesis using country-level data on the waste trade. Kellenberg (2015) shows that the relative stringency of environmental regulations is associated with trade flows—materials move from jurisdictions with stricter regulations to more unregulated locales. Thakur (2026) uses a gravity model to show how the China waste ban increased waste flows to low-income countries. With the exception of Tanaka et al. (2022), however, none of these studies can directly measure changes in externalities associated with these trade flows or location decisions. A few papers have looked at the impacts of the China waste ban on air pollution in China (Guo et al., 2023; Sigman and Strow, 2024), but our novel data collection effort allows us to examine this understudied outcome worldwide and provide evidence for the pollution havens hypothesis on a global scale. In Indonesia we find that this influx of waste led to more fires in open dumps and a large increase in pollution, and simple calculations suggest that the value of these waste imports is low relative to the health costs of the resulting pollution.

The rest of the paper proceeds as follows—in Section 2, we describe relevant background, including existing research on open dumps, and the context of the Chinese waste import ban. Section 3 describes our statistical framework for data collection, including how to estimate efficiency gains with model miscalibration. In Section 4, we describe how these methods are applied to build a . Section 5 gives our descriptive results on global time trends, the Environmental Kuznets Curve, population density, and fires. Section 6 describes our cross-country analysis of the China waste import ban. Section 7 conducts more detailed analyses on the universe of dumps in Java, California, and Southeast Brazil, and estimates the consumer surplus and lives lost from the import shock in Indonesia. Finally, Section 8 concludes.

2 Background

Solid waste generation is expected to increase by more than 70% by 2050 as the world continues to get richer and more urbanized (Kaza et al., 2018). This trend will pose challenges for cities in low-income countries, where waste management is already one of the largest line items in some local budgets, comparable to spending on policing and education (Hoornweg et al., 2013). Despite these expenditures, an estimated 90% of solid waste in low-income countries is disposed of in unregulated open-air dumps or burned, though the authors of this fact warn that the underlying data, "should be considered with a degree of caution due to undefined words or phrases, inconsistent or omitted units, or estimates made without basis"

(Kaza et al., 2018). A major goal of this paper is to collect the data to begin to understand the prevalence and consequences of these practices.

Case studies have shown that open dumping can generate severe externalities. Runoff and leachate from dumps can lead to surface and groundwater contamination (Siddiqua et al., 2022). This may also contribute to the increase in ocean plastics that disrupt ecosystems and fisheries (Jambeck et al., 2015; Beaumont et al., 2019). Dumps may also attract disease-spreading vermin and other pests (Krystosik et al., 2020). The decomposition of organic waste generates methane emissions which contribute to global warming, and the open burning of waste generates air pollution, including particulate matter and black carbon (Pathak et al., 2023).

Existing research has argued that the mismanagement of solid waste is mostly an issue in low-income countries (Jambeck et al., 2015; Lebreton et al., 2017). For example, Meijer et al. (2021) find that most ocean plastics come from a relatively small number of river systems, mostly located in low-income countries. In the absence of data on mismanaged waste, these papers use simple models of waste emissions that depend on population, GDP, and rough estimates of the fraction of mismanaged waste. For the most part, these models ignore waste traded across international borders. In trying to estimate the United States' contribution to ocean plastic pollution, Law et al. (2020) write "no quantitative estimates exist of the proportion of material exported for recycling that is ultimately discarded as waste."

Yet these trade flows are significant. Prior to 2018, around 50% of plastic waste collected for recycling was traded internationally, and China accounted for close to 50% of global imports, including 70% of US exports and 95% of EU exports (Brooks et al., 2018; Ritchie, 2022). Importers pay for this waste to use it in industrial and recycling processes, though a large share of the low-quality imports may be disposed of in environmentally damaging ways (Greenpeace, 2021). As the quotes above emphasize, no good estimates exist on how much of this material is used productively and how much is discarded, a question that we study in Section 6.

Complaints about contamination levels in these shipments and the externalities associated with processing the waste led to a series of progressively stricter restrictions in China. In February 2013, China announced the Green Fence Policy, which increased inspections of waste shipments. Shipments that exceeded prescribed contamination levels could lead to the revocation of import licenses and the forced return of the shipment. Then in July 2017, the Chinese government announced the National Sword Policy, which took effect in January 2018. The policy banned or restricted imports of 24 categories of scrap material, including

plastic scrap and unsorted paper, and lowered contamination thresholds for other materials to levels generally considered unachievable for waste exporters in the US and Europe. At the same time, the Chinese government cracked down on the domestic informal recycling industry. Guo et al. (2023) find that air pollution went down after the ban in Chinese cities that had a large recycling industry, but due to a lack of data on disposal, we do not know whether this is due to a decline in productive recycling activities or a reduction in the open burning of discarded scraps.

Appendix Figure A2 shows that Chinese waste imports began to decline sharply as soon as the ban was announced, falling to nearly zero shortly after implementation.[3] The loss of this market disrupted the global recycling market, leading to pileups of scrap in the US and the diversion of waste exports to other countries. Appendix Figure A2 shows an increase in waste imports to India, Indonesia, Malaysia, and Turkey around the time of the ban, even though total global trade flows decreased due to the loss of the Chinese market. Much of the material that would have otherwise been exported ended up in landfills in the US and Europe (Sigman and Strow, 2024).

Questions about the impacts of the international waste trade have become important in multilateral policy. Recent discussions under the Basel Convention have debated changing the classification of plastic waste, which would severely restrict or ban trade across borders.

3 Active Statistical Inference with Satellite Data

This section introduces the methods we use to collect data on open dumps. The approach combines satellite data, machine learning, and active statistical inference to measure the prevalence of a rare land use. It is designed for settings with little or no representative training data, where the predictions of a machine learning model may be informative about outcomes of interest, though not necessarily perfectly calibrated, and where ground-truth labeling is possible but expensive.

The procedure has four steps. First, we collect a sample of training examples. This paper focuses on settings where selection into the training sample depends on unobservables; therefore, the training examples are not representative of the broader population of interest. Second, we use these examples to train a machine learning model that predicts, from satellite

[3]We focus on plastic (HS 3915) and paper (HS 4707) scrap. These two categories of banned material account for the largest share of imports and are the most likely to end up in open dumps, though waste textiles and some types of metal slag were also banned. The graphs also show that the increased inspections of the 2013 Green Fence Policy had a much smaller impact on imports than the 2017-18 import ban.

data, the probability that each pixel belongs to the land-use class of interest. Since the model's predictions may be biased due to the unrepresentative training data, we do not use them directly as measurement. Instead, in the third step, we use them to design an approximately optimal sampling strategy, which selects a new sample of pixels for verification that is designed to be representative and to minimize the variance of a downstream estimator. We also show how the resulting efficiency gains depend on model miscalibration. Finally, we combine these labels with the model's predictions to obtain unbiased estimates of prevalence.

This section describes the approach in general terms to demonstrate the broader applicability of the methods. Section 4 then shows how we apply the methods to detect open dumps.

3.1 Setup and Estimator

Define $D_i \in \{0, 1\}$ as an indicator for whether pixel i in a population indexed $\{1, ..., N\}$ contains the land use of interest. Our estimand is $\mu = \mathbb{E}(D_i)$, the prevalence (which may be a conditional prevalence) of the land use class. We start with a set of labeled training examples $\mathcal{T}$. These can come from any source, (in our application we will use crowdsourcing, case studies, and various selected databases), and they need not be representative of the population of interest.

The next step is to train a machine learning model that maps any satellite features k_i to a predicted probability $p(k_i)$ that the pixel belongs to the land-use class. Any model that produces such a score can be used—our downstream estimator will remain unbiased regardless of model accuracy, but higher accuracy will give more precise estimates. Define the model's prediction errors as $\nu_i = p(k_i) - D_i$.

A naive estimator of μ in a population of N pixels could be a simple average of the $p(k_i)$. This estimator will give an unbiased estimate of μ if $\mathbb{E}(\nu_i) = 0$ (or in the conditional case, the conditional expectation must be zero). This is unlikely in most settings—a large and growing literature documents non-classical measurement error in machine-learned proxies for environmental and social science variables of interest (see e.g. Jain, 2020; Proctor et al., 2026; Gordon et al., 2026). In our setting, using non-representative training data exacerbates the likelihood of biases, since we put no requirements on the joint distribution of k_i and D_i in $\mathcal{T}$. If the accuracy of the model changes over time or across regions, it would be impossible to determine whether estimated changes are due to changes on the ground or differential prediction accuracy, which could result from something as simple as year-to-year changes in average cloud cover.

Work in statistics and computer science has developed methods to address this problem

(Angelopoulos et al., 2023; Kluger et al., 2025; Egami et al., 2023; Lu et al., 2025; Carlson and Dell, 2025). Methods emerging from different disciplines have adopted slightly different terminology and approaches, but the fundamental idea is to make use of a subset of labeled or ground-truth observations in which D_i is known to correct the biases in the model predictions.

We call this subset the verification sample $\mathcal{S}$. Let $s_i \in \{0, 1\}$ be an indicator for whether $i \in \mathcal{S}$, with $\pi_i = \Pr(i \in \mathcal{S})$. An augmented inverse probability weighted (AIPW) missing-data estimator of μ is:

$$\widehat{\mu} = \frac{1}{N} \sum_{i=1}^{N} \left[p(k_i) + \frac{s_i}{\pi_i} \left(D_i - p(k_i) \right) \right]. \tag{1}$$

In words, the estimator corrects the model's predictions with a Horvitz–Thompson estimator of the average population prediction error using the labeled pixels. The correction can also be applied within subgroups when a conditional mean is of interest. One requirement, however, is that the verification sample inclusion probabilities π_i must be correctly specified or known by design.[4] In some settings, researchers may be able to use $\mathcal{T}$ as the verification sample, by estimating π_i as a function of k_i. As stated above, we assume that the selection process may depend on unobservables, which would lead to misspecification in estimating π_i. Thus, we generate $\mathcal{S}$ by labeling a new set of observations. These labels could come from expert interpretation of high-resolution imagery or field visits, for example.

One further observation, which we will use in some of our analyses, is that the verification sample can also be used on its own without the model predictions. Setting $p(k_i) = 0$ reduces Equation 1 to the standard Horvitz–Thompson estimator using the verification sample. The key benefit of adding the predictions is that they may reduce the variance of $\widehat{\mu}$.

Since collecting the verification sample needed to implement this estimator is costly, we now turn to the question of how to set inclusion probabilities to minimize the variance of $\widehat{\mu}$, subject to a limited labeling budget. By carefully setting inclusion probabilities that are known by design, $\widehat{\mu}$ is finite-population unbiased, regardless of model accuracy.

3.2 Approximately Optimal Active Sampling

Assuming independent labels with fixed probabilities, and recalling that $\nu_i = p(k_i) - D_i$, the variance of $\widehat{\mu}$ is:

$$\mathrm{Var}(\hat{\mu}) = \frac{1}{N^2} \sum_i \frac{1 - \pi_i}{\pi_i} \nu_i^2. \tag{2}$$

[4] In fact, the estimator is doubly robust; it is consistent if either $p(k_i)$ or π_i is correctly specified. However, because of the selected nature of $\mathcal{T}$, we do not want to assume $p(k_i)$ is correctly specified.

This increases when the labeled sample is small, since it is decreasing in π_i.

One option is to construct the verification sample through uniform random sampling. In fact, this could be done without needing a machine learning model or a training sample at all. Doing so would result in an unbiased estimate at the cost of a large variance. In cases where the land use class of interest is rare, the sample size of labeled pixels required to obtain a reasonably precise estimate of the prevalence could be enormous, since samples of a few hundred, or even a few thousand, pixels chosen at random will likely contain few or even zero cases of $D_i = 1$.

Instead, we consider how to use our machine learning model predictions to make verification more efficient. The methods described here have precursors in two-stage 'proxy-Neyman' sampling techniques (Neyman, 1934; Cochran, 1977), and recent advances in the computer science and remote sensing literatures where it is called *Active Statistical Inference* (Stehman, 2012; Bai et al., 2025; Zrnic and Candès, 2024; Hamilton et al., 2025). The key idea can be seen directly in Equation 2—the variance of $\hat{\mu}$ depends on the joint distribution of the sampling probabilities and the squared prediction errors.

Given a limited budget for verification, we can improve the efficiency of the estimator in Equation 1 by choosing π_i according to:

$$\min_{\pi_i} \text{Var}(\hat{\mu}) \quad \text{such that} \quad \sum_i \pi_i = S. \tag{3}$$

The constraint in this setup is an expected sample size constraint. In practice, realized sample sizes may vary. Taking first-order conditions and solving for π_i gives the familiar Neyman (1934) optimal sampling allocation rule: $\pi_i \propto \sqrt{\mathbb{E}(\nu_i^2)}$. This rule states that we should give the highest verification probabilities to the pixels with the largest expected model residuals. These residuals can come from two sources: fundamental uncertainty remaining after conditioning on the predictors, and bias from the selected training sample.

To see this clearly, let $p_i^* = \mathbb{E}[D_i|k_i]$ denote the population expectation of D_i conditional on the satellite data and other predictors k_i, such that $\mathbb{E}(p_i^* - D_i) = 0$. These p_i^* may differ from our model's estimated $p(k_i)$, not least because our model only estimates $p(k_i)$ from the joint distribution of k_i and D_i in a selected, non-random subset of the data. Expanding the squared prediction errors gives $\mathbb{E}(\nu_i^2) = p_i^*(1 - p_i^*) + (p(k_i) - p_i^*)^2$. The first term reflects the irreducible uncertainty of pixel i, after conditioning on satellite features, and the second term captures the model's misspecification bias.

Since we cannot know p_i^*, one way to approximate this optimal verification rule is to simply

plug in the model-predicted probabilities, which gives the sampling rule:

$$\pi_i \propto \sigma_i \equiv \sqrt{p(k_i)(1-p(k_i))}. \tag{4}$$

If our model is miscalibrated, this will not exactly solve Equation 3, but if $p(k_i)$ is close to p_i^*, it will get close.

This proxy-Neyman rule results in sampling probabilities peaking for the pixels that the model is most uncertain about, when $p(k_i)$ is close to 0.5, with π_i shrinking toward zero as $p(k_i)$ approaches 0 or 1. Intuitively, this rule can be especially effective for rare land uses because the vast majority of pixels have a very low probability of containing the class of interest.

Once the sampling probabilities are set, the verification sample can be drawn and labeled, and Equation 1 can be used to estimate μ. The machine learning model supplies a prediction for every pixel, and the labeled pixels, weighted by $1/\pi_i$, correct the average bias of those predictions. Note that conditional means can also be estimated using a conditional version of Equation 1. Standard errors can be calculated using a plug-in estimator for Equation 2 (Zrnic and Candès, 2024).[5]

3.3 Efficiency Gains with Miscalibration

A natural question is what proxy-Neyman sampling gains, or costs, in the presence of model miscalibration. Using the labeled sample, we can perform some calculations to show the precision gains, relative to a uniform random sample. Plugging in our approximately optimal π_i from Equation 4, the variance of $\widehat{\mu}$ becomes:

$$\mathrm{Var}(\widehat{\mu}) = \frac{1}{N^2 S}\sum_{i=1}^{N}\frac{\mathbb{E}(\nu_i^2)}{\sigma_i}\sum_{i=1}^{N}\sigma_i - \frac{1}{N^2}\sum_{i=1}^{N}\mathbb{E}(\nu_i^2). \tag{5}$$

Now let $\sigma_i^* = \sqrt{\mathbb{E}(\nu_i^2)} = \sqrt{p_i^*(1-p_i^*) + (p(k_i)-p_i^*)^2}$ denote the root-mean-squared error of our model's predictions. The design effect (DEFF) is defined as the ratio of the variance of $\widehat{\mu}$, relative to estimating μ under a uniform random sampling design (where π_i is constant c). Up to a finite sample correction, this can be written as:

$$\mathrm{DEFF} := \frac{\mathrm{Var}(\widehat{\mu}|\pi_i \propto \sigma_i)}{\mathrm{Var}(\widehat{\mu}|\pi_i = c)} \approx \frac{\left(\sum_{i=1}^{N}\frac{\sigma_i^{*2}}{\sigma_i}\right)\left(\sum_{i=1}^{N}\sigma_i\right)}{N\sum_{i=1}^{N}\sigma_i^{*2}}. \tag{6}$$

[5]In our applications, we use a more complicated two-stage sampling design detailed below. For this reason, we use a clustered bootstrap procedure to estimate standard errors.

Given our verification subsample, this design effect can be estimated ex-post (see Appendix Section A for the estimator). Decisions about whether to use proxy-Neyman sampling must be made ex-ante, however. While we do not provide a formal decision rule in this paper, we show how the realized design effect depends on model calibration in an intuitive way.

Define $R_i = \frac{\sigma_i^*}{\sigma_i}$ as the model miscalibration ratio—it is the model's root mean squared error divided by the implied model uncertainty. Appendix Figure A6 plots R_i as a function of $p(k_i)$ and p_i^*. Values greater than one imply that the model is 'overconfident'—the true root mean squared error for a pixel, conditional on the satellite data, is greater than implied by the model's estimate. This implies that our proxy-Neyman strategy is less likely to sample this pixel than is optimal. Conversely, values less than one imply the model is 'underconfident'—shrinking predictions toward 0.5 relative to p_i^*. In this case, the proxy-Neyman strategy is more likely to sample this pixel than is optimal. If $p(k_i) = p_i^*$ (perfect calibration), then $R_i = 1$. Substituting R_i into Equation 6, the design effect can be rearranged to show:

$$\text{DEFF} \approx \frac{1 + CV_{\sigma R}^2}{1 + CV_{\sigma^*}^2}, \tag{7}$$

where $CV_{\sigma^*}^2$ is the squared coefficient of variation of σ_i^*, and $CV_{\sigma R}^2$ is the squared coefficient of variation of R_i weighted by σ_i (see Appendix Section A for details).

To gain some intuition for this formula, first consider a perfectly calibrated model, so that $R_i = 1$ for all i. Then $CV_{\sigma R} = 0$, and the numerator simplifies to 1. This case represents the optimal lower bound—there is no miscalibration, so the gains from sampling depend on the denominator, which captures variation in the amount of predicted uncertainty in the data. We use DEFF* to denote this optimal lower bound.[6] When most pixels have predicted probabilities near zero, as is typical for rare land uses, DEFF* can be far below one.

In general, however, the model will be miscalibrated, and this miscalibration will vary across predictions, reducing efficiency gains. In these cases, the coefficient of variation of the weighted R_i becomes important. In an extreme case where our predictions were random noise, so σ_i^* and σ_i are independent, a second order Taylor approximation gives $1 + CV_{\sigma R}^2 \approx (1 + CV_{\sigma^*}^2)(1 + CV_{\sigma}^2)$. In this case, DEFF $\approx 1 + CV_{\sigma}^2$: variation in the weights inflates the variance, and we do much worse than a random sample for the same sample size.[7] Finally, if the model predicts the same value for every pixel, π_i is constant and the method collapses

[6] In this case, the formula simplifies to DEFF$^* \approx \frac{1}{N} \frac{\left(\sum_{i=1}^N \sigma_i^*\right)^2}{\sum_{i=1}^N \sigma_i^{*2}}$, which will always be ≤ 1.

[7] This formula relies on a second-order Taylor approximation of the dispersion. The exact DEFF is unbounded; the approximate formula is simply shown here for intuition. See Appendix Section A for details.

to uniform random sampling, so DEFF = 1. When that constant is $p(k_i) = 0.5$, both the numerator and the denominator of Equation 7 equal one, since $R_i = 1$ and $\sigma_i^* = 0.5$ for every p_i^*.

Thus, the size of the efficiency gain is an empirical question that depends on both model calibration and the distribution of uncertainty in the population. When deciding whether a proxy-Neyman strategy is appropriate, however, studying the distribution of σ_i can be informative about the potential gains.

4 Measuring Open Dumps

We now discuss how the methods described in Section 3 apply to our estimand: the prevalence of open dumps. We define an open dump as any aggregation of waste covering an area of at least 100 m^2, where waste is directly exposed to the surrounding environment.[8] In the notation of Section 3, $D_i = 1$ if pixel i contains exposed waste meeting this definition.

Several countries have statutory definitions of open dumps that refer to waste disposal sites that do not comply with regulations.[9] Since regulations vary across countries, we adopt a definition that is consistent globally, can be measured with our data, and captures the key externalities associated with improper waste disposal. Our definition excludes small, transitory piles of litter and buried waste, and therefore most sanitary landfills. We also include discard pits next to recycling facilities, for example, the ones that commonly accompany paper mills in Indonesia, which are a source of frequent fires and plastic leakage into the environment. We exclude construction debris and automobile junkyards, which differ from the dumps in our data; however, we cannot otherwise distinguish the waste's material composition, which we leave for future research.

We measure the surface area of dumps rather than the volume of waste disposed, as it is difficult to measure the height or density of waste piles using satellite data. Many informal dumps have low density; without capital-intensive construction equipment, it is difficult to pile waste very high or compress it to a significant degree. Even where height and density matter, locating dumps from the air is a necessary first step. In any case, dump area determines how much waste is directly exposed to the environment. We return to the question of what changes in area imply for changes in volume in Section 6.

[8] This definition roughly corresponds with the spatial resolution of the satellite data we use, which is 10 meters by 10 meters at the equator.

[9] For example, in the Philippines, an open dump is legally defined as, "a disposal area wherein the solid wastes are indiscriminately thrown or disposed of without due planning and consideration for environmental and Health standards" (Congress of the Philippines, 2001).

4.1 Training Examples and Satellite Data

Training Examples. Our non-representative training sample $\mathcal{T}$ combines a crowdsourced data collection effort with existing public datasets. We worked with an international NGO, the Basel Action Network (BAN), to create a web portal that allows individuals worldwide to submit the locations of open dump sites and contribute to a training dataset.[10] We received more than 270 submissions from dozens of individuals and organizations across 24 countries. We complement this effort with several other publicly available datasets on landfills and informal dumps that cover particular regions of the world or periods of time (D-Waste, 2014; Greenpeace, 2018, 2021).[11] In particular, we include nearly 1,000 additional unique sites from Earthrise's Global Plastic Watch, which also used satellite imagery to find dumps in a number of countries (Kruse et al., 2023).

We verify the extent of every site using the catalog of historical high-resolution satellite imagery in Google Earth Pro. Figure 1 and Appendix Figures A1 and A4 show examples of open dumps in this imagery, which are typically easy to identify. For each site, annotators drew a polygon around the extent of any exposed waste visible for each year between 2017 and 2022, along with nearby control sites covering other land uses. Given the paucity of initial training data, we expanded the sample using an iterative process based on early models' predictions. We trained a small model on the example sites and then used the model to make out-of-sample predictions. We verified some subset of these predictions to assess accuracy, and incorporated new waste and control sites discovered into the next round of model training. After several iterations, we have nearly 37,000 validated site-year observations across 93 countries and more than 84,000 control site-year observations. Appendix Figure A3 shows the distribution of our training data worldwide.

Satellite Data. For our machine learning model training features (k_i), we use annual summaries of satellite imagery from Sentinel-1 and Sentinel-2, both available with consistent coverage since 2017. Sentinel-2 is an optical satellite constellation that allows us to capture reflectance in the visible and near-infrared wavelengths, which have been previously used to differentiate between plastics and other materials (Biermann et al., 2020; Kruse et al., 2023; Cózar et al., 2024). Sentinel-1 is a radar satellite that provides all-weather and day-and-night coverage and gives information on the "texture" and "smoothness" of land cover, which helps differentiate between waste sites and other objects made of similar materials, such as plastic

[10] Submission form available at: https://www.ban.org/plastic-waste-transparency-project-hub/the-atlas-of-plastic-waste.

[11] Our complete training dataset and list of sources are available from: https://github.com/atlas-of-plastic-waste/atlas-of-plastic-waste.

roofs and greenhouses. Extracting these features for every training polygon in each year from 2017 to 2021 results in over 7.7 million 10-meter pixels. Appendix Section B.1 contains more details on the satellite data cleaning and features used in the model.

4.2 Machine Learning Model

We train a gradient-boosted tree model (Chen and Guestrin, 2016; Guillame-Bert et al., 2023) on our training examples $\mathcal{T}$. The model estimates the function $p(k_i)$, the probability that $D_i = 1$, conditional on the satellite features. We follow standard practice by grouping training sites into 500 spatial clusters and setting aside 20% of the clusters to calculate out-of-sample accuracy metrics. The rest are used for training. This restricts the model from overfitting on similar dumps that are geographically close together. On the held-out sites, the final model has an AUC of 0.96 out of the maximum possible value of 1. Setting a classification threshold that maximizes accuracy, we calculate balanced accuracy at 90%, precision at 92%, and recall at 85%. Appendix Section B.2 provides more details on model training and describes which satellite variables are the most important predictors.

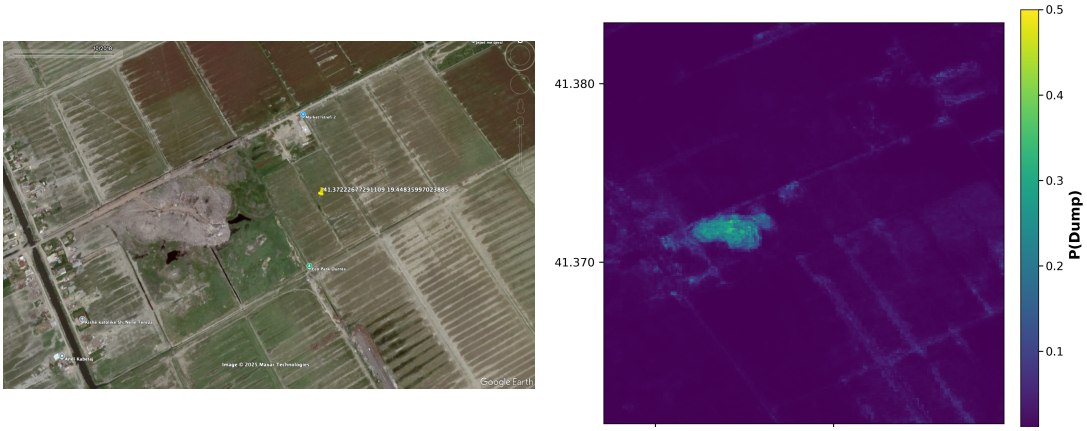
Figure 1 shows high-resolution Google Earth Pro imagery of a dump near Durrës, Albania, alongside our model predictions for the years 2018 and 2021. In 2018, the model puts elevated probability on the pixels corresponding to the exposed waste at the site. The site was subsequently capped and redeveloped as a park. By 2021, exposed waste is no longer visible in the imagery, and the model's predicted probabilities decline accordingly. This suggests that the model captures not only the locations of dumps, but also changes in the amount of exposed waste at a site over time.

Since many of our research questions deal with changes in the prevalence of dumps over time, we can also adapt Equation 4 to maximize power to detect changes. This can be done using the same machinery described above, while simply replacing D_i with an indicator that pixel i changed status between periods, and using a model that predicts changes instead of prevalence. Define p_i^{Δ} as the estimated probability that pixel i changed land-use status during a given period. If we want to precisely estimate changes, we set $\pi_i \propto \sigma_i^{\Delta} = \sqrt{p_i^{\Delta}(1 - p_i^{\Delta})}$, simply substituting p_i^{Δ} for $p(k_i)$ in Equation 4. Rather than train a dedicated change model, we construct p_i^{Δ} from our prevalence model, as the difference between the maximum and minimum predicted probabilities across 2017–2021.[12] As discussed above, miscalibration will

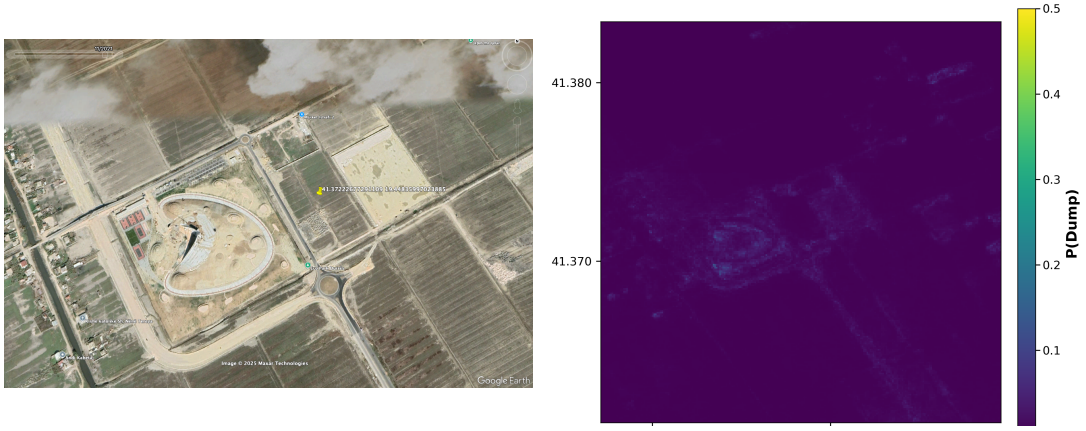
[12] We use this simple calculation because our training data are not temporally balanced, making it difficult to train a change model. The calculation assumes that a pixel's prediction errors are strongly correlated over time: if the pixel contains the land use in its lowest-probability period, it does so in all periods, and if it does not in its highest-probability period, it never does.

not bias our downstream estimates—it will only matter for efficiency.

Figure 1: Example Site in Durrës, Albania



(a) 2018



(b) 2021

Notes: Figure shows Google Earth Pro historical satellite imagery (image dates October 2018 and November 2021) and machine learning predictions (probability of each pixel being part of a waste site) for a small dump site near Durrës, Albania that was converted into an eco-park in 2020.

Despite the promising accuracy metrics reported above, open dumps are rare enough that small biases, or differences in accuracy across years and places, could confound real changes. Furthermore, since our training data are not representative of the broader population, true out-of-sample accuracy is unknown. We therefore use both $p(k_i)$ and p_i^{Δ} to design a verification sample following the approach in Section 3.

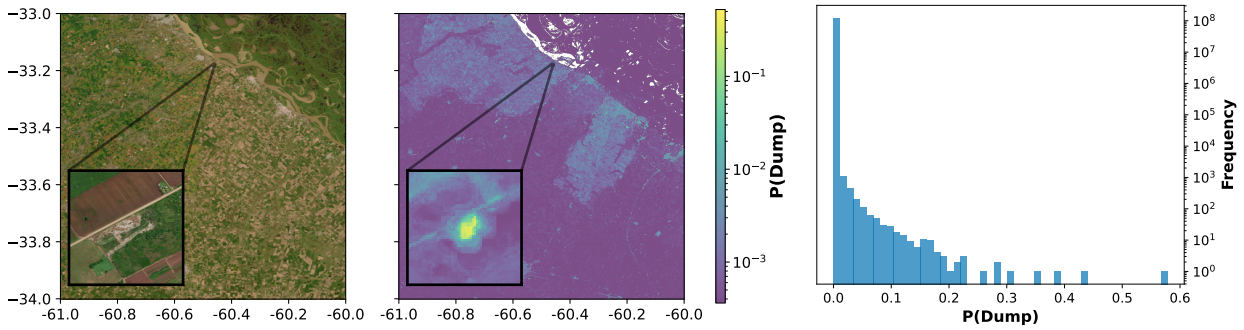
4.3 Sample Design

Predicting every 10-meter pixel on Earth for multiple years would be prohibitively expensive, so we deploy the model in two stages. We first randomly select 500 $1° \times 1°$ tiles (approximately

111 km × 111 km at the equator). We stratify tiles by a proxy for where we expect to see the biggest changes in our outcome variable of interest (open dumps), and then we sample within strata with probability proportional to population density.[13] Our model makes predictions $p(k_i)$ and we derive p_i^{Δ} for all 10-meter pixels within the selected tiles.

As discussed above, we should expect that our $p(k_i)$ are somewhat miscalibrated due to the selected training examples. While miscalibration could erode the expected efficiency gains from proxy-Neyman sampling, this possibility needs to be weighed against the potential efficiency gains. To assess the magnitude of these benefits, consider the tile shown in Figure 2 covering an area in north-east Argentina. The left and middle panels show the high-resolution imagery and the model predictions for the tile, respectively, with a small open dump highlighted in the inset. The right panel shows the histogram of model-predicted probabilities on a log scale. The vast majority of pixels have predicted probabilities very close to zero, because the model easily distinguishes common land uses, like forests and crops, from open dumps. Far fewer pixels have probabilities even exceeding 10%. If we calculate the optimal lower bound defined in Section 3.3 based on the distribution of σ_i in this scene, we find $\text{DEFF}^* = 0.0012$. This means that if the dispersion in our predicted probabilities reflected genuine pixel-level variation in uncertainty, the active sampling method would reduce the variance by 99.88% relative to uniform random sampling. While this ex-ante estimate represents the best-case scenario for what we can hope to achieve, the vast potential gains suggest that even if the model is somewhat miscalibrated, it should help to improve the allocation of verification samples relative to uniform random sampling.

Figure 2: A Sample Tile and Histogram of Pixel Probabilities



Notes: A sample tile near Rosario, Argentina. Left panel: high-resolution satellite imagery. Inset shows a small open dump. Middle panel: model predictions for the same region. Right panel: histogram showing the model-predicted probability distribution for all pixels in the scene (log-scale y-axis).

[13] We use the per-capita change in plastic waste imports from 2017 to 2019, multiplied by Kaza et al. (2018) estimates of waste mismanagement, allocated to tiles within countries by population. See Appendix Section B.4 for more details and Appendix Figure A5 for the strata and selected tiles.

We therefore apply the sampling rules in Section 3 within each selected tile. We set our expected sampling budget to 2.5 $\times 10^{-7}$ of valid land area, with half the sample allocated proportional to σ_i (based on predictions for the year 2018) and half proportional to σ_i^{Δ}. A pixel's overall inclusion probability accounts for the full design by multiplying tile selection probabilities by within-tile pixel selection probabilities (see Appendix Section B.4). This results in a sample of 14,086 verification pixels—on average fewer than 30 pixels in each of our 500 tiles. Thus, our estimates are highly uncertain within tiles, which span more than 10,000 square kilometers near the equator, but become more precise at higher levels of aggregation.

4.4 Verification Set

For each chosen pixel, we ask two experts trained in visual interpretation of satellite imagery to independently determine whether the pixel is an open dump using the high-resolution Google Earth Pro historical imagery for all years from 2011 to 2023. This window is deliberately wider than the 2017–2021 period covered by the model's predictions, since once a pixel is sampled, labeling additional years costs little. The longer time series is helpful for our analysis of the China waste ban in Section 6. As described in Section 3.1, we can use a simple Horvitz–Thompson estimator on the labels for years when model predictions are not available.

A key complication is that high-resolution images are not available for all pixels in all years, so some labels in our verification set are interpolated. We fill each annotator's unlabeled years from the years they did label using rules detailed in Appendix Section B.4. Year-to-year changes are rare, so in practice the interpolation carries a single observed state across the years an annotator did not label. We test the sensitivity of our empirical results to these interpolations in Section 6. While the two annotators agreed in almost all cases, a small set of sampled pixels was either never reviewed by either annotator due to lack of high-resolution satellite imagery or had disagreements between the annotators. The authors reviewed these disagreements and missing values manually. Appendix B.4 and Appendix Table A2 report the full details of these processes.

4.5 Efficiency Gains

We can reassess the model's classification performance in the verification sample in order to measure accuracy at the point of deployment rather than in the training hold-out set, which was not representative. For dump levels, the model achieves a weighted AUC of

0.97. At the population-optimal classification threshold, weighted balanced accuracy is 95%. Performance is even stronger for changes: the AUC is 0.99, with balanced accuracy of 97%. These strong results give reassuring evidence that our model remains accurate outside of the selected training data.

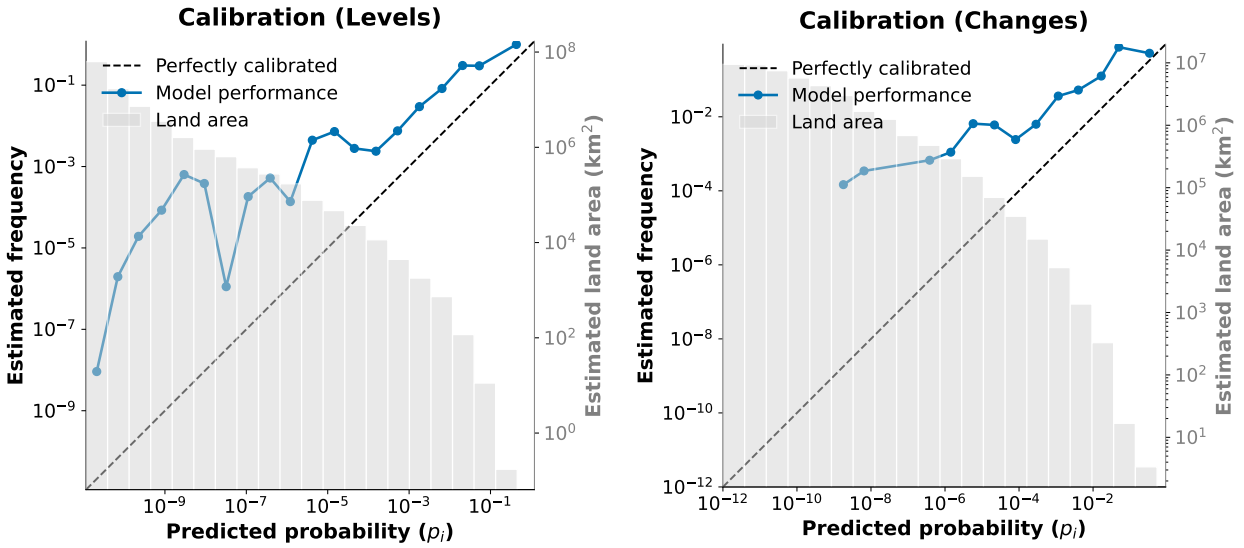
However, we do see evidence of biases. Appendix Table A6 shows the results of regressions of prediction errors on year dummies, world region dummies, a quadratic of log GDP, and log population density. Relative to 2017, our model under-predicts dumps in 2018 and 2019—the two years after the waste ban—meaning that the naive method would underestimate an increase in dumps that occurred during those years. Our model tends to predict too many dumps in Europe, Central Asia, and Sub-Saharan Africa compared to East Asia & the Pacific, and too few dumps in areas of high population density. The magnitudes of these coefficients are on the same order as our estimated prevalence of dumps, meaning that the differential bias between regions can be just as large as the actual dump prevalence.

What is crucial for our sampling strategy, however, is not necessarily accuracy or biases, but calibration. Figure 3 shows model calibration plots. The x-axis shows model-predicted probabilities and the left y-axes show the estimated frequency of pixels in that probability bin that were verified as containing an open dump (on the left) or changing status at least once (on the right), using the inverse probability weights. If our probabilities were well calibrated, the blue line would lie the 45-degree line. Clearly, this is not exactly the case—at very low probabilities, we underestimate the probability of dumps. Still, it is clear that our model's predictions contain useful information, which helped to appropriately allocate our verifications. A similar story holds for our predicted changes, as seen in the right panel. In both plots, the gray bars show a histogram of the number of pixels at each predicted probability estimated for the entire world, with the axes in log scale on the right.

We can use the verifications to estimate the ex-post efficiency of our proxy-Neyman strategy, as described in Appendix Section A. For estimating the prevalence of dumps, we find a design effect of 0.044, indicating that we can estimate mean dump prevalence with a variance 95.6% smaller than if we had used uniform random sampling. The corresponding calculation for our power to detect changes gives DEFF = 0.041. In other words, when estimating the prevalence of dumps, our 14,086 observations give us the same effective sample size as if we had verified approximately 318,000 observations using uniform random sampling. On the other hand, this DEFF is nearly 37 times *larger* than the DEFF* estimated from the Argentina scene in 4.3 above, indicating that model miscalibration likely reduced the value of our sampling strategy relative to what a perfectly calibrated model could have achieved.[14]

[14]The ex-ante estimate of DEFF* could be overly optimistic relative to the true feasible optimal lower bound

Figure 3: Model Calibration



Notes: Figure shows calibration plots in blue for $p(k_i)$ and p_i^{Δ} against dashed black 45-degree line. Model-estimated probabilities are on the x-axis, and relative open dump prevalence and change frequencies are on the left y-axis in black. Gray histograms and right y-axes show the estimated number of pixels at each estimated $p(k_i)$ and p_i^{Δ}. All axes are on a log scale.

Still, the potential gains were so large that the proxy-Neyman strategy proved valuable, despite some degree of miscalibration.

5 Global Dump Descriptives

This process allows us to generate a novel set of descriptive facts about the global distribution of open dumps. We estimate aggregate and conditional means using the bias-corrected estimator (Equation 1), which combines the labels with the model's predictions. In some cases, we also use a Horvitz–Thompson estimator on the verifications for comparison, or for estimating prevalence in years when the model's predictions are unavailable.

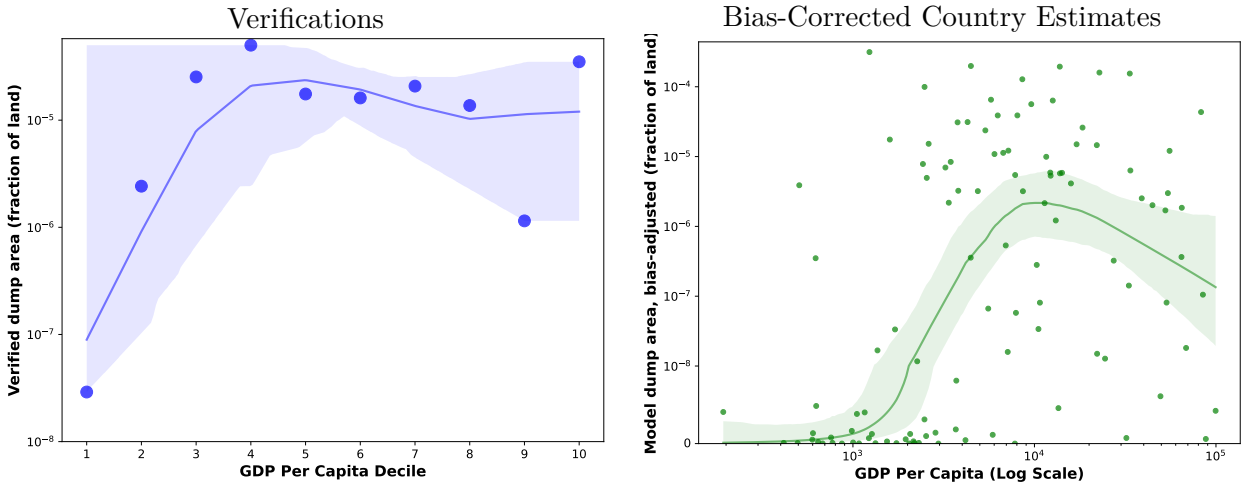
We estimate that the fraction of global land area covered with open dumps in 2018 was 3.2×10^{-5} (95% CI: $[1.5,\ 5.5] \times 10^{-5}$). This corresponds to an area of more than 3,000 square kilometers, larger than Yosemite National Park. For another point of reference, this is about 0.3% of estimated urban land surface area in 2020.[15] Note that, as discussed above, this

if σ_i is overly dispersed relative to σ_i^*. In Appendix Section A, we discuss how to estimate the ex-post feasible DEFF* using the verification sample. While these estimates are somewhat sensitive to assumptions about how to model p_i^*, the lower end of the estimates in Table A1 are comparable to the ex-ante estimate from the Argentina tile.

[15] Estimates of urban land area from Gao and O'Neill (2020).

estimate does not include covered waste, for example, buried in sanitary landfills, thus, it is not intended to measure the total footprint of human waste, just open dumps as defined in Section 4.

Figure 4: Dump Prevalence vs. GDP Per Capita



Notes: Estimates of dump prevalence vs. country GDP per capita (World Bank $2023). Left panel shows average prevalence using only verified pixels within country-GDP per capita deciles. Right panel shows AIPW estimates from Equation 1 within deciles. Plots show LOESS smoothed lines and 95% confidence intervals from a stratified tile-cluster bootstrap with 1,000 replicates.

Figure 4 shows the prevalence of open dumps plotted against GDP per capita.[16] The plots clearly show an 'Environmental Kuznets Curve' (EKC) in both the verified points (left) and the AIPW bias-corrected country-level estimates (right). Low-income countries have the lowest prevalence, presumably because they generate less waste, though it is also plausible that littering is more prevalent, and this more dispersed disposal method is harder for us to detect in the satellite record. Dump prevalence rises sharply for middle-income countries, then declines slightly for high-income countries, though the right side of the EKC is fairly flat. This decline could reflect increased use of sanitary landfills and other waste disposal techniques that are less visible to satellites and likely less environmentally harmful.

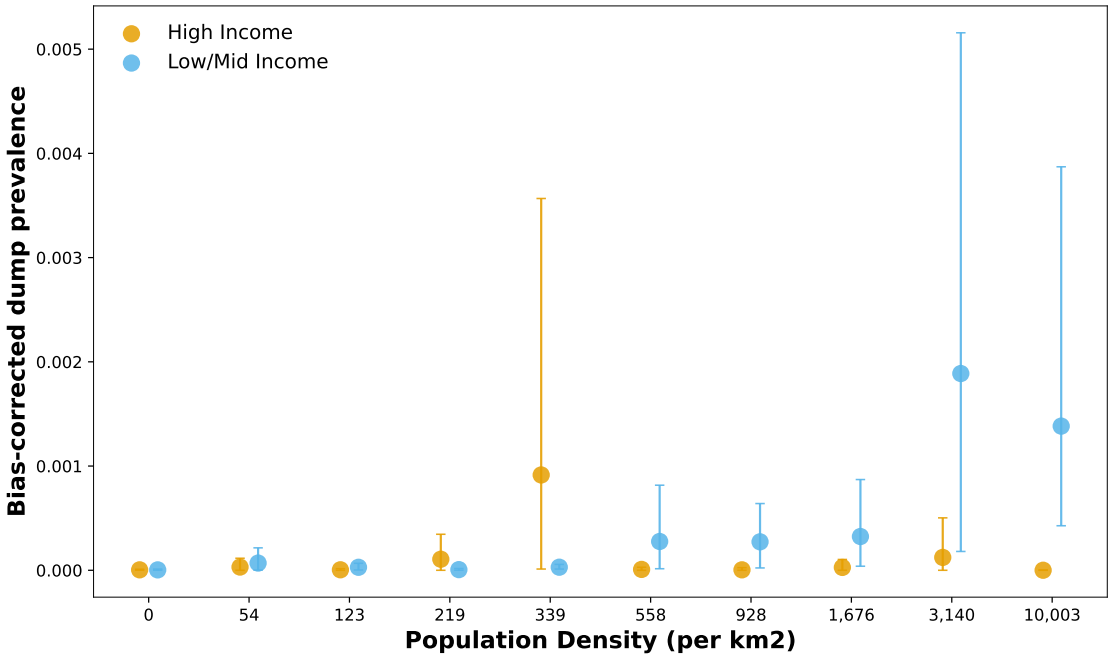
Figure 5 shows the prevalence of open dumps with respect to population density.[17] In low- and middle-income countries, dumps are concentrated in areas at the 80–100th percentile of population density. Waste is generated in population centers and is costly to transport, facts which are sufficient to explain this pattern. However, given the externalities associated with

[16] GDP data from the World Bank in 2023 USD.

[17] We measure population density at the pixel level using the Gridded Population of the World dataset, which provides population estimates on a 1 km grid for 2010, 2015, and 2020 (Earth Science Data Systems, NASA, 2024).

open dumps, optimal policy might involve reducing human exposure by locating dumps in less-dense areas. We see some evidence of this in high-income countries represented by the dots in orange. In these countries, dumps are less prevalent overall, but, more importantly, we see no open dumps in the densest cities.

Figure 5: Dump Prevalence vs. Population Density



Notes: Estimates of the prevalence of open dumps, by population density decile using Equation 1. Deciles defined such that each includes 10% of the world's population. Orange points show estimates for high-income countries, and blue points show combined low- and middle-income country estimates (World Bank classification threshold of $13,935 GDP per capita). Whiskers show 95% confidence intervals from a stratified tile-cluster bootstrap. Population density data are based on a 1 km grid from Earth Science Data Systems, NASA (2024).

We now turn from describing the location of dumps to showing evidence of externalities—the prevalence of fires in open dumps. Dump fires may be set intentionally, but they can also occur naturally due to the heat generated by decomposing organic waste. We use the NASA VIIRS Active Fire Detection product, which uses satellite-based sensors to detects fires globally at 375 m resolution (Schroeder et al., 2025). We overlay fire detections with our verified and predicted dumps to estimate the prevalence of global garbage fires.[18] Appendix Table A7 shows the results by income and population density.

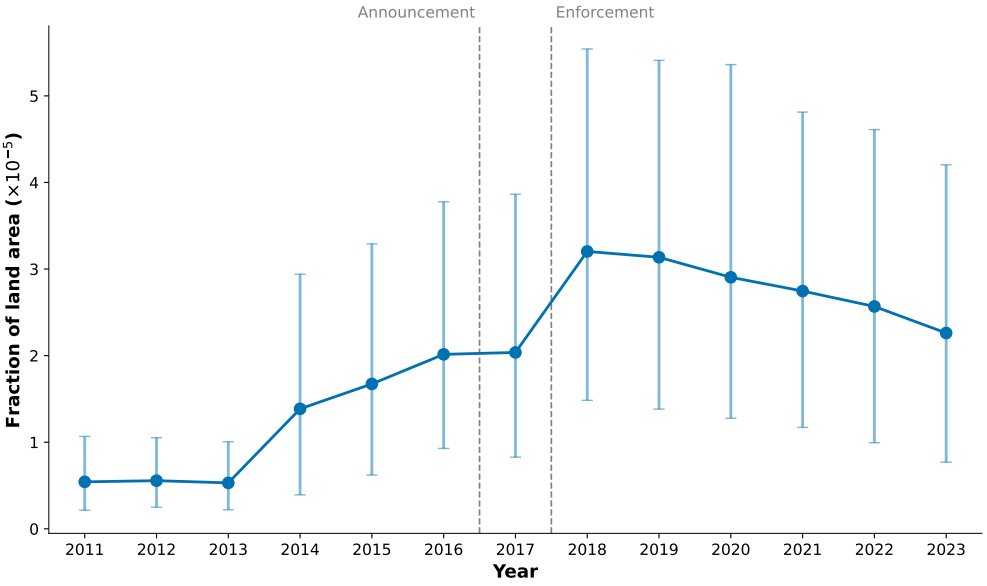
Strikingly, 16.7% of dump pixels in low- and middle-income countries have at least one fire

[18]These products differ in resolutions (375 m vs 10 m), so one fire detection can intersect multiple dump pixels.

detection in a given year. In high-income countries, the number is 3.6%. In China, the share of dump pixels ever on fire is even lower, at 2.4%, averaged across years. Dumps in low- and high-density locations are on fire at similar rates. However, because dumps are much more prevalent in dense areas, burning dumps are concentrated where people live. These facts highlight the importance of waste mismanagement as a crucial public-health issue in low- and middle-income countries, which we examine directly in the case studies in Section 7.

Finally, Figure 6 shows the time trend of our global estimates.[19] A large jump between 2017 and 2018, the year the Chinese waste import ban took effect, is immediately visible. Point estimates indicate a 58% increase in global dump area, though confidence intervals are large. While we do not claim causality based on this figure alone, the results strongly suggest that the ban led to a global increase in dump area, despite an overall decrease in international trade flows. Another notable jump in the time series occurs between 2013 and 2014, corresponding to the Green Fence Policy, when China increased inspections of waste imports (see Section 2).

Figure 6: Global Results: Share of Land Area Covered by Open Dumps



Notes: Horvitz–Thompson estimates using the 14,086 verified pixels. 95% confidence intervals from a stratified tile-cluster bootstrap. Dashed lines show China's waste ban announcement (2017) and enforcement (2018).

These increases are not a priori obvious. If a large fraction of the pre-ban traded waste ended up in open dumps, we should expect dumps to decrease after the ban, especially if

[19] These time-series figures use the Horvitz–Thompson estimator on only the verified pixels, since satellite data, and therefore model predictions are only available for 2017–2021.

the alternative is sanitary landfills in high-income countries. Instead, Panel A of Appendix Figure A7 shows that dumps increased after the ban, even within China itself. This striking pattern suggests that the recyclers that were shut down after the National Sword Policy was enacted may have been processing Chinese domestic waste as well as foreign imports. After the ban, open dumping in China increased because domestic waste was no longer recycled. These results conflict with previous work showing that reduced waste imports led to lower air pollution in Chinese recycling cities (Guo et al., 2023; Sigman and Strow, 2024). One way to reconcile the findings is if pollution from waste imports came from recycling activities rather than open burning. Another possibility is that the increase in dumps occurred outside Chinese recycling centers; consequently, this spillover effect may contaminate regressions using other Chinese cities as controls. These are important questions for future research that will require more high-resolution data on the distribution of dumps within China.

Panel B of Appendix Figure A7 shows that the increase in dumps after 2018 was not confined to China. Thus, another source of the increase is that some of the waste imports previously destined for China ended up in the open dumps of other countries. We explore this further in the following section.

6 Dumps and Trade

We use the natural experiment induced by the Chinese waste import ban to estimate how much open-dump area each additional ton of traded waste creates. On the one hand, if imported waste is fully recycled, we might not see any increase in open dumps. However, if the raw material used in recycling is contaminated and a some fraction is discarded, or if imported material replaces local waste inputs in recycling processes, we should see a rise in dumps in countries that saw their imports increase. Our data are uniquely suited to study this question.

6.1 Empirical Strategy

We use BACI trade data, which gives information on the value and mass of waste traded between countries for 2012–2023 (Gaulier and Zignago, 2010), with corrections for misreporting. Our main measure of waste imports covers plastic waste and scrap (HS 3915) plus waste paper (HS 4707), the two categories at the center of the import ban, which together made up over 93% of the tonnage China imported across all banned categories in 2016. In alternative

specifications we consider additional categories of banned waste, including slag and textiles.[20] Our independent variable of interest is constructed as each country's change in *net* waste imports (imports minus exports) from before the ban (2016) to after the ban (2019).[21]

Because the treatment is a change in *net* imports, it increases both for countries that received increased imports (e.g. Vietnam and Indonesia) and for countries that reduced their exports (e.g. the United States). Appendix Figure A8 maps the distribution of the treatment. Pooling the two assumes that a ton kept at home in an exporting country and a ton received by an importing country add to open dump area at the same rate. We examine heterogeneity below, and report results without the US (Appendix Figure A12 and A13).

Our independent variable of interest is the area of open dumps. As discussed in Section 4, we only measure the surface area of dumps, so we can only detect extensive margin increases in waste mismanagement. To the extent that imported waste is piled up or compressed in landfills, it will bias downward our estimates of the fraction of mismanaged waste. We primarily use the manual verifications, as described in the previous sections, since they give a longer time series (2011–2023) than the machine learning predictions, which only cover 2017 to 2021. In supplemental analyses, we show how it is possible to incorporate these predictions in the years in which they are available. We exclude China to focus on displacement effects—as discussed above, Appendix Figure A7 shows that dumps *increased* after the waste ban in China). Our final sample covers 115 countries, however we only verify greater than zero dump pixels in all years for 31 of these countries.

We leverage changes in imports around the timing of China's National Sword Policy as a quasi-exogenous shock. Because the treatment varies at the country level, we first estimate a continuous two-way fixed effects model on a country-by-year panel:

$$\text{DumpArea}_{it} = \beta \Delta\text{NetImports}_i \times \text{Post}_t + \gamma_i + \delta_t + \varepsilon_{it}, \tag{8}$$

[20] We use three alternative definitions, one with all banned categories without slag (which adds textile and fiber waste, HS 5103–5104, 5202, 5505, 6310), one with all banned categories including slag (also adds HS 2619–2620), and a narrower definition only including plastic scrap and unsorted paper (HS 3915, 4707.90). See https://www.laregionalagency.us/comprehensive-list-of-banned-materials/ for the list of banned materials. Values are available for most countries in all years. To create a complete panel of trade *quantities*, we impute missing quantity observations by applying the exporter-specific average quantity-to-value ratio within HS heading and year, and fill the very few remaining gaps with the global ratio for that heading and year.

[21] $\Delta\text{NetImports}_{i,2016-19} = (\text{Imports}_{i,2019} - \text{Exports}_{i,2019}) - (\text{Imports}_{i,2016} - \text{Exports}_{i,2016})$, measured in thousands of tons.

and its corresponding event-study specification:

$$\text{DumpArea}_{it} = \sum_{k=2011, k\neq 2016}^{2023} \beta_k \Delta\text{NetImports}_i \times \mathbb{1}[t = k] + \gamma_i + \delta_t + \varepsilon_{it} \tag{9}$$

where DumpArea_{it} is verified open dump area (km^2) in country i and year t.[22] The continuous treatment variable, $\Delta\text{NetImports}_i$, captures a country's change in waste trade exposure and is calculated as the difference in net waste imports between 2019 and 2016, as described above. With a continuous treatment, β is a weighted average of per-kilogram responses and interpreting it as a single per-kilogram effect assumes the response is linear (Callaway et al., 2024). In Equation 8, we interact this time-invariant continuous measure with Post_t, an indicator variable for years after the policy's 2017 announcement. Because the policy was announced in July 2017 and enforced from January 2018, 2017 is neither a clean pre- nor a clean post-ban year, so we exclude it from Equation 8. The event study lets us check for differential pre-ban movements in dump area among the countries that received large shocks, and shows how the effect evolves after the ban. Both models include country (γ_i) and year (δ_t) fixed effects.

We estimate the same designs on the un-aggregated verified pixels themselves. The outcome is verified pixel p's dump status in year t multiplied by the same country land-area multiplier used for the country outcome, so that coefficients are in the same units as the country-level model. The treatment is still the pixel's country-level import shock. We replace country fixed effects with sampling-tile fixed effects. Each pixel enters with its sampling weight, and standard errors remain clustered at the country level, the level of treatment. The two aggregations answer slightly different questions. The country model weights every country equally and asks whether countries hit by larger trade shocks saw their dumps grow more. The pixel model weights countries by land area and estimates the dump area added per kilogram for the average square kilometer of land.

For β to identify the causal change in dump area per kilogram of net imports, $\Delta\text{NetImports}_i$ must be uncorrelated with unobserved drivers of trends in open dumping. This is plausible in our setting—not only was the timing of the ban unanticipated, but the magnitude of the import shock was mostly determined by the structure of pre-ban international trade patterns.[23] Open dump area, by contrast, is a slow-moving stock. The suddenness and scale

[22] This is the weighted prevalence estimate of Section 4.3 multiplied by the country's land area inside the sampling frame, estimated from the same design weights.

[23] Since China has a significant trade surplus with the US in manufactured goods, shipping from the US to China is cross-subsidized by the need for container ships to make a round trip, thus making it affordable to ship recycled materials from the US to China (Kellenberg, 2010). Since these ships continued to make

of the ban therefore work in our favor as a discrete, externally timed jump in imports is easy to distinguish from the smooth background trends that country and year fixed effects absorb.

However, it is still plausible that realized trade flows after 2017 were partly chosen in response to the ban. Countries with weak environmental enforcement, for example, may have attracted more of the diverted waste and also seen dumps expand for domestic reasons. We therefore also implement a shift-share instrumental variables version of the model above that uses only the component of the import change predicted by the pre-ban structure of the trade network. For each exporting country j we measure the flow of banned waste it sent to China before the ban, and we allocate that soon-to-be-displaced flow across destinations in proportion to j's pre-ban exports to destinations other than China:

$$z_i = \sum_c \sum_j s^c_{ji}\, g^c_j, \tag{10}$$

where g^c_j is exporter j's average 2015–2016 exports of waste category c to China (i.e., the shift) and s^c_{ji} is destination i's share of j's 2015–2016 exports of c to non-China destinations (i.e., the shares), constructed separately for each HS code and summed. The instrument z_i is thus the diversion *predicted* for country i had every exporter redirected its stranded China volumes along its pre-existing non-China routes. In the IV specification we instrument $\Delta\text{NetImports}_i \times \text{Post}_t$ in Equation 8 with $z_i \times \text{Post}_t$. The shifts all stem from a single policy shock, so identification rests on the exogeneity of the pre-ban *shares* (Goldsmith-Pinkham et al., 2020): countries' positions in the pre-2017 waste-trade network must be unrelated to counterfactual *trends* in open dumping. We provide evidence on this assumption below.

Standard errors are clustered by country throughout. Because treatment varies only across countries and its distribution is skewed, we also report randomization-inference (RI) p-values for the OLS coefficients, which permute the exposure vector across countries. For the IV coefficients we also calculate Anderson–Rubin p-values (Anderson and Rubin, 1949), which are valid at any first-stage strength (Andrews et al., 2019).[24]

the round trip journey after the waste ban, the most convenient destinations for the recycled goods were the places that were on the way back to China, or only required a short detour.

[24] AR p-values invert the Anderson–Rubin test on a grid. RI p-values permute the country-level exposure vector 2,000 times and compare the studentized coefficient to its permutation distribution. The stratified version permutes within four strata of pre-ban dump area: countries with none, and terciles of the rest.

6.2 Results

Table 1 shows our main estimates, with our country-level aggregation in Panel A and our pixel-level version in Panel B. Column (1) shows the OLS version, while Column (2) shows the shift-share IV version. In both panels the coefficients can be interpreted as m^2 of additional dump area per kg of additional net waste imports. Both estimators give positive, statistically significant effects of similar size. The OLS estimates imply 0.054 m^2 (95% CI [0.015, 0.093]) or about a 9-inch-by-9-inch square of additional dump area per kg of additional annual net imports at the country level and 0.078 ([0.064, 0.092]) at the pixel level, while the shift-share IV estimates are 0.064 ([0.006, 0.122]) and 0.076 ([0.048, 0.104]). The reduced-form coefficients are statistically significant and are shown in Appendix Table A9 along with the complete first stage results. The implied elasticities are large. Evaluated at sample means relative to pre-ban dump area, elasticities are 1.4–1.7 at the country level and 1.8 at the pixel level.

Figure 7 plots the event study version of Column (2) in Table 1, while Appendix Figure A9 shows the OLS, reduced-form, and the first-stage event studies. The dynamics line up with the ban's timeline as the pre-ban coefficients are flat across 2011–2015 and turn positive within the first two years after the 2017 announcement.

The first stage is moderate at the country level (cluster-robust F of 11.3) and weak at the pixel level (3.8). Because conventional standard errors are unreliable when the first stage is weak, Table 1 also reports Anderson–Rubin p-values. For the country-level estimates, these are similar to the Wald p-values. For the pixel-level estimates, the Anderson–Rubin p-values are more conservative than the Wald p-values (0.088 against $p < 0.001$), but still statistically significant at the 10% level. As a different check of the weak and moderate first stage, we re-estimate the main specification under four different definitions of banned products in Appendix Table A12. The narrowest definition includes plastic scrap and only unsorted recovered paper (HS 3915 and 4707.90). This version produces a far stronger instrument, with first-stage F of 40.3 at the country level and 18.6 at the pixel level, because the shock it isolates is the outright ban rather than the contamination-threshold tightening that applied to the rest of HS 4707. The first-stage strength comes at the trade-off of some variation, so some of the estimates are correspondingly less precise. However, the point estimates are, if anything, larger, and the IV coefficients are very stable across all of the trade definitions.

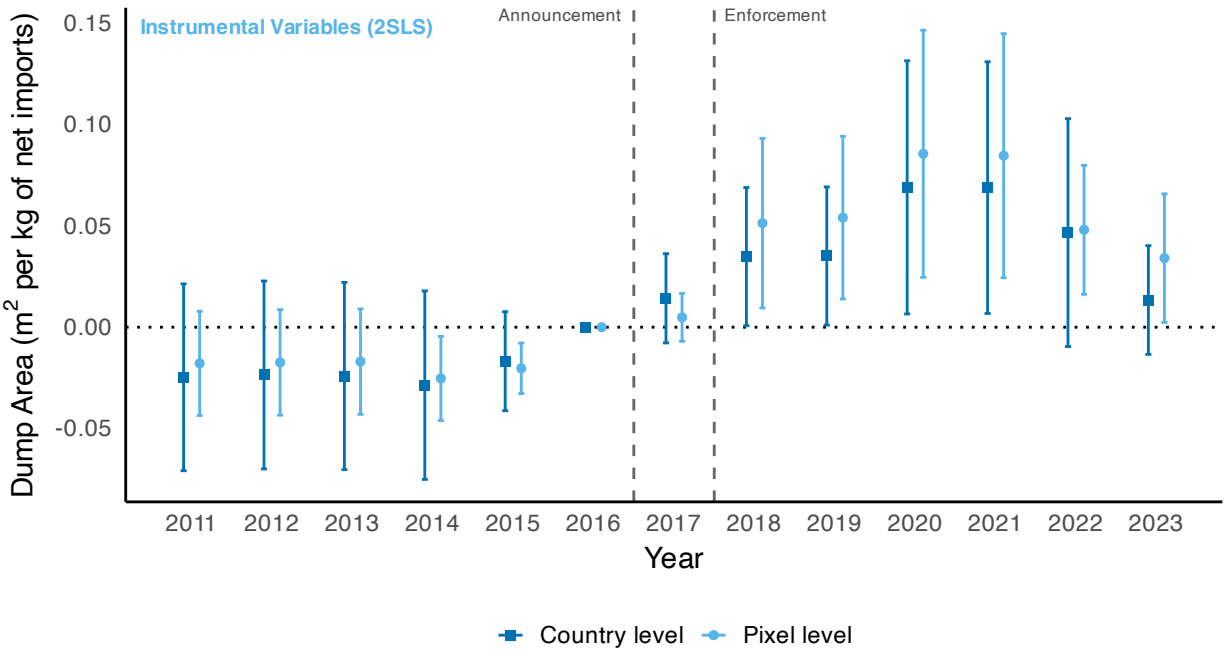
We next discuss evidence supporting the main identifying assumption that pre-ban trade network position of countries is unrelated to their counterfactual trends in dumping. The assumption cannot be tested directly, but three kinds of evidence are consistent with it.

Table 1: Effects of Change in Net Imports on Open Dump Area (Pre/Post Ban)

	Dependent variable: Dump area (km^2)	
	OLS	IV
	(1)	(2)
Panel A: Country-Level		
Δ Net Imports (kt)	0.054***	0.064**
× Post-2017	(0.020)	(0.029)
	[0.008]	[0.030]
Implied elasticity	1.445	1.722
First stage		0.473*** (0.140)
First-stage F		11.3
RI p-value	[0.002]	
RI p-value, stratified	[0.062]	
AR p-value		[0.025]
Observations	1,380	1,380
Countries	115	115
Panel B: Pixel-Level		
Δ Net Imports (kt)	0.078***	0.076***
× Post-2017	(0.007)	(0.014)
	[0.000]	[0.000]
Implied elasticity	1.812	1.751
First stage		0.789* (0.407)
First-stage F		3.8
RI p-value	[0.003]	
RI p-value, stratified	[0.024]	
AR p-value		[0.088]
Observations	146,304	146,304
Countries	115	115

Notes: Table shows results of estimating equation 8. Coefficients in both panels are m^2 of dump area per kg of increase in net imports (numerically identical to km^2 per kt). Treatment is change in net imports of plastics and recovered paper (HS 3915 + 4707) × post (post = year > 2017). China notified the WTO in July 2017 and enforced the ban from January 2018, so 2017 is neither a clean pre- nor a clean post-ban year; it is excluded from the estimation sample, so the pre-ban baseline is the years through 2016. Appendix Table A8 reports the same specification with 2017 kept in the sample and given its own interaction. Instrument is predicted diversion from pre-ban bilateral trade shares (per-HS-code shift-share, equation 10). Panel A: country × year panel, outcome = verified dump area. Panel B: verified-pixel × year panel, outcome = pixel dump status × country land area, weighted by inverse sampling probabilities. Implied elasticity: coefficient × $\bar{X}/\bar{Y}$, with $\bar{X}$ the estimation-sample mean net-import change and $\bar{Y}$ the pre-ban (year ≤ 2016) mean outcome — the percent increase in dump area relative to its pre-ban mean implied by a one-percent larger rise in net imports. First stage: realized change × post on predicted diversion × post (coefficient, SE, and cluster-robust F). Full reduced-form and first-stage estimates are shown in Appendix Table A9. SEs clustered by country in parentheses, p-values in brackets; China excluded. The AR p-value inverts the Anderson–Rubin test. RI is a randomization-inference p-value; the stratified version permutes only within four strata of pre-ban dump area (countries with none, and terciles of the rest). *p< 0.1; **p< 0.05; ***p< 0.01.

Figure 7: Effects of Change in Net Imports on Open Dump Area, Event Study Version (IV)



Notes: Figure shows year-by-year IV estimates of Equation 9 where each Δ net imports × year interaction is instrumented by predicted diversion (z_i, Equation 10) × year, with 2016 as the omitted reference year. Darker squares show the country-level panel (country and year fixed effects, unweighted), while lighter circles show the pixel-level panel (tile and year fixed effects, weighted based on sampling probabilities). Coefficients are m^2 of dump area per kg of net imports; dashed vertical lines mark the ban's announcement (July 2017) and enforcement (January 2018). Bars are 95% confidence intervals from standard errors clustered by country. OLS, reduced-form, and first-stage versions of the same figure appear in Appendix Figure A9.

First, in Appendix Table A10 we note that while the instrument correlates with some baseline country characteristics (levels of which are absorbed by country fixed effects), it is uncorrelated with the pre-ban trend in dump area on its own. Second, the pre-ban coefficients in Figure 7 and Appendix Figure A9 are flat across 2011–2015.[25] Finally, we run two placebo exercises. The first is a commodity placebo that rebuilds the shift-share instrument using primary-form plastics (HS 3901–3914), an upstream commodity China never banned. This "predicted diversion" captures generic trade-network position with no actual shock attached. When we include this placebo instrument in the reduced-form regressions, the real instrument

[25]The joint test that the five pre-ban IV coefficients are all zero rejects at both the country level ($p = 0.007$) and the pixel level ($p < 0.001$). A common level difference from the reference year also produces that rejection. The five coefficients are statistically indistinguishable from one another ($p = 0.25$ at the country level, $p = 0.58$ at the pixel level) and show no linear drift across relative years ($p = 0.67$ and $p = 0.64$), and the OLS event studies show the same pattern. The rejections therefore reflect 2016 sitting above a flat 2011–2015 plateau rather than a differential trend.

retains its effect while the placebo is close to zero (Appendix Figure A10).[26] Our second placebo is a timing one that moves the date of the ban. We re-estimate the design on pre-ban years only (2011–2016) with a counterfactual ban in 2014 and reassuringly find a pooled coefficient near zero and a flat year-by-year series (Appendix Figure A11). Overall, these tests show that exposure is uncorrelated with pre-ban dump trends on its own, no effect is visible before the ban, none appears at a placebo date, and none flows through a placebo commodity.

Separate from identification, measurement is a potential concern given that annotators were not required to label every year for every pixel (and roughly 45% of label cells are filled in from the same pixel's recorded years, as shown in Appendix Table A2). Filling smooths each pixel's series, so it shrinks the measured response to the ban. If filling were more common in some countries than others, the common post-ban rise in dump area would be measured differently across countries. If that difference were correlated with exposure, it could bias β in Equation 8. Appendix Table A11 shows that label density is indeed positively correlated with exposure measures: more exposed countries were labeled more densely, so their series were smoothed less, which could bias β upward. This correlation largely reflects income and population density. Two checks suggest that these correlations do not drive our estimates. First, we rerun our main specification with the two-thirds of countries with the densest labels. Second, we control directly for label density $\times$ post. Both of these checks leave the OLS and IV estimates essentially unchanged (Appendix Figure A12).

We further test the robustness of our findings. Appendix Figure A12 re-estimates the main OLS and IV results under fourteen alternatives: a yearly-dose version in which annual net imports are instrumented by the time-varying predicted diversion; replacing year with land-area-quartile-by-year fixed effects; restricting to the 31 countries with verified dumps in every year; restricting to the two-thirds of countries whose pixels are most densely labeled by the annotators; controlling for each country's label density interacted with post; substituting the debiased machine learning area for the verified area; equalizing pixel design weights within countries; keeping 2017 in the sample, either with its own interaction or folded into the pre-ban baseline; a pooled-network version of the instrument; adding baseline GDP per capita, population density, and land area $\times$ post controls; comparing countries only within baseline-dump-area quartiles interacted with year; rebuilding the instrument without the largest Rotemberg-weight cell (US waste paper)[27]; and dropping the United States, the

[26] We run the reduced form version of the regression as the two exposures are highly correlated and a placebo regression alone would simply proxy the real effect.

[27] We note that the Rotemberg weights underlying the instrument are nearly all positive although concentrated, as shown in Appendix Table A13. Rebuilding the instrument without the largest weight

country with the largest change in net imports. Point estimates are stable across every specification and remain statistically significant across most.[28]

Finally, Appendix Figure A13 re-estimates the main specification within groups of countries, defined by pre-ban trade position and by income. Among net-importing and low- and middle-income countries, which are the main destinations of diverted waste, the IV estimate is close to the pooled estimate but less precise, and the OLS estimate is smaller. For net exporters and high-income countries, the estimates are driven by the United States.[29] We also split the pixel-level outcome by population density across land. Almost all of the newly added dump area appears in the densest 5% of the land, where 61% of the population lives. This is consistent with the descriptive results shown in Section 5 that show that dumps are concentrated in densely populated areas.

To summarize, the core finding of an increase in dumping area in response to increases in net imports is stable in magnitude across two aggregation levels, two estimators, and the majority of the fourteen alternative specifications of Appendix Figure A12, while both placebo exercises find no effect where none should exist.

Are the large magnitudes presented above realistic? Although we measure only the surface area of dumps, our estimates imply a depth once combined with typical waste densities. Importantly, these simple calculations assume that *all* of the imported waste is dumped. While this is unlikely to be true, it is nevertheless a helpful exercise in interpreting the magnitude of the coefficient we estimate. We can estimate depth $= T/(\beta \times \rho)$, where T is the number of years the extra waste has been flowing, β is the regression coefficient (m^2 per kg of annual net imports), and ρ is the density of waste (kg per m^3). With $T = 3$ years (the average post-ban horizon in our sample) and $\rho = 175$ kg per m^3 for loose plastic and paper waste, the country-level IV point estimate implies a layer 27 cm deep, and the ends of the confidence interval bracket 14 cm to 2.9 m. These depths correspond to different kinds of dumping that our back-of-the-envelope calculation cannot separate, but none is implausible. A 14–27 cm layer is consistent with the informal roadside dumping typical of the main receiving countries, while depths of a couple of meters point to larger, more consolidated sites, again with the caveat that these numbers assume all imported quantities are dumped.

leaves the estimate unchanged.

[28] The exception is the OLS version without the United States. However, the IV estimates here too are similar in magnitude to the main version, although somewhat less precise.

[29] The US's net imports rose more than any other country's as its exports to China collapsed. Excluding it, the exporter and high-income estimates fall to approximately zero. We note that this matches the Drop US specification in Appendix Figure A12: without the United States, the IV estimate is similar in magnitude to the main estimate, though less precise, while the country-level OLS estimate is not.

7 Case Studies: Java, California, Southeast Brazil

The analyses above are globally representative and well-suited to analyzing the cross-country determinants of open dumps. Even with our optimal sampling methods, our 14,086 observation verification sample, which contains mostly zeros, is not sufficiently well powered to be able to detect the local externalities associated with dumps, however. To more precisely study the disamenities of open dumps, we choose a select subset of sub-national geographies and use our model to find as many dumps as we can.

We select three regions based on a combination of factors. First, we select the island of Java in Indonesia. Java is not only one of the most densely populated regions on Earth, but it was also a major recipient of diverted scrap materials after the 2018 China import ban, due to its convenient location along the US-China trade route (see discussion in previous section). Furthermore, Indonesia was a major source of training data for our model, as early attempts to use satellite data to detect dumps focused on Indonesia and found a large number of informal sites that we incorporated into our training pipeline (Kruse et al., 2023). For this reason, we expect our model to perform well there.

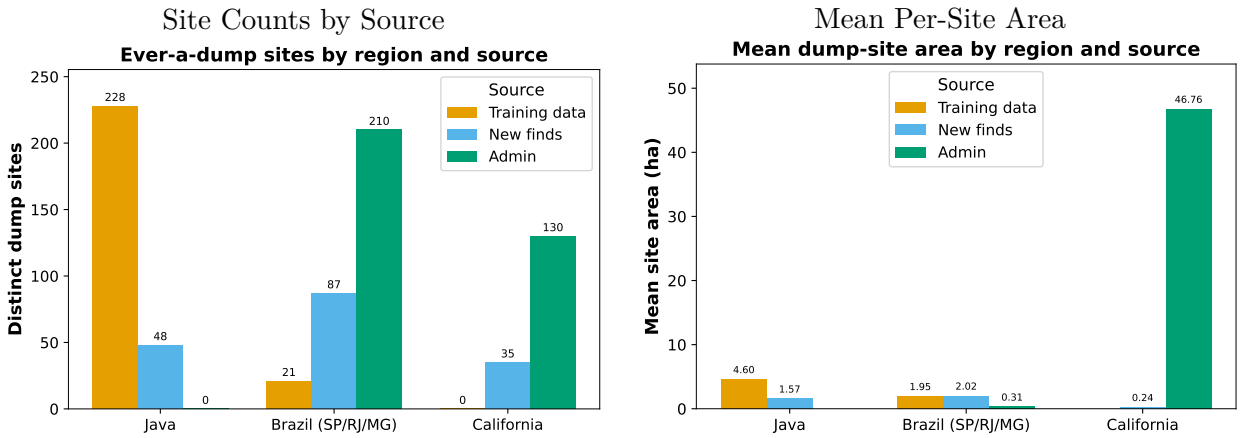
Second, we choose the state of California in the United States. While Indonesia is a middle-income importer of waste, California is a high-income exporter. Analysis of the dumps in that state allows us to see whether reduced exports after the China waste ban increased landfill-related pollution in the US. Furthermore, we can compare and contrast the open dumps that our model finds to the list of formal sanitary landfills tracked by the EPA Landfill Methane Outreach Program (LMOP). This database reports the location and area of municipal solid waste landfills.

Finally, we look at three densely populated states in Southeast Brazil: Rio de Janeiro, São Paulo, and Minas Gerais. Brazil is not a noteworthy importer or exporter of waste, and saw only a very small increase in imports after 2018 due to its distance from major trade routes. Brazil is one of the few countries, however, that attempts to administratively track both open dumps and sanitary landfills through the government registry of the National Sanitation Information System (Nacional de Informações sobre Saneamento, SNIS). This allows us to compare our model predictions to known open dumps in a country where we did not have much training data.

In each region, we use our existing model to predict the probability of a dump for every pixel. Second, we set a region-specific threshold probability, and use a clustering algorithm (DBSCAN) to draw a polygon around pixels with p_i greater than or equal to the threshold.

Then we have expert annotators review all of these polygons and mark which ones were open dumps. The thresholds were set to give a manageable number of polygons to review in each region. In Java, we reviewed 348 polygons and found 48 new open dumps that were not previously in our training data. In Brazil, we reviewed 1,091 potential sites and found 87 new, undocumented open dumps. In California, we reviewed 931 potential sites and found 35 dumps not listed in the EPA LMOP data.

Figure 8: Case Studies: Dump Sites by Source—Known Sites and Model New Finds



Notes: Dump sites by source. Admin based on 2018 data, and includes SNIS registered dumps (Brazil), and EPA LMOP landfills (California). Training data described in Section 4.1. New finds are expert-verified in high-resolution satellite imagery after reviewing model predictions, greater than 100 m from known open dumps.

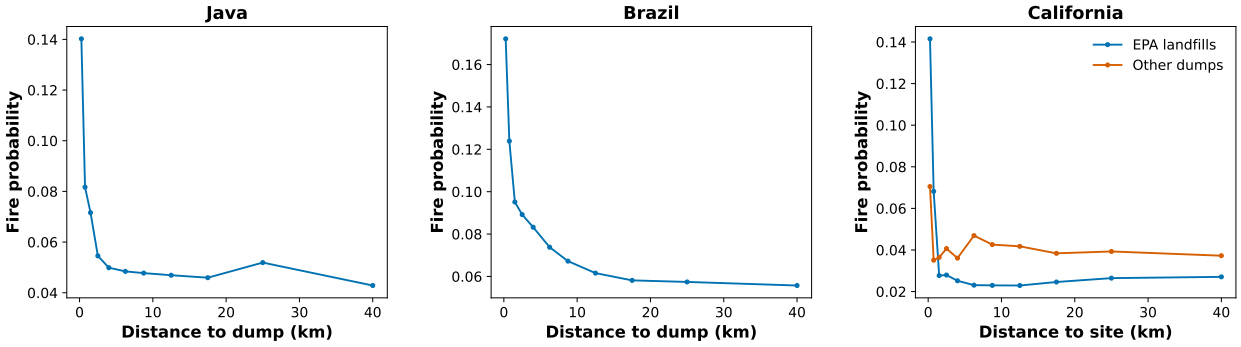
This exercise shows that, even when official data on open dumping exist, our model is able to locate dumps that are not known to authorities. Undoubtedly we are still missing some dumps in each region, however the relatively low success rates obtained by our verification efforts suggest that further lowering our thresholds in order to find more dumps would have resulted in many more potential polygons to investigate, the vast majority of which would be false positives. Figure 8 shows the number of sites in each region by source on the left, and the mean site area by source on the right. When we sum the area of all found dumps from any source, and compare it to the area of the region, we can compare these estimates to our estimates of the global prevalence of dumps from our representative global data in the previous sections. This is done in Appendix Figure A14. Our estimate for Java is somewhat higher than the global estimate, plausibly because of the high population density of the island. On the other hand our estimate for the three Brazilian states is somewhat lower—possibly because our model is less accurate in Brazil, and thus there are more dumps that we have yet to find. California is not directly comparable, since it includes the sanitary landfills in the EPA LMOP data that our model is not designed to capture. Figure 8 shows

that these landfills are much larger than the typical open dump.

Next, we again use the NASA VIIRS active fire detections and overlap them with our known dumps in each region. Figure 9 plots the probability of at least one fire detection in a given year as a function of distance to the nearest dump. In all three regions, we see elevated detections in the pixels closest to verified dumps, with 14–16% of dump pixels in Java and Brazil showing at least one fire detection in any given year, consistent with the estimates from the global analysis in Table A7. This probability drops off sharply as distance from the dump increases. In California, we see a high probability of fire detections at EPA landfills, due to methane flaring, while the probability of a fire at an open-dump is much lower, though still somewhat elevated relative to background rates. Appendix Figure A15 shows the expected number of fire detections conditional on at least one detection as well as fire intensity. We see that dump fires are likely to be relatively low intensity, consistent with smoldering waste. Conditional on one dump fire, between 2 and 4 fires are likely to be detected annually—an underestimate of the true number of fires, given gaps in the satellite coverage.

Figure 9: Case Study: Garbage Fires by Distance to Dump

Extensive Margin: Probability of Any Fire



Notes: VIIRS active-fire detections in grid cells around verified dump sites, by distance to the nearest dump (pooled 2012–2023). Appendix Figure A15 plots the expected number of detections in pixels with at least one detection and the mean fire radiative power per detection (MW).

An alternative possibility is that, since dumps tend to be in dense areas, this pattern could be picking up other urban heat sources besides trash fires. Appendix Figure A16 shows fire detections as a function of distance to other sites of interest including hospitals and markets. While detections tend to be elevated at intermediate distances from these places, reflecting a variety of urban heat sources, fires are much rarer in the closest pixels to these type of sites, in contrast to the monotonic decrease of fire probability with distance that we see with open dumps.

7.1 Pollution from Waste Imports

These smoldering fires in open dumps generate a lot of particulate pollution. As discussed in Section 5, these fires may occur naturally from the heat generated by the decomposition of organic waste, or they may be set intentionally to dispose of low-quality materials. Poorly managed plastic and paper waste is highly flammable, and may catch fire, whether intentionally or not. This raises the question of whether waste imports contribute to these garbage fires, thus raising pollution levels around open dumps in importing countries. Indeed, in Java, population-weighted $PM_{2.5}$ increased substantially between 2017 and 2019, despite having fallen for the previous three years (Appendix Figure A17).

As in Section 6, we use the China waste ban as an exogenous shock, in this case studying whether the diverted waste imports induced by the ban lead to increased fires in open dumps and increased particulate pollution. To study the effects of these fires on air pollution, we measure fine particulate matter ($PM_{2.5}$) using the satellite-derived surface concentration estimates of van Donkelaar et al. (2019), which combine satellite aerosol optical depth retrievals with chemical transport modeling and ground-monitor calibration on a 0.01° (∼1 km) global grid. This allows for estimates of particulate matter concentrations in regions where ground monitors are sparse.

We use a basic event study framework, estimating the following specification:

$$\log PM_{it} = \sum_{d} \sum_{q} \beta_{d,q} \left(\text{Ring}_i^d \times \mathbb{1}[q(t) = q] \right) + \alpha_{i,m(t)} + \lambda_{a(i),t} + \varepsilon_{it}, \tag{11}$$

where i indexes grid cells, t indexes months, and cells are assigned to distance rings $d \in \{0\text{–}2, 2\text{–}5, 5\text{–}10, 10\text{–}20, 20\text{–}30\}$ km from the nearest dump, with a 30–40 km ring as the omitted reference. Pixels greater than 40 km from any dump are dropped from the analysis.

The specification includes $\alpha_{i,m(t)}$, cell × month-of-year fixed effects to absorb pixel-specific seasonality in air pollution, and $\lambda_{a(i),t}$, administrative district by month fixed effects to absorb time varying shocks common to all pixels in a district. Because $\alpha_{i,m(t)}$ absorbs each ring's average level in every calendar month, the $\beta_{d,q}$ are identified only up to a constant for each ring and calendar quarter. We normalize each ring × calendar-quarter series to average zero over the pre-announcement quarters (2012Q1–2017Q2), so that the coefficients $\beta_{d,q}$ can be interpreted as the change in the gap between ring d and the 30–40 km reference ring in quarter q, relative to the pre-announcement average of the same gap in the same calendar quarter. The key identifying assumption is therefore that in the absence of the 2018 China waste ban, the pollution—distance-to-dump gradient would have remained similar to what

it was before the policy was announced.

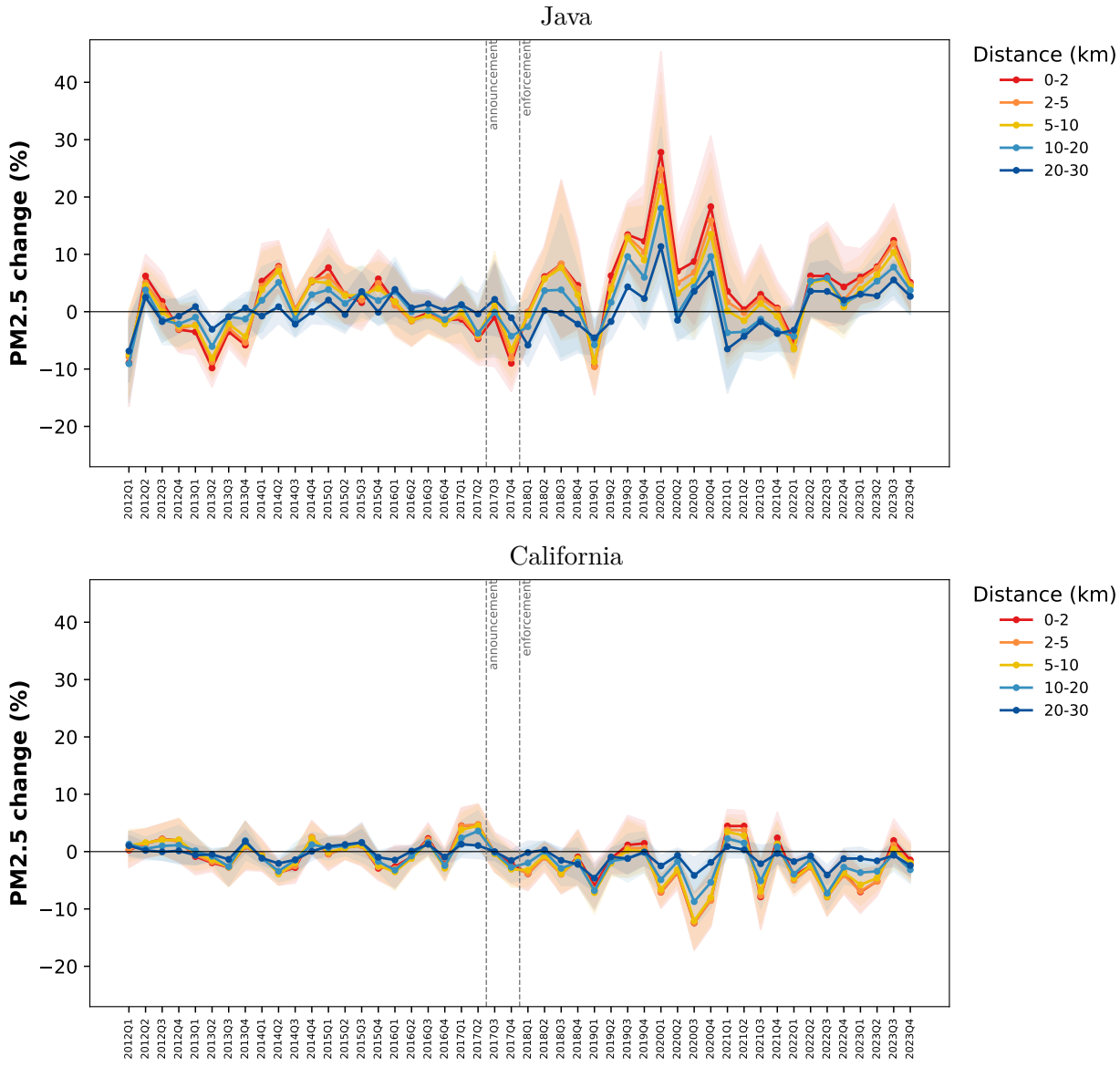
Figure 10 plots the coefficients of these event studies for Java and California.[30] Vertical lines mark China's announcement of the National Sword Policy (July 2017) and the beginning of the enforcement of the import ban (January 2018). In Java we see flat pre-trends in different distance buckets before the announcement, although there is some volatility. We test for pre-trends formally by fitting a separate linear time trend to each distance ring, relative to the omitted 30–40 km ring, over the pre-announcement months—and fail to reject the F-test null that all five ring-specific slopes are zero ($p = 0.27$).

In contrast, most of the time-periods after 2018 show a clear increase in pollution in the cells closest to dumps, with the increases diminishing monotonically as distance from the dumps increases. There is some volatility here too, with the effect apparently peaking in early the first quarter of 2020 at a nearly 30% increase in the cells closest to dumps, relative to the far-afield grid cells. On the other hand, the lower panel of Figure 10 shows the results of the same specification for California, including both EPA dumps and our newly found open dumps. We see no such increase in pollution near dumps—in fact if anything, we see a slight decrease, though this effect may be confounded with the effects of COVID and a particularly bad wildfire season.

Appendix Tables A14 (for Java) and A15 (for California) pool time periods and estimate a single coefficient for each bin in the post-enforcement period, relative to the pre-announcement period, thus excluding the anticipatory period between July 2017 and January 2018, when the ban had been announced but not yet enforced. In the main specification in Column (1), we see a 6.09% increase in pollution in the 0–2 km bin for Java, which fades to a 2.49% increase by the 10–20 km cells. In California, estimated effects are negative and significant across all distance bins, though smaller in magnitude than the Java estimates. We can clearly rule out large increases in pollution as a result of the increased landfilling that occurred after 2018. These estimates are robust across a range of specifications, in columns (2) and (3) we test for the impact of seasonality, by replacing the cell by month-of-year fixed effect with a simple cell fixed effect in column (2) and aggregating the panel to annual resolution in Column (3). In Column (4) we drop border pixels, since they might be contaminated by out-of-region dumps (e.g. in Nevada). Column (5) drops 2020, the year of Covid lockdowns and large California wildfires, which attenuates the estimates a bit in both geographies, though they remain significant at all distance bins.

[30] We focus on these two countries due to Indonesia's increase in imports after 2018 and the US's decrease in exports. Brazil was not a major importer or exporter of scrap either before or after 2018. A set of results for our dumps in Southeast Brazil is shown in Appendix Figure A19, showing little change in outcomes of interest.

Figure 10: PM2.5 Around Dumps and the China Ban: Event Studies



Notes: Event-study estimates from Equation 11, in percent ($100 \times (e^{\hat{\beta}_{d,q}} - 1)$). Estimates give the gap between the indicated distance ring and the omitted 30–40 km ring in that quarter, measured relative to the average of the same gap in the same calendar quarter over the pre-announcement period (2012Q1–2017Q2). All specifications include cell-by-month-of-year and district-by-month fixed effects. Shaded areas give 95% confidence intervals with standard errors clustered by district. Vertical lines mark China's WTO notification announcing the import ban (July 2017) and the ban taking effect (January 2018).

Appendix Figure A18 gives evidence for the mechanism behind these pollution increases. We use the same event study specifications, replacing the dependent variable with a dummy for a VIIRS fire detection as the dependent variable. This time in both Java and California, we see elevated fires after 2018, but only in the pixels closest to dumps. Appendix Figure A20 shows estimates of a single coefficient for each distance bin for the post-enforcement period relative to the pre-announcement period. In Java the coefficient on the 0–2 km distance bin is positive and significant. In California, the point estimate is positive, but not significant. Since the detected fires in California are most likely methane flaring, which does not generate large amounts of particulate pollution, this explains the contrast with the results for Java.

7.2 Waste Trade and Welfare

Taken together, these results suggest that much of the diverted waste was disposed of in open dumps where it caught fire and generated large amounts of pollution. On the other hand, trade data shows that Indonesian importers paid a positive price for these scrap imports suggesting some amount of private benefits. Therefore, we conduct a simple, back-of-the-envelope calculation to compare the magnitude of these private benefits to the external costs of the pollution generated from improper disposal.

To calculate the benefits to importers, we simply compute the change in consumer surplus in each banned waste category directly from the observed pre- and post-ban import prices and quantities in the BACI trade data. We assume that China's exit shifted the supply of scrap toward the remaining importers, so Indonesia's change between its pre-ban (2012–2016) and post-ban (2018–2023) average prices and quantities traces its import-demand curve.[31] Assuming linear demand, we can calculate consumer surplus as the area of the trapezoid between these two points (see Figure A22). While the linear demand assumption is conservative, since constant-elasticity demand curves would give lower estimates of consumer surplus, two other caveats are worth noting. First of all, this exercise assumes that Indonesia's demand curve remained constant over the period. This is a reasonable approximation—net imports spiked around the time of the ban, but do not show much of a time trend otherwise (see Appendix Figure A21 for plastic imports). Second, it assumes the quality of the imported waste was not changed by the ban. Scrap quality can vary (e.g. contamination levels), but assuming constant quality is probably a reasonable approximation in this case.

[31] For prices we use annual unit prices from the BACI data, converted to Indonesian rupiah, and then deflated to constant 2016 prices using exchange rates and Indonesian CPI from the World Bank. Prices do not reflect a PPP adjustment since these are tradable goods, but we convert back to 2016 USD using market exchange rates for ease of interpretation.

Making these assumptions allows us to calculate increases in consumer surplus of $21 million for plastic scrap (HS 3915), $34 million for paper waste (HS 4707), $3 million for textiles (HS 5103, 5104, 5202, 5505, 6310), and $11 million for slag (HS 2619 and 2620) covering the major categories of materials banned by China. This gives a headline estimate of $69.2 million for all categories, or $55.5 million if we just count paper and plastic, which are the materials most likely to end up in dumps (slag is more valuable for industry and more easily recyclable).

We compare this consumer surplus to one particular external cost, the air pollution close to dumps generated from the diverted imports, noting that water pollution, pests, and other nuisances are neglected in this analysis. To measure air pollution precisely, we estimate the average monthly pollution plume induced by the diverted waste imports as a flexible function of distance and wind direction, following Schlenker and Walker (2016).[32] We use the following specification:

$$\log \mathrm{PM}_{it} = f(d_i, w_{it}) \times P_t + \alpha_{i,m(t)} + \lambda_{a(i),t} + \varepsilon_{it}. \tag{12}$$

In this regression, P_t denotes a vector with dummies for the announcement period (post July 2017) and the enforcement period (post January 2018). We allow the plume in each period to change flexibly as a function of distance to the nearest dump d_i, and monthly prevailing wind direction relative to the nearest dump, w_{it}. We use a cubic polynomial of downwind distance and a quadratic polynomial of crosswind distance for $f(d_i, w_{it})$. Appendix Figure A23 plots the predicted change in pollution for the post-enforcement period as a function of distance and direction to the nearest dump, relative to far upwind pixels.

To translate the estimated pollution changes into health impacts, we overlay the fitted plume surfaces on our dumps. For each cell within 40 km of a dump, we evaluate the enforcement-period plume surface at the cell's relative wind position in each month, and subtract off our estimated change (which is defined relative to changes at the far upwind pixels) to construct a counterfactual pollution path in the absence of China's waste import ban. This gives us the predicted change in exposure for each grid cell.

We then combine this exposure change with the gridded population data and a concentration-response function from Burnett et al. (2018) to estimate excess mortality. We construct confidence intervals by conducting 1,000 Monte Carlo draws from the plume coefficient

[32] We use monthly mean wind fields (10m u- and v-components) from the ERA5 reanalysis, available on a 0.25°grid from the Copernicus Climate Data Store, to measure the prevailing wind direction around dump sites (Hersbach et al., 2020).

covariance matrix and the sampling distribution of the CRF parameter. Several caveats apply. First, as stated previously, the plume identifies pollution changes near dumps relative to far upwind pixels. If the waste import shock changed pollution levels in these far upwind pixels, that change is not identified in our analysis. This is possible, for example if recycling factories generate pollution that is not co-located with dumps. In this case, we will underestimate mortality impacts. Second, overlaying these plumes on population data abstracts from post-ban migration or additional defensive actions that households may take in response to the increase in pollution. Finally, these plumes are average, both over time and across dumps. If the pollution effect varies significantly with respect to population, that could affect our conclusions.

Conditional on these caveats, we estimate an additional 4,123 annual premature deaths in Java (95% CI: [1,433, 7,055]) as a result of the post 2018 waste import shock. Dividing our estimated Indonesian consumer surplus from paper and plastic imports by this number gives us an estimate of the welfare gains per life lost of $13,465 (95% CI: [$7,626, $38,745]). This goes up to $16,779 if slag and textiles are included. These numbers are orders of magnitude lower than standard estimates of the value of a statistical life. Furthermore, it should be noted that the mortality estimates only reflect a subset of Indonesia, whereas the consumer surplus is at the country level.

These numbers do not reflect a complete welfare analysis, since they do not reflect producer surplus in exporting countries. Indeed, if we conduct an analogous exercise to measure the lost producer surplus corresponding to the collapse in demand for US plastic and paper waste exports after the China import ban, we estimate $190.4 million in lost producer surplus in the US, although this number is not directly comparable to the Indonesian figure, since the US sends waste to multiple export destinations. Still, it highlights the distributional issues arising from this trade.

8 Conclusion

In this paper, we develop a methodology to generate globally representative time-series estimates of the prevalence of open dumps. We use crowdsourced, non-representative training data to train a biased machine learning model that predicts dumps based on satellite imagery. We then use active learning methods to choose an approximately optimal subset of predictions to verify using high-resolution imagery. We choose these predictions to make our estimates of open dump prevalence and changes as precise as possible, and we come up with a simple formula to characterize the ex-post efficiency gains, relative to uniform sampling,

in the presence of model miscalibration. These methods could be applied to other types of untracked and difficult-to-detect phenomena.

We use our data on dumps to present new descriptive facts about the global distribution of dumps, as well as causal evidence on the relationship between waste imports and open dumps. We show that dumps are concentrated in places with high population density, especially in low-income countries. Furthermore, a large fraction of dumps in low- and middle-income countries (16.7%) experience frequent fires—at least one a year.

Turning to our analysis of trade flows, we see a large increase in dumps globally in the year that China banned waste imports. Event studies and a shift-share IV show that this increase occurs in countries that received an influx of imports diverted by the China waste ban. This provides evidence that a meaningful share of exported recycled materials are ultimately discarded in poorly managed dumps in low-income countries. We also find that the import shock increased $PM_{2.5}$ levels close to dumps, and show evidence that increased garbage fires are the mechanism behind this pollution. A back-of-the-envelope welfare calculation for Indonesia estimates that the ratio of consumer surplus from increased imports to lives lost from the resulting pollution is $13,465, orders of magnitude lower than standard VSL estimates.

Our work opens the door to further research on this topic. Our data and methods should be useful to more precisely measure other externalities associated with informal waste sites, including water pollution and ocean plastic emissions. An improved understanding of the impacts of waste management practices on human health and ecosystem services is an important research agenda, given the long lifetimes of plastic scrap and the rapid growth rate of waste production.

Furthermore, future research should strive to better understand the economic impacts of dumps and investments in urban sanitation infrastructure. Dumps may provide jobs to informal workers, but they also have a high opportunity cost to the extent that they are located in dense and rapidly growing cities. Research should study cost-efficient ways to minimize the harms from waste disposal, including foreign imports. On the latter topic, import quality standards may reduce the amount of scrap material that ends up in dumps, and better regulations and infrastructure could limit the harms from disposal. This would be relevant for multilateral environmental and trade policy. In 2019, 187 countries signed an amendment limiting exports of plastic waste to developing countries, but several major economies, including the US, did not participate.

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Appendix for "Open Dumps and the Global Trade in Garbage"

Matthew Gordon, Anna Papp, and Monica Shandal

A Design Effect with Miscalibration

This appendix derives the design-effect expressions in Section 3.3 and shows how the design effect can be estimated using a representative verification sample.

Recall that $\sigma_i = \sqrt{p(k_i)(1-p(k_i))}$, and $\sigma_i^{*2} = \mathbb{E}(\nu_i^2) = p_i^*(1-p_i^*) + (p(k_i)-p_i^*)^2$, and $R_i = \frac{\sigma_i^*}{\sigma_i}$. Define weights as $w_i = \frac{\sigma_i}{\sum_j \sigma_j}$, and $\sum_{i=1}^N \pi_i = S$ is the expected sample size.

The exact finite-population design effect is

$$\text{DEFF} = \frac{\left(\sum_{i=1}^N \frac{\sigma_i^{*2}}{\sigma_i}\right)\left(\sum_{i=1}^N \sigma_i\right) - S\sum_{i=1}^N \sigma_i^{*2}}{(N-S)\sum_{i=1}^N \sigma_i^{*2}}. \tag{A1}$$

When the sampling fraction S/N is small, this is approximately

$$\text{DEFF} \approx \frac{\left(\sum_{i=1}^N \frac{\sigma_i^{*2}}{\sigma_i}\right)\left(\sum_{i=1}^N \sigma_i\right)}{N\sum_{i=1}^N \sigma_i^{*2}}. \tag{A2}$$

Since $\frac{\sigma_i^{*2}}{\sigma_i} = R_i^2\sigma_i$ and $\sum_i R_i\sigma_i = \sum_i \sigma_i^*$, equation (A2) can be written as

$$\text{DEFF} \approx \left[\frac{\sum_i w_i R_i^2}{\left(\sum_i w_i R_i\right)^2}\right]\left[\frac{\left(\sum_i \sigma_i^*\right)^2}{N\sum_i \sigma_i^{*2}}\right] = \frac{1+CV_{\sigma R}^2}{1+CV_{\sigma^*}^2}, \tag{A3}$$

where $CV_{\sigma R}$ is the coefficient of variation of R_i using weights w_i, and CV_{σ^*} is the coefficient of variation of σ_i^*. If $R_i = 1$ for all observations (perfect calibration) then we obtain the Neyman optimal design effect of:

$$\text{DEFF}^* \approx \frac{1}{1+CV_{\sigma^*}^2}. \tag{A4}$$

More generally, exact calibration is not necessary: any proportional distortion $\sigma_i^* = c\sigma_i$ leaves the sampling allocation unchanged and also attains the optimum. A useful benchmark is the case in which σ_i^* and σ_i are independent across pixels (e.g. random predictions). Equation

(A2) then becomes

$$\text{DEFF} \approx \frac{\mathbb{E}_i[\sigma_i^{*2}/\sigma_i]\mathbb{E}_i[\sigma_i]}{\mathbb{E}_i[\sigma_i^{*2}]} = \mathbb{E}_i[\sigma_i]\,\mathbb{E}_i\left[\frac{1}{\sigma_i}\right]. \tag{A5}$$

When the dispersion in σ_i is small, a second-order expansion gives

$$\mathbb{E}_i[\sigma_i]\,\mathbb{E}_i\left[\frac{1}{\sigma_i}\right] \approx 1 + CV_\sigma^2.$$

In general, however, DEFF can be arbitrarily large.

A.1 Ex-post Design Effect Estimator

Suppose that a representative verification sample exists, denoted by $i \in \mathcal{S} \implies s_i = 1$, in which the true binary outcome D_i is observed for sampled pixels, with known inclusion probabilities π_i. This verification sample can be used to estimate the ex-post design effect. Conditional on the realized prediction errors, the sampling variance of $\widehat{\mu}$ is

$$V_{\text{sample}} = \frac{1}{N^2}\sum_{i=1}^{N}\frac{1-\pi_i}{\pi_i}\nu_i^2. \tag{A6}$$

Under uniform sampling, $\pi_i = S/N$ for all i, so the corresponding variance is

$$V_{\text{unif}} = \frac{N-S}{N^2 S}\sum_{i=1}^{N}\nu_i^2. \tag{A7}$$

The realized finite-population design effect is therefore

$$\text{DEFF}^{\text{R}} = \frac{V_{\text{sample}}}{V_{\text{unif}}}. \tag{A8}$$

The prediction errors are observed only for pixels in the verification sample $\mathcal{S}$. Under sampling independence,[33] we can estimate the two variances using Horvitz–Thompson estimators:

$$\widehat{V}_{\text{sample}} = \frac{1}{N^2}\sum_{i\in\mathcal{S}}\frac{1-\pi_i}{\pi_i^2}\nu_i^2, \tag{A9}$$

[33] In the paper we use a two-stage design, first selecting tiles, then pixels within tiles. Therefore we estimate these variances using a tile-clustered bootstrap.

and

$$\widehat{V}_{\text{unif}} = \frac{N-S}{N^2 S} \sum_{i \in \mathcal{S}} \frac{\nu_i^2}{\pi_i}. \tag{A10}$$

We therefore estimate the realized design effect as

$$\text{DEFF}^{\text{R}} = \frac{\widehat{V}_{\text{sample}}}{\widehat{V}_{\text{unif}}}. \tag{A11}$$

Under standard conditions for design consistency of the Horvitz–Thompson estimator,

$$\frac{1}{N}\left[\sum_{i \in \mathcal{S}} \frac{1-\pi_i}{\pi_i^2}\nu_i^2 - \sum_{i=1}^{N} \frac{1-\pi_i}{\pi_i}\nu_i^2\right] \xrightarrow{p} 0, \tag{A12}$$

and

$$\frac{1}{N}\left[\sum_{i \in \mathcal{S}} \frac{\nu_i^2}{\pi_i} - \sum_{i=1}^{N} \nu_i^2\right] \xrightarrow{p} 0. \tag{A13}$$

Provided that $N^{-1}\sum_{i=1}^{N} \nu_i^2$ is bounded away from zero, the continuous mapping theorem implies

$$\text{DEFF}^{\text{R}} \xrightarrow{p} \text{DEFF}^{\text{R}}. \tag{A14}$$

A.2 Estimating the Feasible Lower Bound

The verification sample can also be used to estimate the design effect of the feasible optimal sampling rule based on $p(k_i)$. Recall from Section 3.3 the optimal lower-bound design effect is

$$\text{DEFF}^{*} = \frac{\left(\sum_i \sigma_i^*\right)^2}{N \sum_i \sigma_i^{*2}}. \tag{A15}$$

The plug-in rule using squared model prediction errors is consistent for the denominator using the same arguments as in the section above. However, the numerator depends on $\sqrt{\mathbb{E}(\nu_i^2)}$, and $\mathbb{E}(\sqrt{\nu_i^2})$ is not unbiased for this. To estimate the pixel-level oracle bound, assume that the true probability is determined by the model score, so that $p_i^* = p^*(p(k_i))$. Define the calibration function

$$p^*(p) = \mathbb{E}[D_i \mid p(k_i) = p].$$

Estimate $p^*(p)$ using, for example, inverse-probability-weighted LOESS on the verification sample, with weights $1/\pi_i$, and denote the resulting estimate by $\widehat{p}^*(p)$.

For each pixel define

$$\widehat{\sigma}_i^* = \sqrt{\widehat{p}^*(p(k_i))\left[1-\widehat{p}^*(p(k_i))\right] + \left[p(k_i) - \widehat{p}^*(p(k_i))\right]^2}. \tag{A16}$$

Recall $\sigma_i^{*2} = p_i^*(1-p_i^*) + (p(k_i)-p_i^*)^2$, and therefore $\widehat{\sigma}_i^{*2} - \sigma_i^{*2} = [1-2p(k_i)][\widehat{p}^*(p(k_i)) - p^*(p(k_i))]$. Since the calibration estimator is consistent under standard regularity conditions, and $|1-2p(k_i)| \leq 1$, we have

$$\left|\widehat{\sigma}_i^{*2} - \sigma_i^{*2}\right| \leq \left|\widehat{p}^*(p(k_i)) - p^*(p(k_i))\right|.$$

Since, $|\sqrt{a}-\sqrt{b}| \leq \sqrt{|a-b|}$, Jensen's inequality gives:

$$\frac{1}{N}\sum_i |\widehat{\sigma}_i^* - \sigma_i^*| \leq \sqrt{\frac{1}{N}\sum_i \left|\widehat{p}^*(p(k_i)) - p^*(p(k_i))\right|} \xrightarrow{p} 0.$$

Therefore if

$$\widehat{\text{DEFF}}^* = \frac{\left(\sum_i \widehat{\sigma}_i^*\right)^2}{N\sum_i \widehat{\sigma}_i^{*2}}, \tag{A17}$$

then

$$\widehat{\text{DEFF}}^* \xrightarrow{p} \text{DEFF}^*$$

by the continuous mapping theorem.

We estimate $p^*(p)$ using LOESS regression with our verification sample, and we find that the resulting estimates are sensitive to the LOESS bandwidth, though whether the estimation is done in logs or levels does not make much difference. The table below summarizes the estimates.

Table A1: Sensitivity of DEFF*

	Prevalence		Changes	
Bandwidth (k/n)	p	$\log p$	p	$\log p$
0.001	0.002	0.002	0.0009	0.0009
0.01	0.021	0.021	0.0075	0.0074
0.1	0.086	0.093	0.0326	0.0405

Notes: Table shows estimates of DEFF*, the feasible lower bound on the design effect that a perfectly calibrated model would have attained, estimated on the verification sample. Estimating it requires the calibration function $p^*(p)$, which we fit by inverse-probability-weighted LOESS. Column indicate whether the smoother is estimated on the score itself (p) or its logarithm. The rows give the bandwidth, the share of the verification sample entering each local fit.

B Satellite Data, Machine Learning Model, and Verification Labels: Additional Details

This appendix gives additional details on the satellite data, machine learning model, and verification process described in Section 4.

B.1 Satellite Data

As described in Section 4.1, the features our model is trained on (k_i) include annual summaries of satellite imagery from Sentinel-1 and Sentinel-2. The European Space Agency has operated both of these satellites with consistent coverage since 2017. We begin our satellite analysis in 2017 when Sentinel-2B was launched. Sentinel-2 is an optical satellite constellation with a spatial resolution of 10 meters and a revisit frequency of 3–5 days. Sentinel-1 is a radar satellite that provides all-weather and day-and-night coverage. For each pixel within the training polygons, we download data from Google Earth Engine for each year from 2017 to 2021, keeping cloud-free, non-water observations. To mask out cloudy pixels, we use Google's Cloud Score+ and define clear pixels as those with a clear-sky score of at least 65% (Pasquarella et al., 2023). We use Dynamic World to mask out water pixels (Brown et al., 2022).

The satellite data features used to train the model include each of the 13 bands of Sentinel-2, indices derived from these bands, and all Sentinel-1 bands collected in interferometric wide swath mode and derived indices. For a complete list of satellite bands and derived indices, see Appendix Tables A3 and A4. For each band and/or derived index, we take the median, mean, minimum, and maximum value across all the images from each year. We also utilize focal statistics, taking the mean and standard deviation of each band and index in a 3 pixel radius around each pixel to include information on nearby areas' shapes and characteristics. This allows the model to use information on surrounding pixels in classification, similar to a convolutional neural network, although we use a model that is simpler and less computationally intensive to train.

In addition to the raw satellite data, we use the Dynamic World predictions of major land-use categories that Google derives based on its own model using Sentinel-2 bands (Brown et al., 2022). We also calculate the population density within 5 km and the distance to the nearest port for each pixel (Earth Science Data Systems, NASA, 2024; National Geospatial-Intelligence Agency, 2019).

B.2 Machine Learning Model

As described in the text, we use a Gradient Boosted Tree model, trained on our training examples $\mathcal{T}$. During training, we assign control pixels a relative weight of 332. This reflects a guess about the relative prevalence of waste pixels we expected to encounter when deploying our model out of the training sample, which we calibrated based on our early iterative experiments. As described in the text, to tune the model, we divide our sites into 500 clusters using a k-means algorithm based on their geographic coordinates. We use 80% of clusters for training and hold out the last 20% to calculate accuracy metrics. Within the 80% training set, we choose our model hyperparameters using 3-fold cross validation to maximize the area under the receiver-operating characteristic curve (AUC) in a held out fold. The final model builds an ensemble of 100 decision trees with a maximum depth of 25 splits. Hyperparameter values tried and chosen are shown in Appendix Table A5. AUC is a metric is robust to heavily unbalanced classes—it incentivizes correctly ranking observations by the probability that they are a dump, but it may lead to miscalibrated probabilities, especially for unlikely cases. We thus apply a sigmoid recalibration to the resulting predictions: $p_i = (1 + \exp(-a \log(\tilde{p}_i/(1-\tilde{p}_i))))^{-1}$ where $\tilde{p}_i$ are the original predictions, and we set the coefficient $a = 3.15$ to reflect a prior about the rarity of dump pixels in the deployment population based on experiments with early versions of the model. This transformation rescales probabilities without changing their relative ranking; in particular it shrinks small probabilities closer to zero, reducing their likelihood of being sampled.

We analyze which features in our data seem to be most helpful in detecting dumps by ranking them by their sum-of-split scores. This metric calculates the total reduction in the model's loss contributed by that feature across all trees in the ensemble. Population, distance to the nearest port, and the Dynamic World labels (Brown et al., 2022) are the most important features. The Dynamic World labels are based on Sentinel-2 imagery and are a reasonable summary of the spectral information contained in a given pixel, so it makes sense that they would be important. Beyond that, the Sentinel-1 radar derived indices rank highly, as do the near infrared bands and red-edge, though there is a long tail of additional features that are important. These findings are broadly consistent with previous studies of the spectral signatures of open dumps (Biermann et al., 2020; Kruse et al., 2023; Cózar et al., 2024), however given the large number of features that seem to contribute to the model, and the fact that in model tuning we find that relatively deep trees improve accuracy, it appears that our model is relying on relatively complex combinations of many features, rather than finding one or a few proxy variables that are highly predictive of open dumps.

After completing model and hyperparameter testing, we train a model using all pixels from

all sites so that our final model can learn from all the examples we have.

B.3 Sampling Frame and Global Deployment

As discussed in the text, Sentinel-2 has a spatial resolution of 10 meters, making predictions for the entire land area of earth for multiple years prohibitively expensive. This section details how we deployed our model and selected pixels to sample for the verification set.

The scaling economics of our deployment strategy may be interesting to some readers, as cost, time, and compute constraints all bind. A single script that downloads satellite data, preprocesses features, and makes predictions using our trained model can be deployed as a 'task' that runs on Google Earth Engine servers. This task scales well within contiguous areas, but is subject to built-in memory constraints. Thus we define our task to run on contiguous $1° \times 1°$ tiles (approximately 111 km $\times$ 111 km at the equator), which are the largest tiles that consistently run without triggering memory errors.

Running this sequence of operations typically takes about 1 hour per tile-year, though several tasks can run in parallel (the exact number of parallel tasks depends on server traffic). Thus we choose 500 tiles, stratified based on a rough estimate of where we expect to see the biggest changes in our outcome variable of interest (open dumps). We calculate the per-capita change in plastic waste imports from 2017 to 2019 at the country level from the BACI trade data, and multiply it by the estimated fraction of mismanaged waste at the country level from Kaza et al. (2018). We then allocate this to tiles within countries by multiplying by tile population.

The bottom panel of Figure A5 shows six strata defined by this variable using the 5th, 25th, 50th, 75th, and 95th percentiles as thresholds. We then sample an equal number of tiles within each of these strata with within strata inclusion probabilities proportional to the tile population density to focus our analysis on areas where the externalities of improper waste management are large due to the proximity to population centers. Our selected tiles are overlaid in red. Since our tile sampling is without replacement, and some tiles, in particular those with large populations, are near certainty units, we compute each tile's inclusion probability using the standard formula for sequential sampling without replacement:

$$\pi_\tau^{tile} = 1 - \exp\left(-A_s\, p_\tau\right), \qquad \text{with } A_s \text{ defined by } \sum_{j \in s} \left(1 - \exp\left(-A_s\, p_j\right)\right) = n_s, \tag{A18}$$

where p_τ is tile τ's share of total population density within its stratum s, and n_s is the number of tiles sampled from each stratum (Gordon, 1983).

Within each tile, we allocate our sample according to the proxy-Neyman scheme described in Section 3.2, with half of the sample allocated proportionately to σ_i, and half proportionately to σ_i^{Δ}. If π_i^L is the inclusion probability for pixel i in the levels arm, and π_i^{Δ} is the inclusion probability for pixel i according to the changes arm, then our final inclusion probability π_i for a pixel is:

$$\pi_i = \pi_\tau^{tile} \times \left(1 - \left(1 - \pi_i^L\right)\left(1 - \pi_i^{\Delta}\right)\right) \tag{A19}$$

One further correction is needed, since tile areas are defined in degrees, and the area of a $1° \times 1°$ tile changes depending on latitude. Therefore final land area weights are constructed as the cosine of the tiles' latitude divided by π_i.

While our sampling frame is global land area, for several classes of pixels, we assume that the probability of containing an open-dump is exactly 0, and thus these pixels are excluded from sampling. The first such class is all pixels within uninhabited tiles. Since tiles are large, completely uninhabited tiles are rare, and mostly found in Antarctica, Greenland, Northern Canada, and the Saharan Desert (see Figure A5, top panel), and assuming no dumps in these regions is reasonable. Secondly, we drop tiles in countries lacking trade data, as this invalidates our stratification variable. The largest areas affected by this drop are Taiwan, Eritrea, Somaliland, Puerto Rico, Gambia, Guinea-Bissau, Kosovo, Western Sahara, and some small island nations. Thus our analysis does not include dumps in these areas, which are highlighted in pink in Figure A5. Third, we exclude inland water from both numerator and denominator of our estimates, by dropping pixels labeled as water by Dynamic World (Brown et al., 2022). We leave lake and river dumping for future research. Lastly, we exclude open-pit mining areas as mapped by Tang and Werner (2023), since these sites tend to be difficult for our model to distinguish from dumps. This risks excluding dumps inside of mines—probably not a major concern in practice. For each tile, we keep one 'stand-in' pixel to count the number of zero probability pixels within the tile.

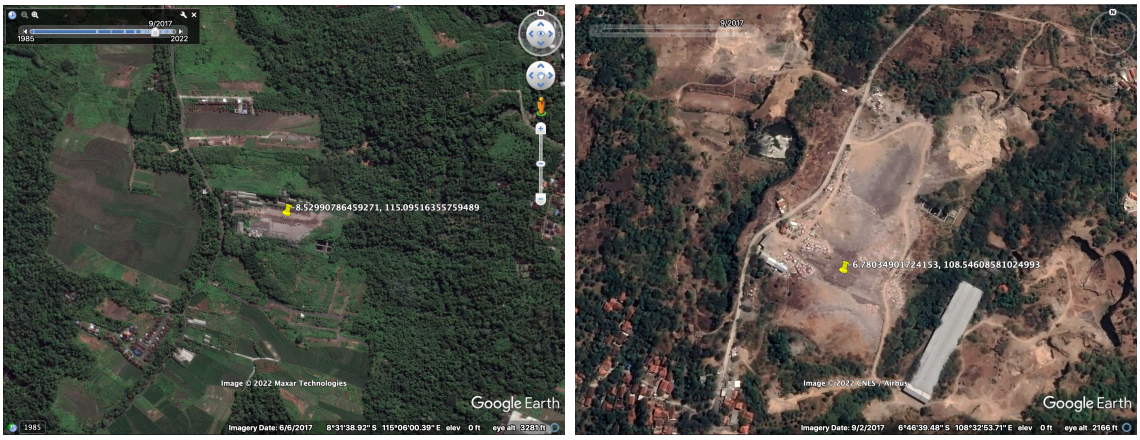
For all other pixels with at least one high-quality, cloud-free Sentinel-2 and Sentinel-1 image for each year between 2017 and 2021, we use our model to estimate p_i and p_i^{Δ}. For pixels with missing imagery (e.g. due to cloud coverage in a particular year), we impute the tile median probability, so that they are included in the sampling frame.

B.4 Verification and Panel Construction

This section details the verification process and describes how we construct our final verification panel. Each of the 14,086 sampled pixels was independently given to two annotators, who recorded whether the pixel showed an open dump in each year from 2011 to 2023 based

on Google Earth high-resolution satellite imagery. Examples of these images are shown in Figure A1.

Figure A1: High-Resolution Satellite Data Verification Examples



Notes: Figures show two examples of waste sites in Google Earth high-resolution satellite imagery.

Given that high-resolution images do not exist for every location in every year, annotators were not able to label every pixel-year. Out of the 183,118 pixel-year observations, both annotators recorded a label in 53.2% of cases. On co-observed cells the annotators agree 99.5% of the time, but raw agreement is less informative because both annotators record "no dump" in 98.4% of these observations. Among cells where either annotator recorded a dump, both did so in 65.6% of cases. Authors reviewed and manually adjudicated the 221 pixels where the annotators disagreed in at least one year, as well as 120 pixels (1,560 pixel-year observations) where the annotators were not able to label a single observation. In these latter cases, we sometimes relied on high-resolution images from other sources or from years outside of the study window. None of these sites showed any evidence of having ever been an open dump. A further 94 pixels (1,222 pixel-year observations) are tile-level stand-ins for pixels scored as zero probability of containing a dump.

For the 13,872 pixels carrying at least one label, we fill unobserved cells within each annotator's series by one of the following mechanisms. The largest category is gaps that fall between two labeled years carrying the same value (43,658 pixel-year observations). For these we assume persistence. Where a gap falls between years carrying different values, we interpolate linearly, which produces fractional labels; this is rare (209 pixel-year observations). The next largest category is extrapolation before an annotators first label or after their last label. We again assume persistence in these cases. These assumptions are reasonable because dumps to non-dumps transitions are rare. We have 13,137 pixels with ≥ 2 high-resolution observations,

and of those, 98.89% do not change classes (99.9956% by weight). Of the remaining series that rest on a single labeled year, the observed value is "no dump" for every single one. The main risk from these interpolation procedures is that we will miss some transitions, and this could attenuate our estimates of the effects of imports on dumps if these changes are correlated with imports. Section 6 tests whether the filling affects the trade estimates and finds no evidence that it does.

Lastly, for a small number of observations (21 pixels), one annotator left the series blank in every year. In these cases, we assign them the other annotator's series. Table A2 summarizes these figures.

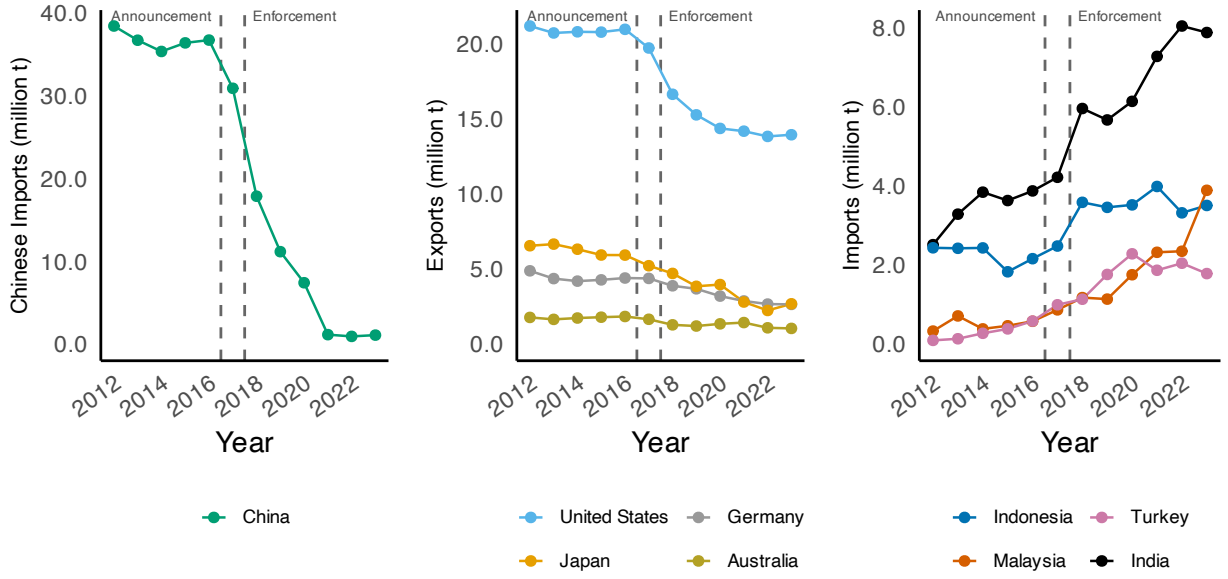
Table A2: Construction of the Verification Panel

	Count	%
Sampled pixels	**14,086**	100.0
Observations to label (pixel × year)	**183,118**	100.0
Labeled by both annotators	97,491	53.2
Raw agreement: 99.5%		
Agreement, conditional on at least one dump label: 65.6%		
Labeled by one annotator	2,057	1.1
Labeled by authors after manual review	1,560	0.9
Set to zero by construction (per-tile stand-ins)	1,222	0.7
Unobserved, filled by the rules below	80,788	44.1
Interpolated between labeled years	**43,776**	23.9
same value on both sides	43,658	23.8
different values (fractional label)	118	0.1
Extrapolated beyond the labeled range	**37,012**	20.2

Notes: Details on the construction of the verification panel, including number of observations interpolated and extrapolated. Each sampled pixel was labeled independently by two annotators for each year 2011–2023, but annotators did not label every year, since cloud cover and imagery gaps leave some years unreadable. Agreement statistics are computed before any filling.

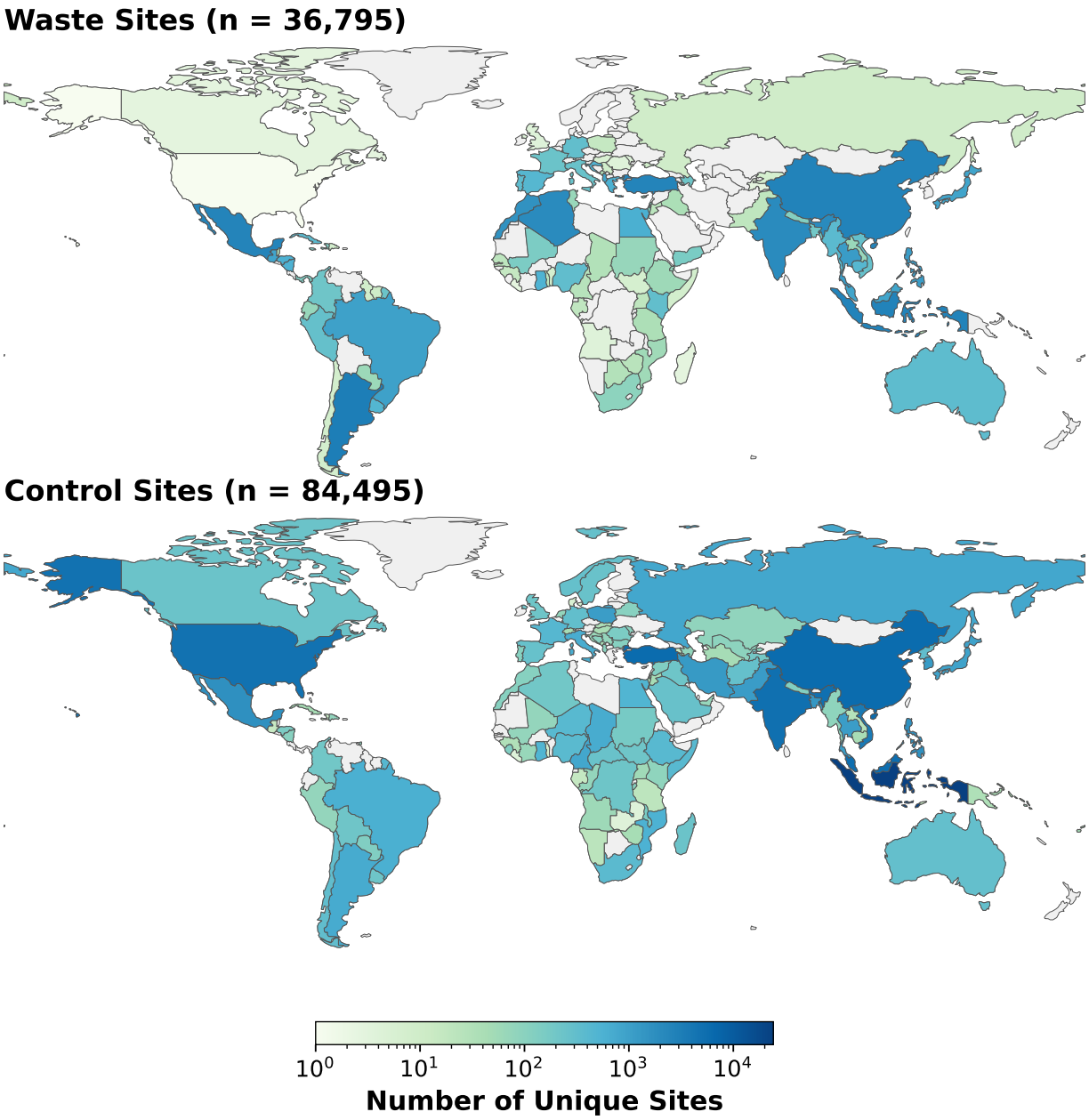
C Additional Figures

Figure A2: Plastic and Paper Scrap Imports and Exports Over Time



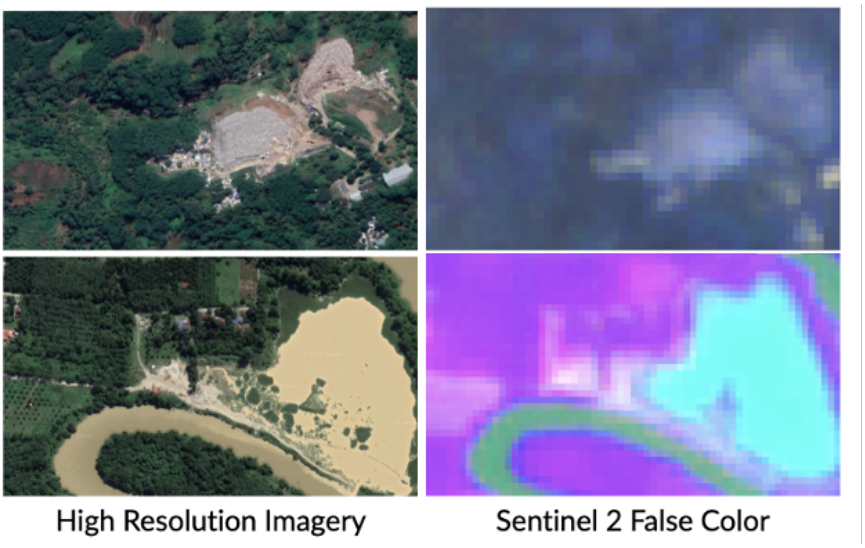
Notes: Figure shows trade in the two major categories of banned waste under the National Sword Policy: plastic waste and scrap (HS 3915) plus waste paper and paperboard (HS 4707), which together account for over 93.7% of China's 2016 (pre-policy) import tonnage in banned categories. Units are millions of tons. Data are from CEPII-BACI, 2012–2023. Left panel shows China's imports; center panel shows exports of four large pre-ban senders (United States, Japan, Germany, Australia); right panel shows imports of four major receiving countries (Indonesia, Malaysia, Turkey, India). We note that almost all of India's imports are in unsorted waste paper, not plastic waste. Dashed vertical lines mark the announcement of the National Sword Policy (July 2017) and its effective date (January 2018).

Figure A3: Distribution of Training Data by Country



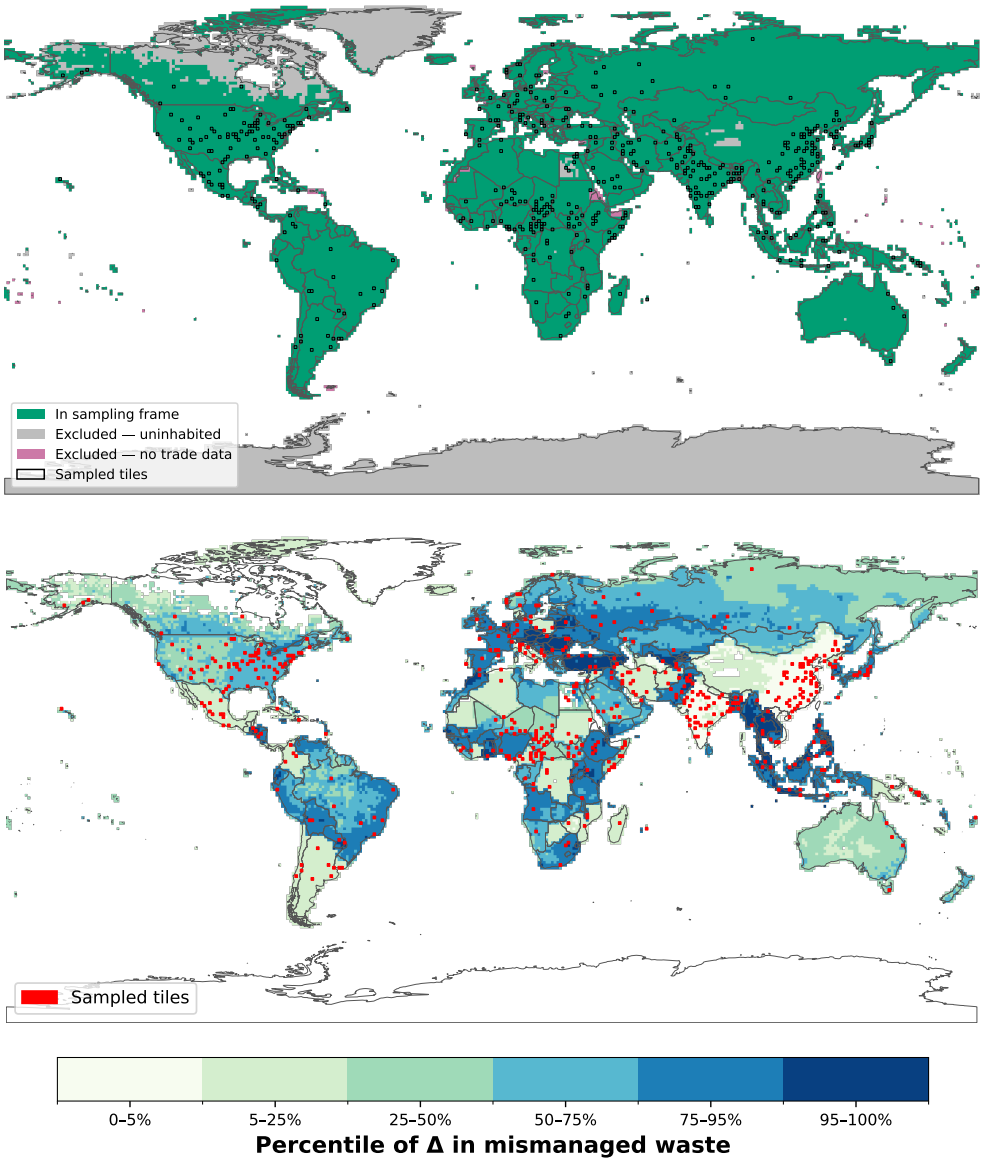
Notes: The distribution of training sites (both waste sites and control sites) by country based on crowdsourced and other data collection. Counts reflect site-year observations, not necessarily unique sites.

Figure A4: Comparison of Google Earth High-Resolution Imagery and Sentinel-2 Imagery



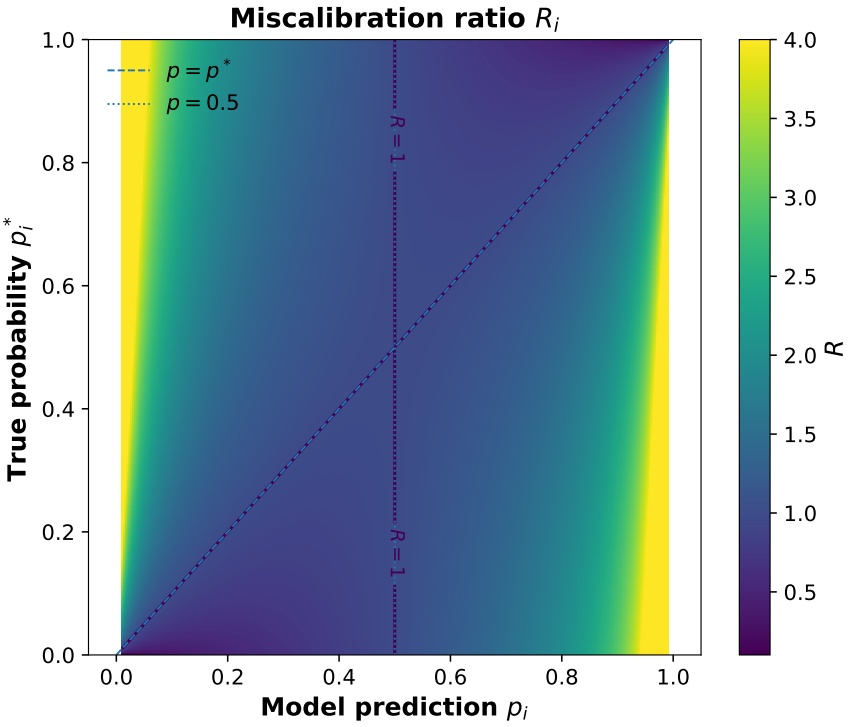
Notes: Figure shows high-resolution historical imagery from Google Earth Pro (left) and the same sites in Sentinel-2 false-color imagery (right). Dumps are able to be easily verified by human annotators using the high-resolution imagery. Models are trained on the right imagery, which includes radar and infrared information that is useful in classifying dumps.

Figure A5: Strata and Tile Sample for Global Analysis



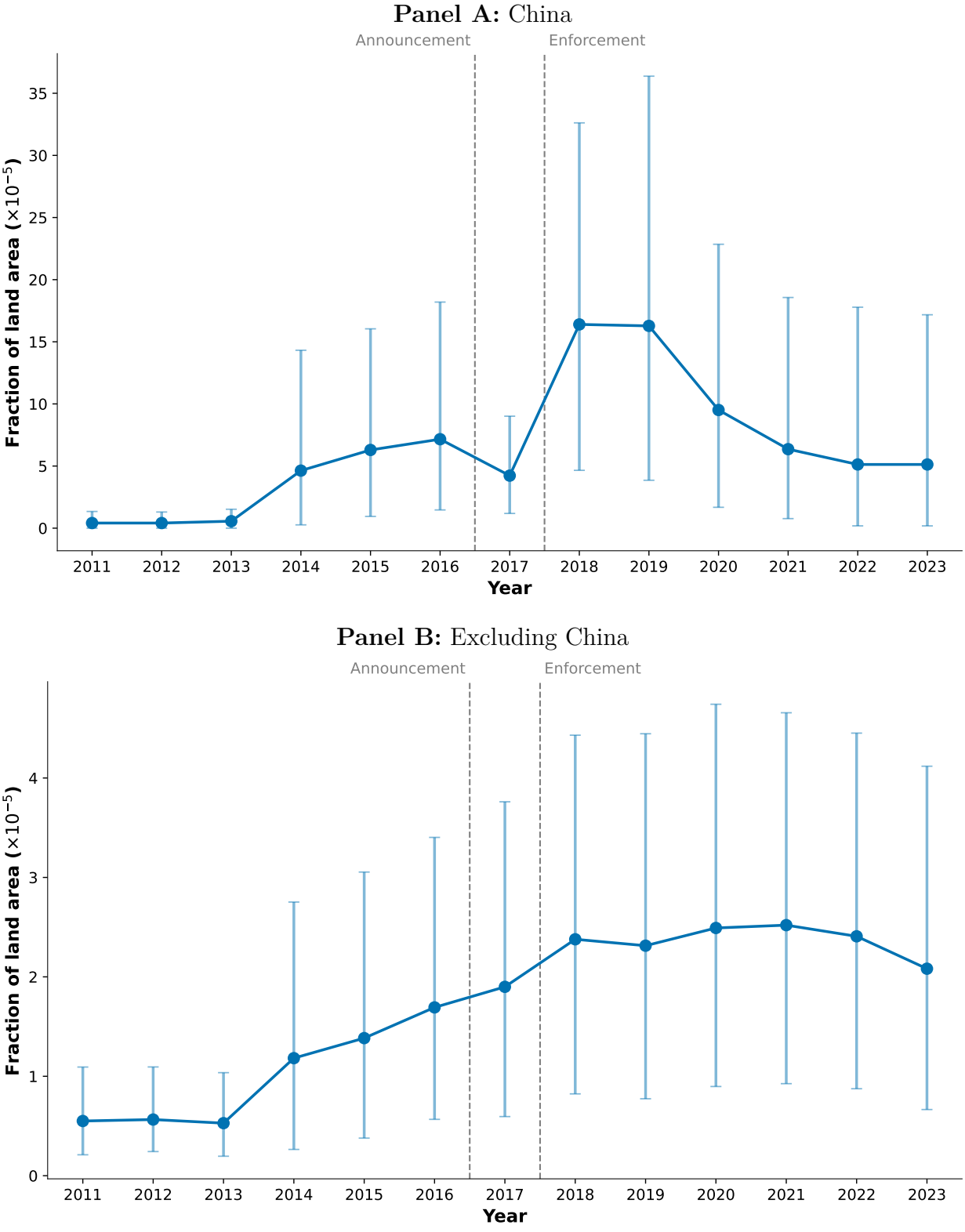
Notes: Figure illustrates global deployment on $1° \times 1°$ tiles. Top panel shows tiles that had zero probability of selection, including uninhabited tiles and tiles for countries missing trade data. Bottom panel shows the six strata based on estimated changes in mismanaged imports 2017–2019, with the selected tiles overlaid in red.

Figure A6: The Miscalibration Ratio R_i



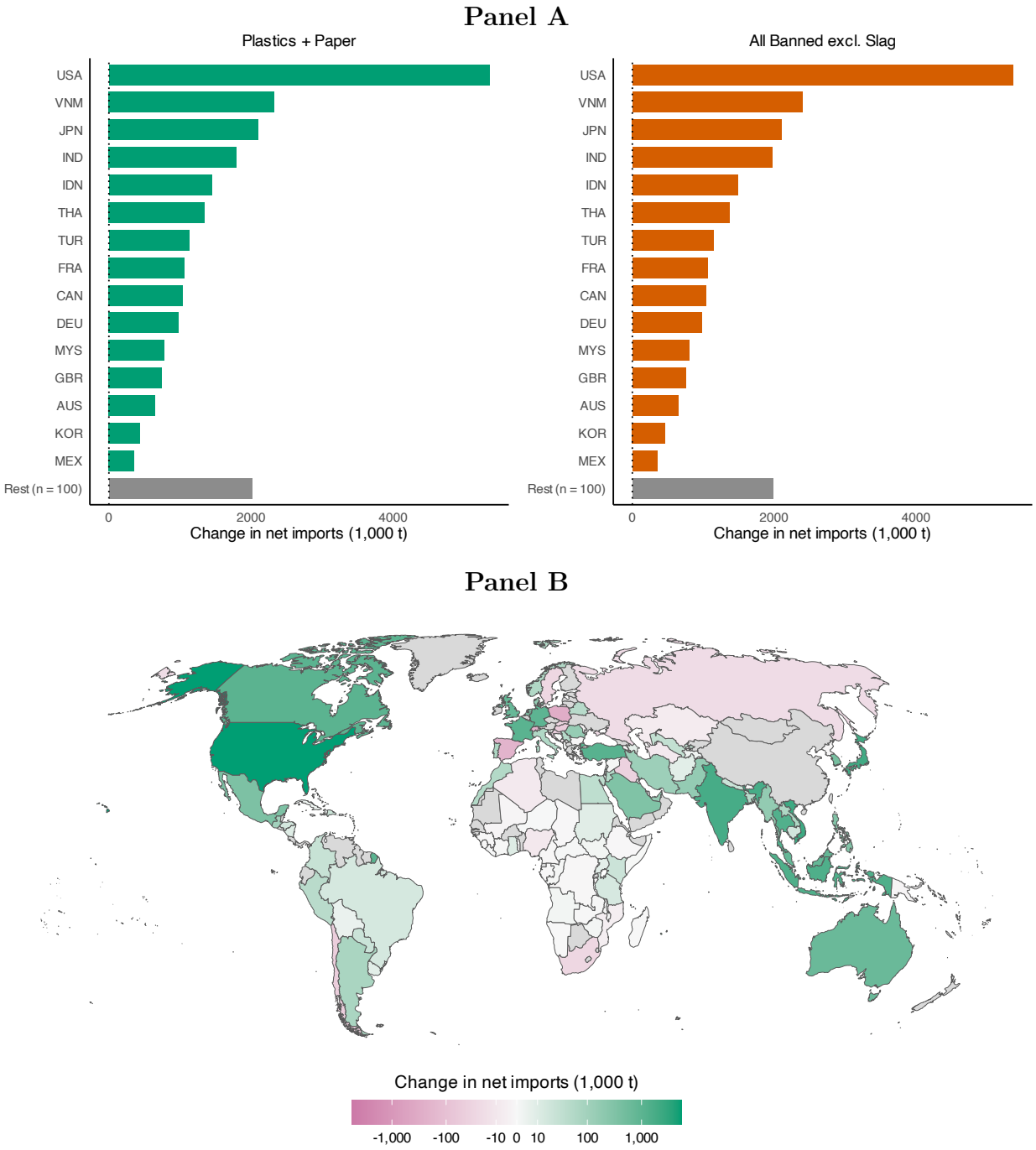
Notes: The miscalibration ratio $R_i = \sigma_i^*/\sigma_i$ as a function of the model prediction p_i (horizontal axis) and the true probability p_i^* (vertical axis). Values are capped at 3 for display. The dashed line marks perfect calibration ($p(k_i) = p_i^*$), along which $R_i = 1$; the dotted line marks $p(k_i) = 0.5$, along which $R_i = 1$.

Figure A7: Share of Land Area Covered by Open Dumps: China vs. Rest of World



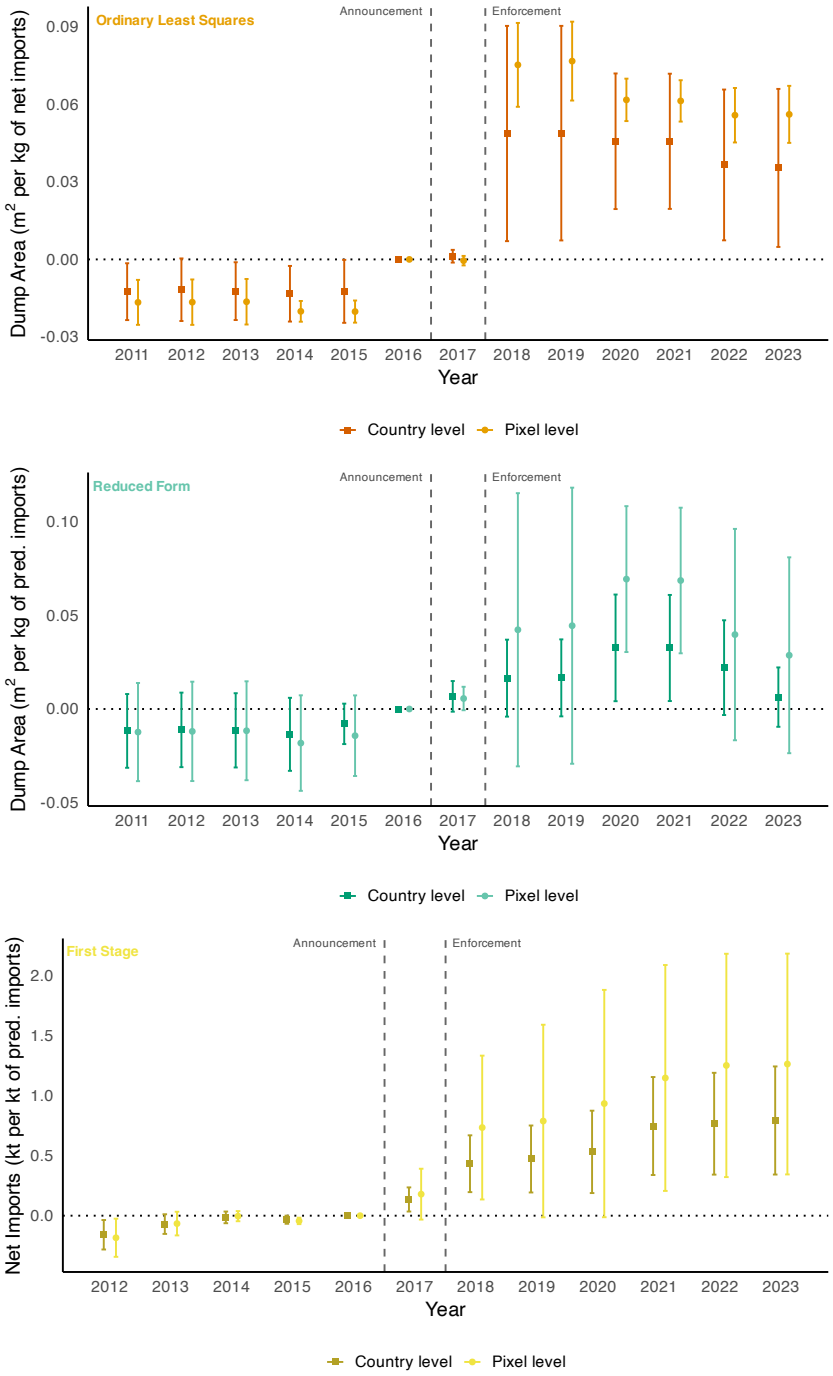
Notes: Estimates based on verified pixels, split into pixels inside and outside China. 95% confidence intervals from a stratified tile-cluster bootstrap. Dashed lines show China's waste ban announcement (2017) and enforcement (2018).

Figure A8: Change in Net Imports Following China's National Sword Policy



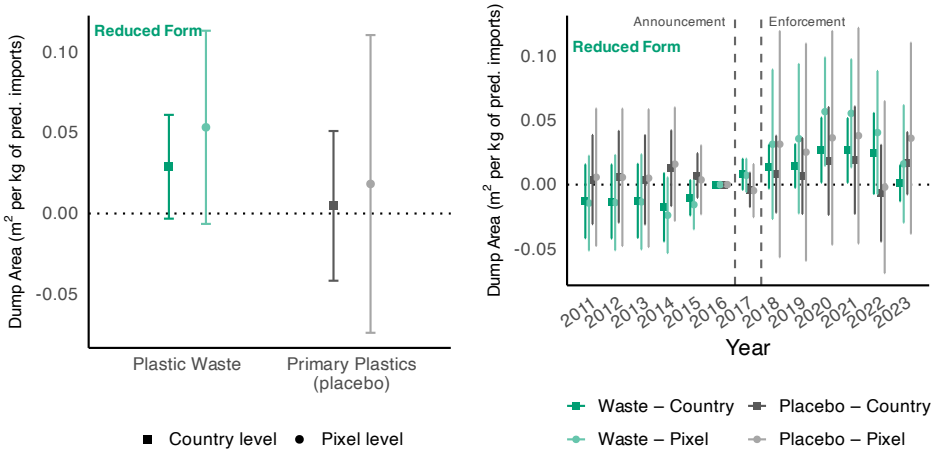
Notes: Figure shows the treatment variable in our analysis of trade and open dumps. Changes in net imports are defined as the change in imports minus exports between 2019 and 2016. Panel A shows the fifteen countries with the largest absolute changes and the pooled remainder for the main categories (left) and all categories (right) of banned materials. Panel B maps the change in net imports for the 115 countries in the estimation sample.

Figure A9: Effect of Change in Net Imports on Open Dump Area, Event Study Version (OLS, Reduced Form, First Stage)



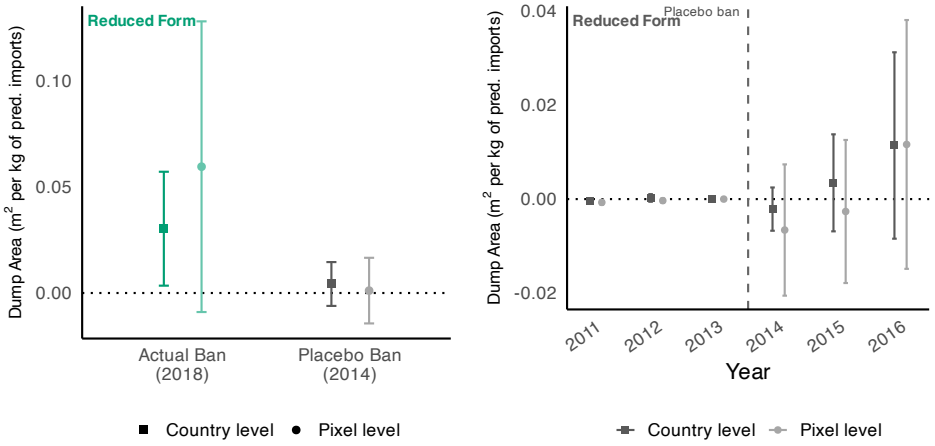
Notes: Figure shows alternative versions of the main instrumental variables event study figure. Top panel shows OLS (Equation 9; m^2 per kg of realized net imports); middle panel shows the reduced form (dump area on predicted diversion, in m^2 per kg of *predicted* diversion); bottom panel shows the first stage of yearly net imports (thousands of tons) on predicted diversion × year. The first-stage panel begins in 2012, the first year of BACI trade data. In each panel, darker squares are the country level and lighter circles the pixel level, with 95% confidence intervals.

Figure A10: Effect of Change in Net Imports on Open Dump Area, Reduced Form, Commodity Placebo



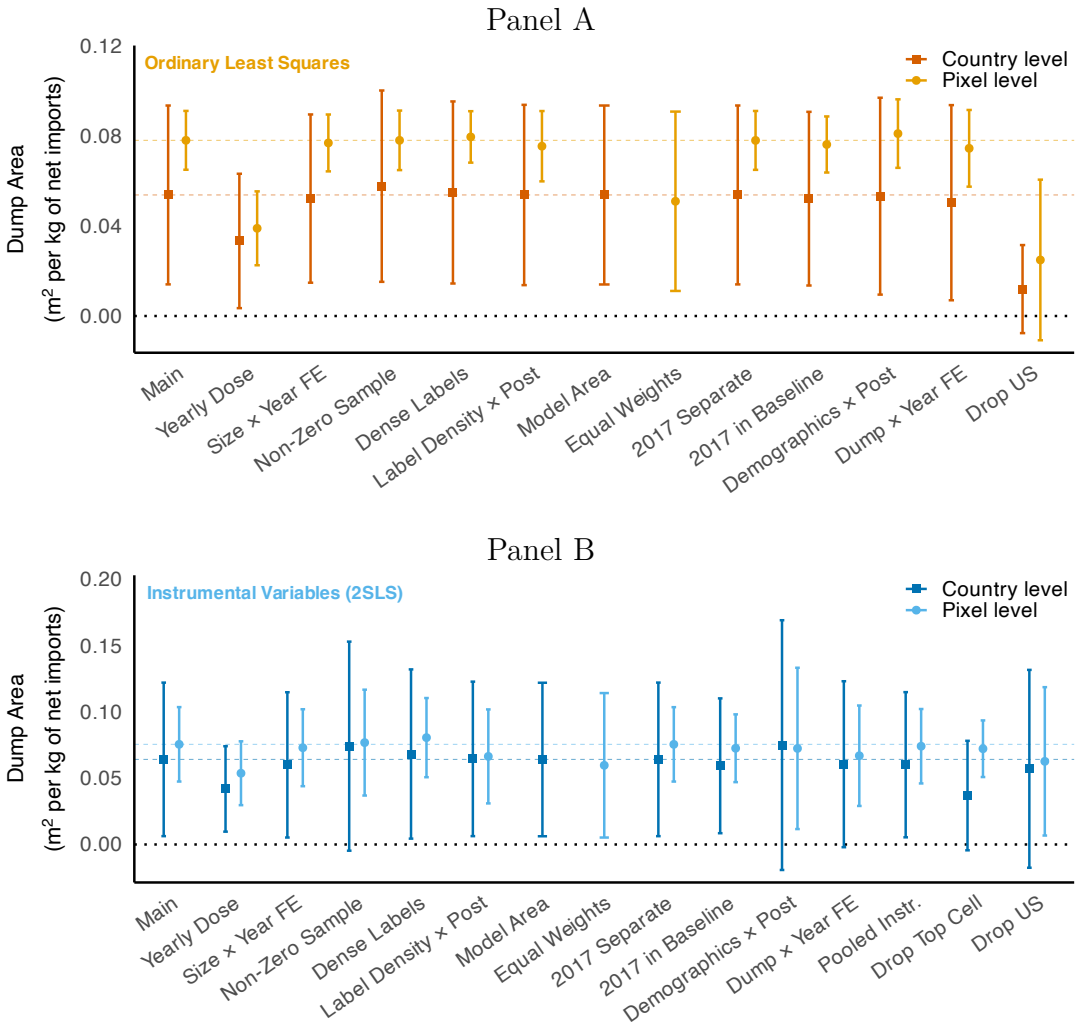
Notes: Figure shows the commodity placebo, where the same shift-share instrument construction is applied to primary-form plastics (HS 3901–3914), a commodity never banned by China. Left panel shows pooled pre/post coefficients for the waste and placebo exposures at each level. Right panel shows the year-by-year version, with both exposures interacted with event year in a single regression; dashed lines mark the ban's announcement and enforcement. Coefficients are in m^2 per kg of predicted diversion; 95% confidence intervals, standard errors clustered by country.

Figure A11: Effect of Change in Net Imports on Open Dump Area, Reduced Form, Placebo Ban Date



Notes: Figure shows the timing placebo, where the reduced form of the main design is re-estimated on pre-ban years only (2011–2016), with a counterfactual placebo ban dated between 2013 and 2014. The left panel shows the pooled reduced-form coefficient on predicted diversion at the actual ban in green, next to the same coefficient at the placebo ban (gray, pre-ban sample). The right panel shows the year-by-year reduced form around the placebo ban on the pre-ban sample, with 2013 as the reference year. Coefficients are in m^2 per kg of predicted diversion; 95% confidence intervals, standard errors clustered by country.

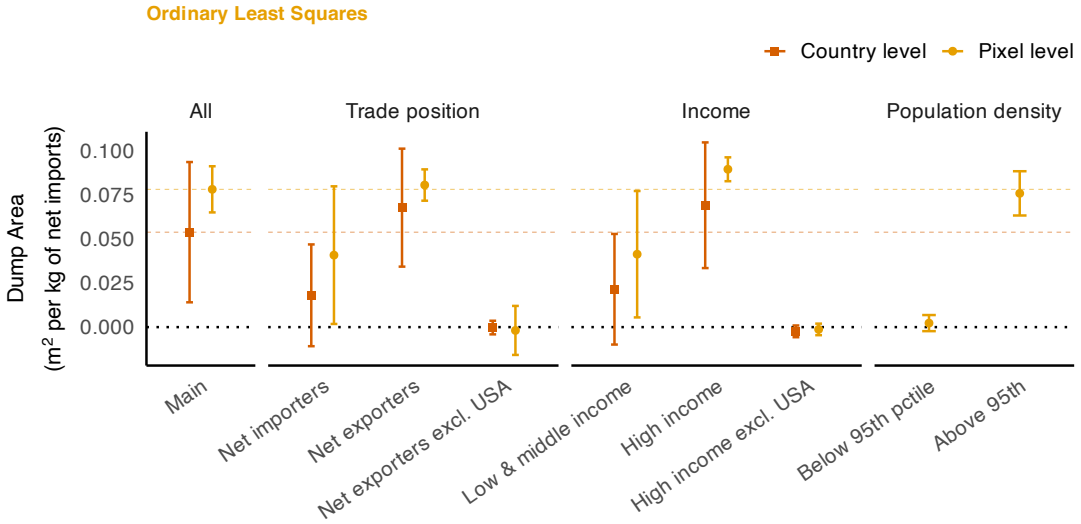
Figure A12: Effect of Change in Net Imports on Open Dump Area, Robustness



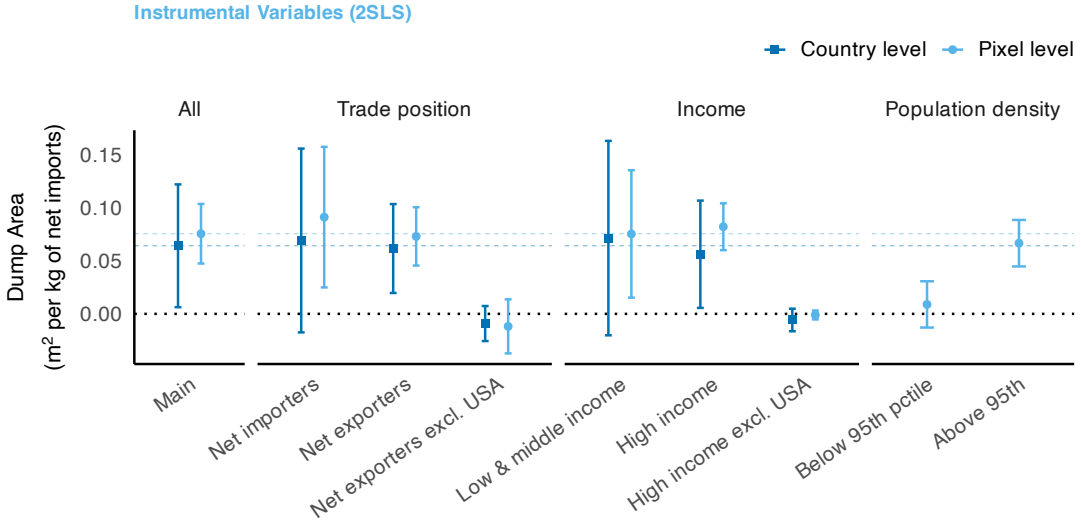
Notes: Figure shows robustness of the main OLS (Panel A) and IV (Panel B) estimates of Table 1, at the country level (darker squares) and pixel level (lighter circles) across alternative specifications. Dashed horizontal lines mark the two main estimates. From left, the specifications are the following: Main (the Table 1 specification); Yearly Dose (annual net imports instrumented by time-varying predicted diversion); Size × Year FE (land-area-quartile × year fixed effects); Non-Zero Sample (the 31 countries with verified dumps in every year); Dense Labels (the two-thirds of countries whose pixels carry the most annotator labels); Label Density × Post (the share of annotator-year cells recorded before filling, interacted with post: country mean at the country level, the pixel's own density at the pixel level); Model Area (debiased machine-learning dump area in 2018–2021, the years it covers, and verified area otherwise; country level only); Equal Weights (pixel design weights renormalized to sum to one per country × year, so each country carries equal total weight; pixel level only); 2017 Separate (the announcement year kept in the sample with its own interaction); 2017 in Baseline (the announcement year kept in the sample and folded into the pre-ban baseline); Demographics × Post (baseline log GDP per capita, log population density, and log land area interacted with post); Dump × Year FE (baseline-dump-area-quartile × year fixed effects); Pooled Instr. (instrument built on the pooled rather than per-HS-code network); Drop Top Cell (instrument rebuilt excluding US waste paper, the largest Rotemberg-weight cell); and Drop US (the United States excluded). Pooled Instr. and Drop Top Cell are properties of the instrument and so appear in Panel B only. Alternative definitions of the banned trade basket are reported separately in Appendix Table A12, because each rescales the treatment and its coefficients are therefore not comparable on this axis. Bars are 95% confidence intervals, standard errors clustered by country.

Figure A13: Effect of Change in Net Imports on Open Dump Area, Heterogeneity

Panel A

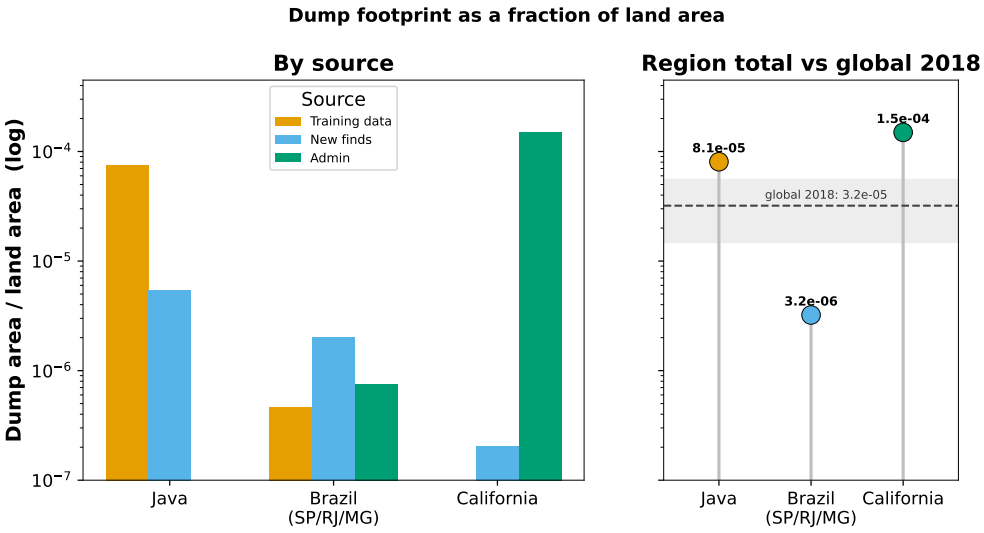


Panel B



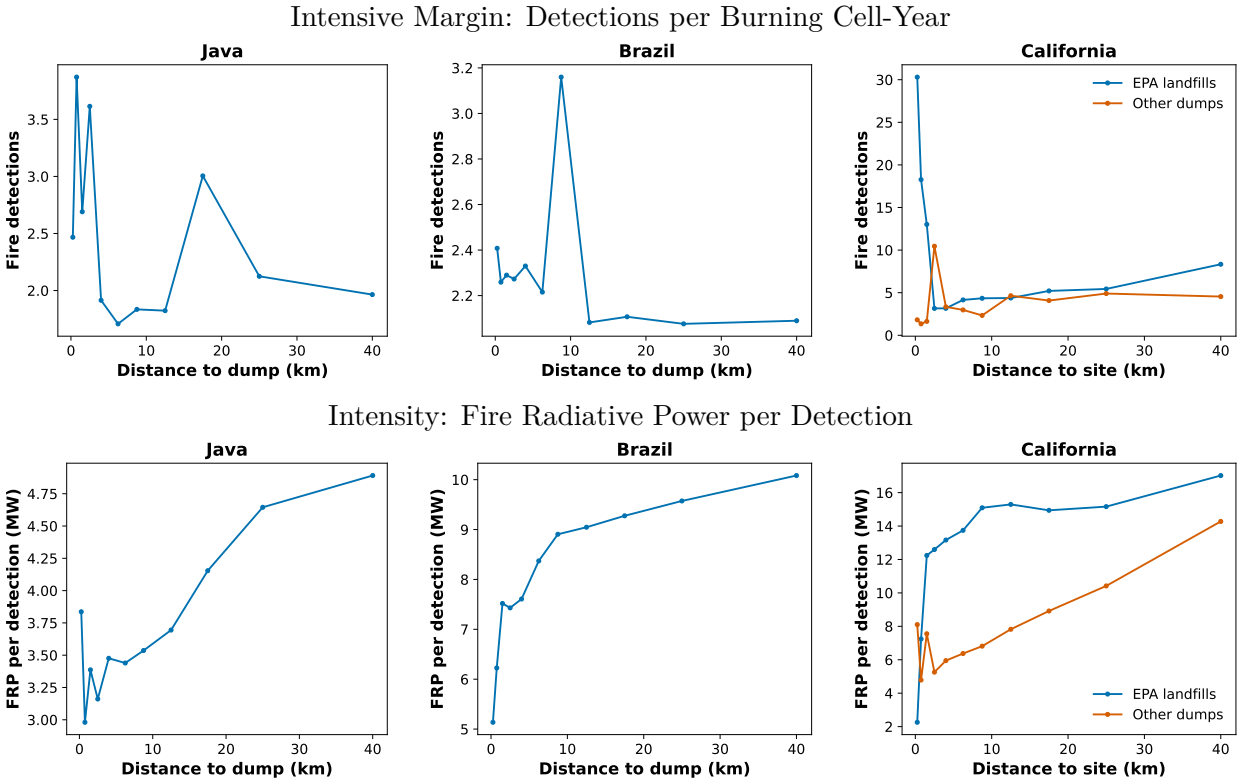
Notes: Figure shows the OLS (Panel A) and IV (Panel B) estimates of Table 1 re-estimated within groups, at the country level (darker squares) and pixel level (lighter circles). Every group keeps the main specification: Δ net imports $\times$ post, instrumented in Panel B by predicted diversion $\times$ post, with country (or tile) and year fixed effects, 2017 excluded, and pixels weighted by their sampling weights. Dashed horizontal lines mark the two main estimates. Trade position classifies countries by mean net imports of plastic waste and waste paper over 2015–2016. Income uses the World Bank high-income threshold of $13,935 GDP per capita. Net exporter and high-income groups are also shown excluding the United States, which has the largest change in net imports in the sample. Population density is pixel level only. It splits the outcome, not the sample, into dump area in pixels above and below the 95th percentile of population density across land, both estimated on the full pixel sample, so the two coefficients sum to the main pixel-level estimate. The densest 5% of land is home to 61% of the population in the sample. Bars are 95% confidence intervals, standard errors clustered by country.

Figure A14: Case Study: Dump Footprint as a Fraction of Land Area



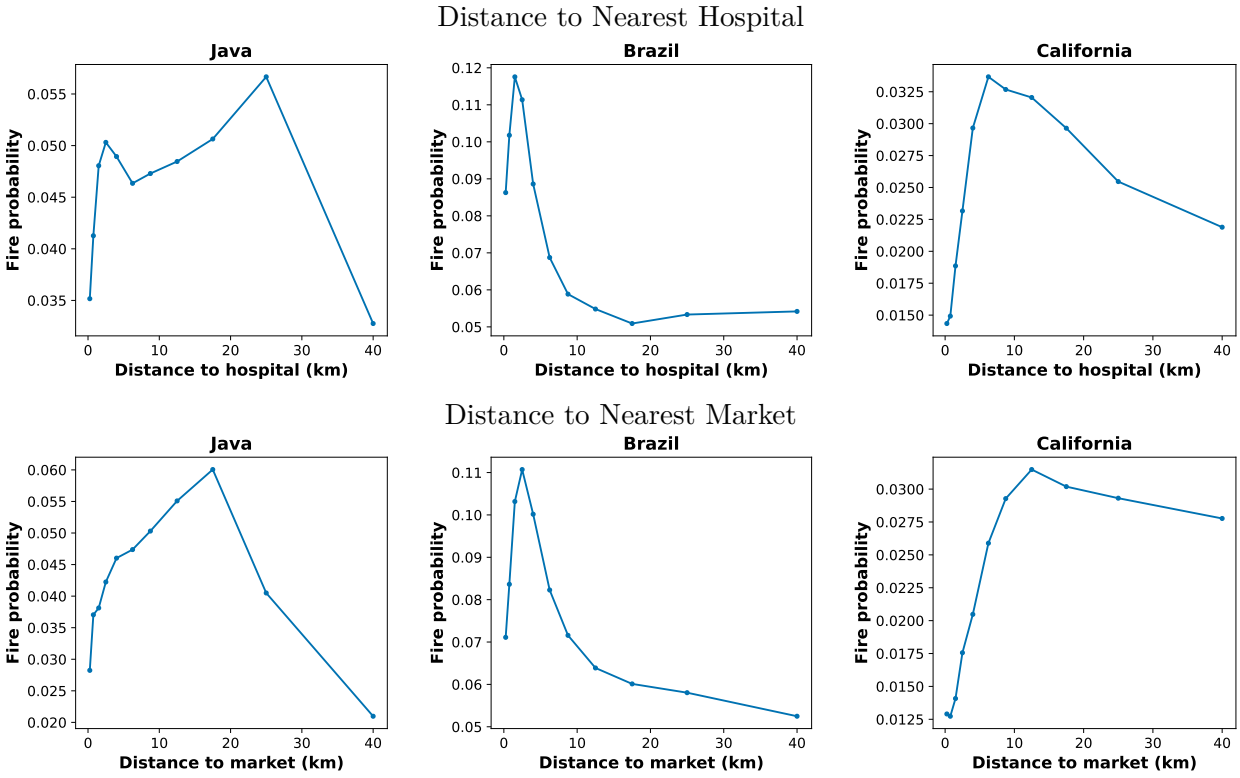
Notes: Total dump area from all sources as a share of each case-study region's land area, by source (left, log scale) and in total (right). The dashed line and shaded band show our globally representative 2018 estimate of verified dump prevalence and its 95% bootstrap confidence interval from Section 5. Note that the California estimates include sanitary landfills which are not comparable to open dumps.

Figure A15: Case Study: Garbage Fires by Distance to Dump



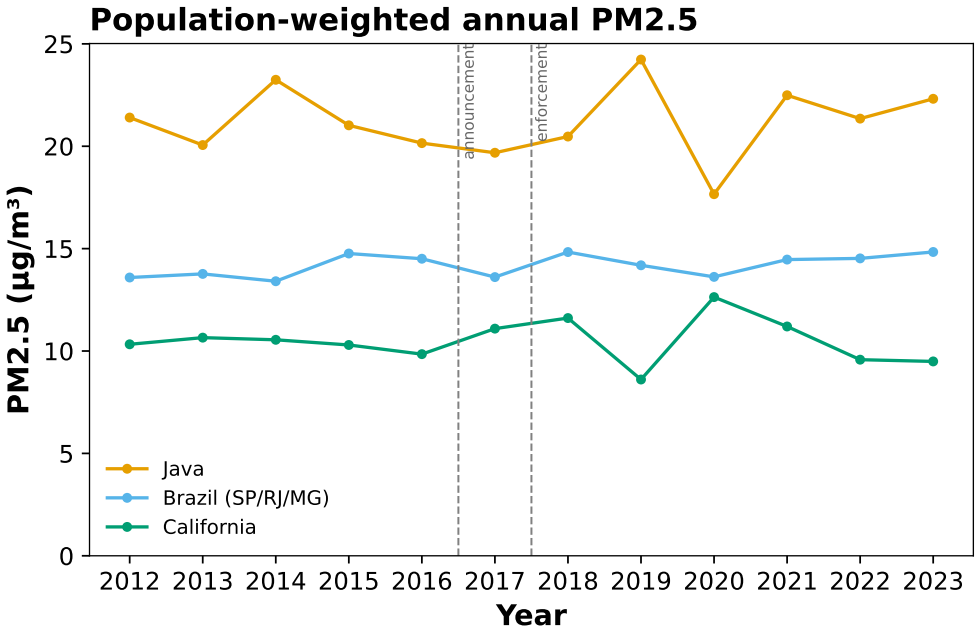
Notes: VIIRS active-fire detections in grid cells around verified dump sites, by distance to the nearest dump (pooled 2012–2023). First row plots the expected number of detections in pixels with at least one detection and the bottom row plots mean fire radiative power per detection (MW).

Figure A16: Case Study: Fire Probability by Distance to Placebo Landmarks



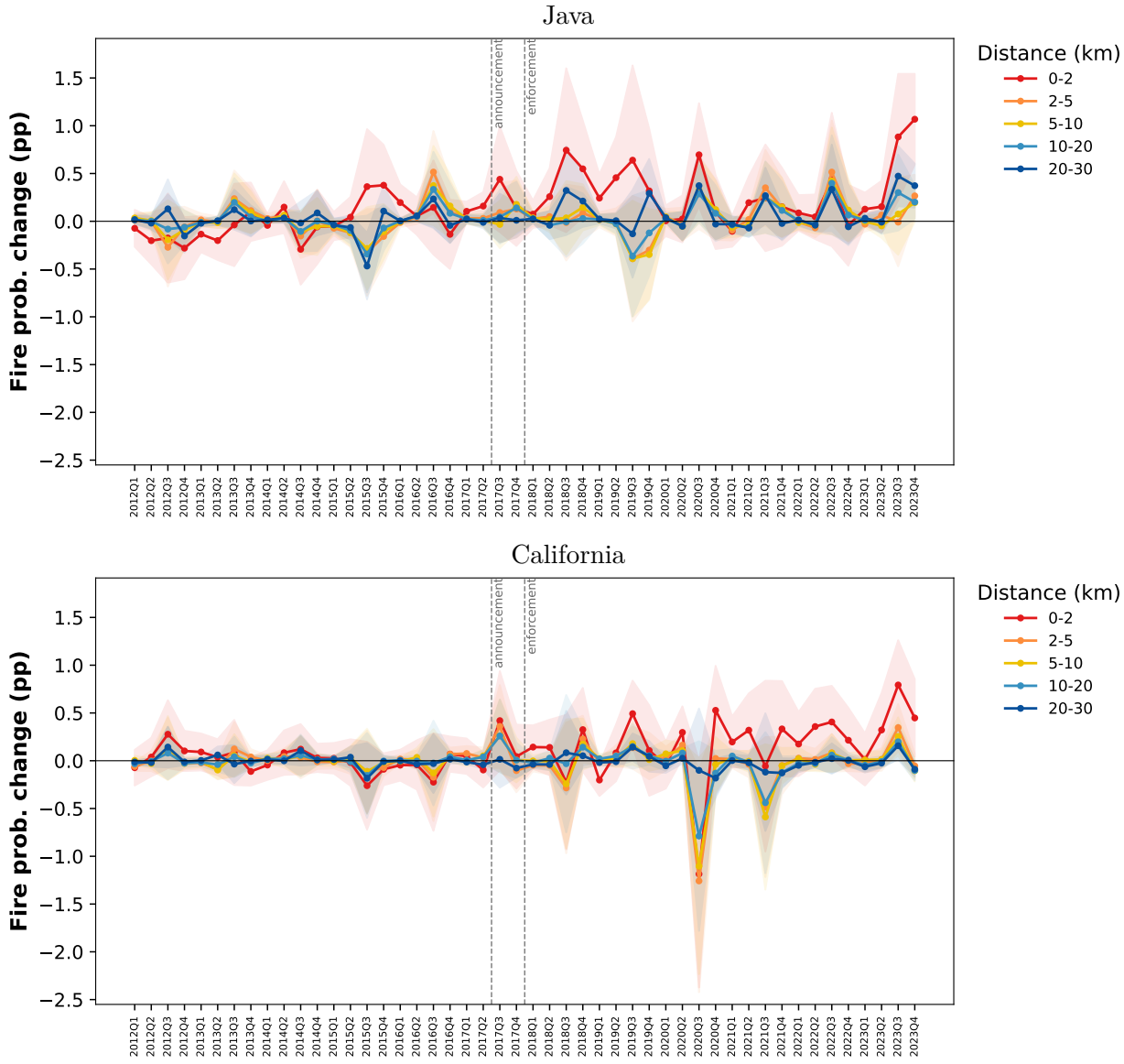
Notes: Probability of at least one VIIRS fire detection per cell-year (pooled 2012–2023), by distance to the nearest hospital (top row) or marketplace (bottom row). Hospital and marketplace locations are from OpenStreetMap (OpenStreetMap, 2026), retrieved via the Overpass API in July 2026.

Figure A17: Population-Weighted $PM_{2.5}$ in the Case-Study Regions



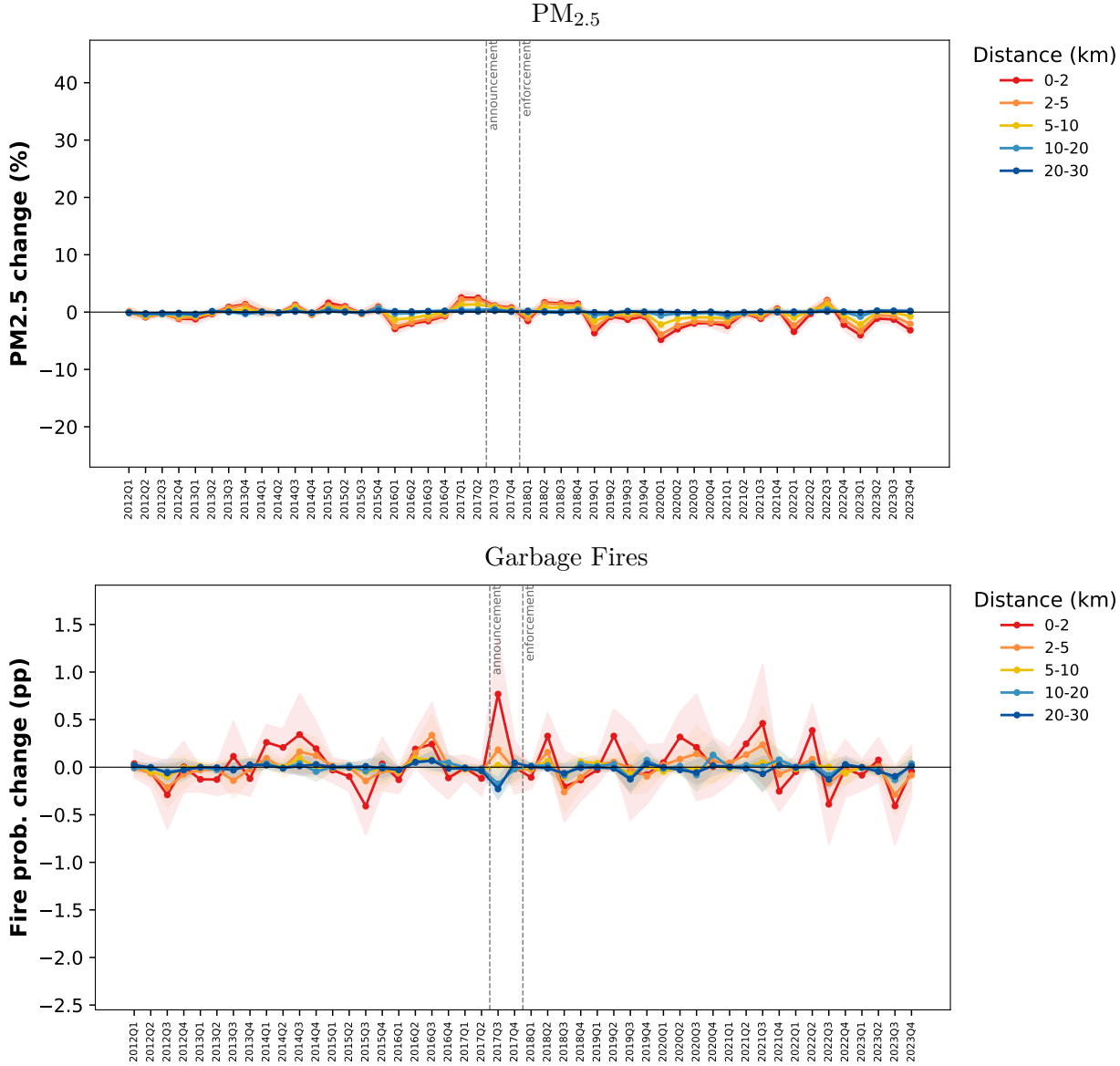
Notes: Annual mean surface $PM_{2.5}$ (van Donkelaar et al.) over all land cells in each case-study region (Java; São Paulo, Rio de Janeiro, and Minas Gerais; California), weighted by GPW v4 population density (2010) at the cell centre. Weights are fixed at their 2010 values, so changes over time reflect pollution rather than population movement. Dashed lines mark the announcement (2017) and enforcement (2018) of the China National Sword policy.

Figure A18: Garbage Fires Around Dumps and the China Ban: Event Studies



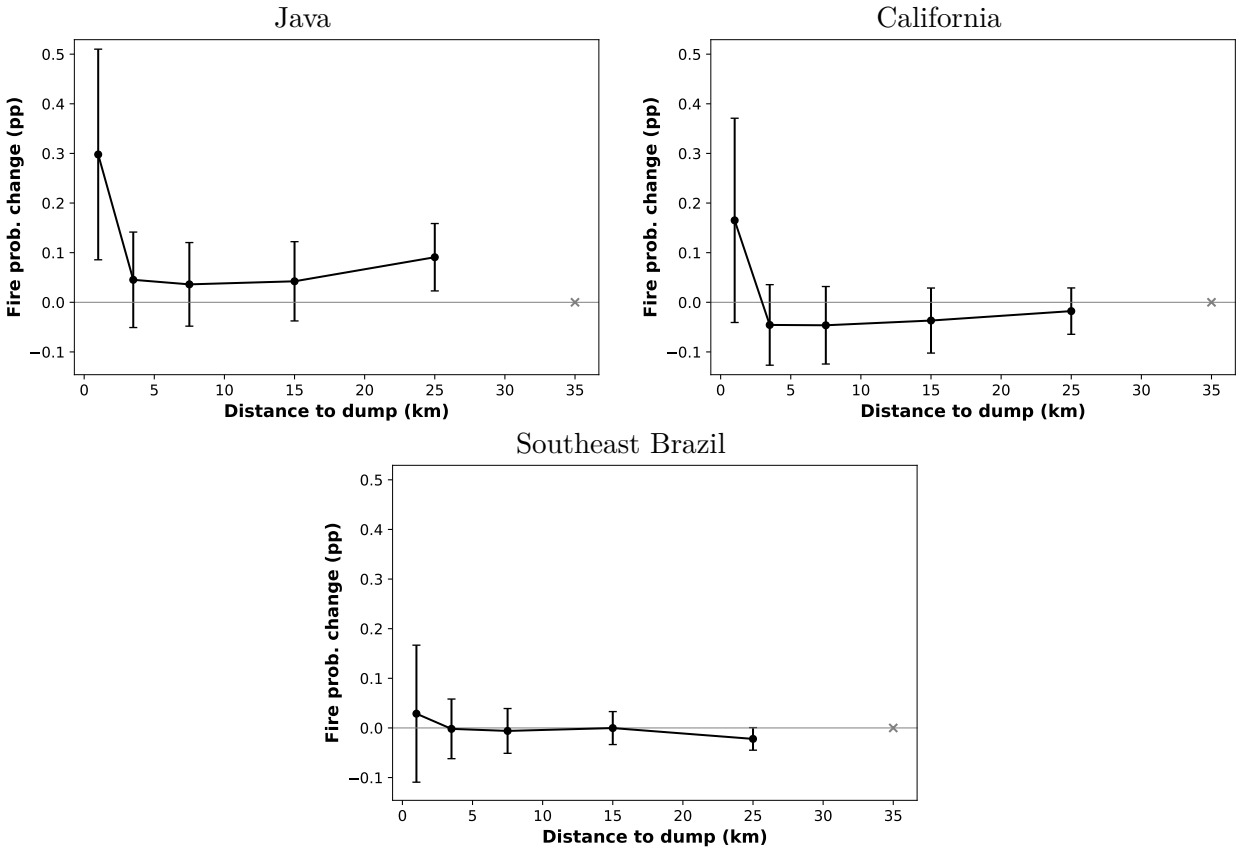
Notes: Same design as Figure 10, with an indicator for at least one VIIRS active-fire detection in the cell-month as the outcome.

Figure A19: Southeast Brazil: $PM_{2.5}$ and Garbage Fires Around Dumps, Event Studies



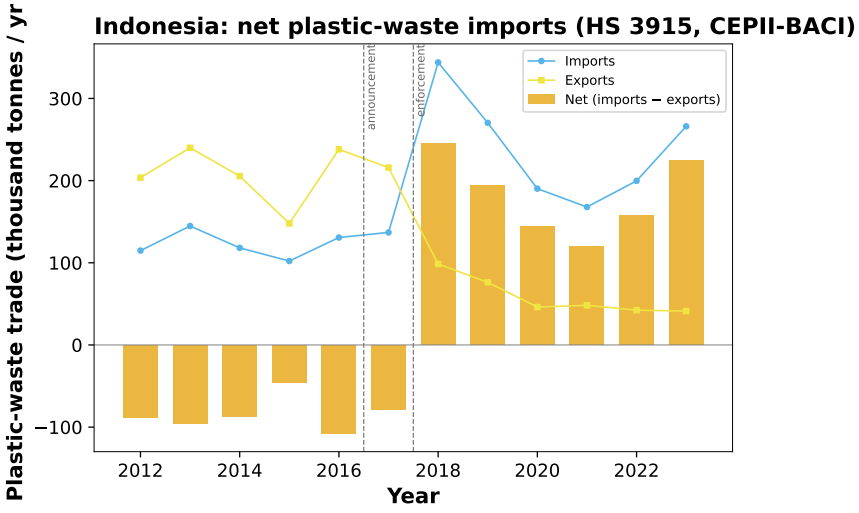
Notes: Event studies for São Paulo, Rio de Janeiro, and Minas Gerais. Top panel uses the same design as Figure 10, with log $PM_{2.5}$ as the outcome. Bottom panel uses the same design as Figure A18, with an indicator for at least one VIIRS active-fire detection in the cell-month as the outcome.

Figure A20: Garbage Fires Near vs. Far From Dumps



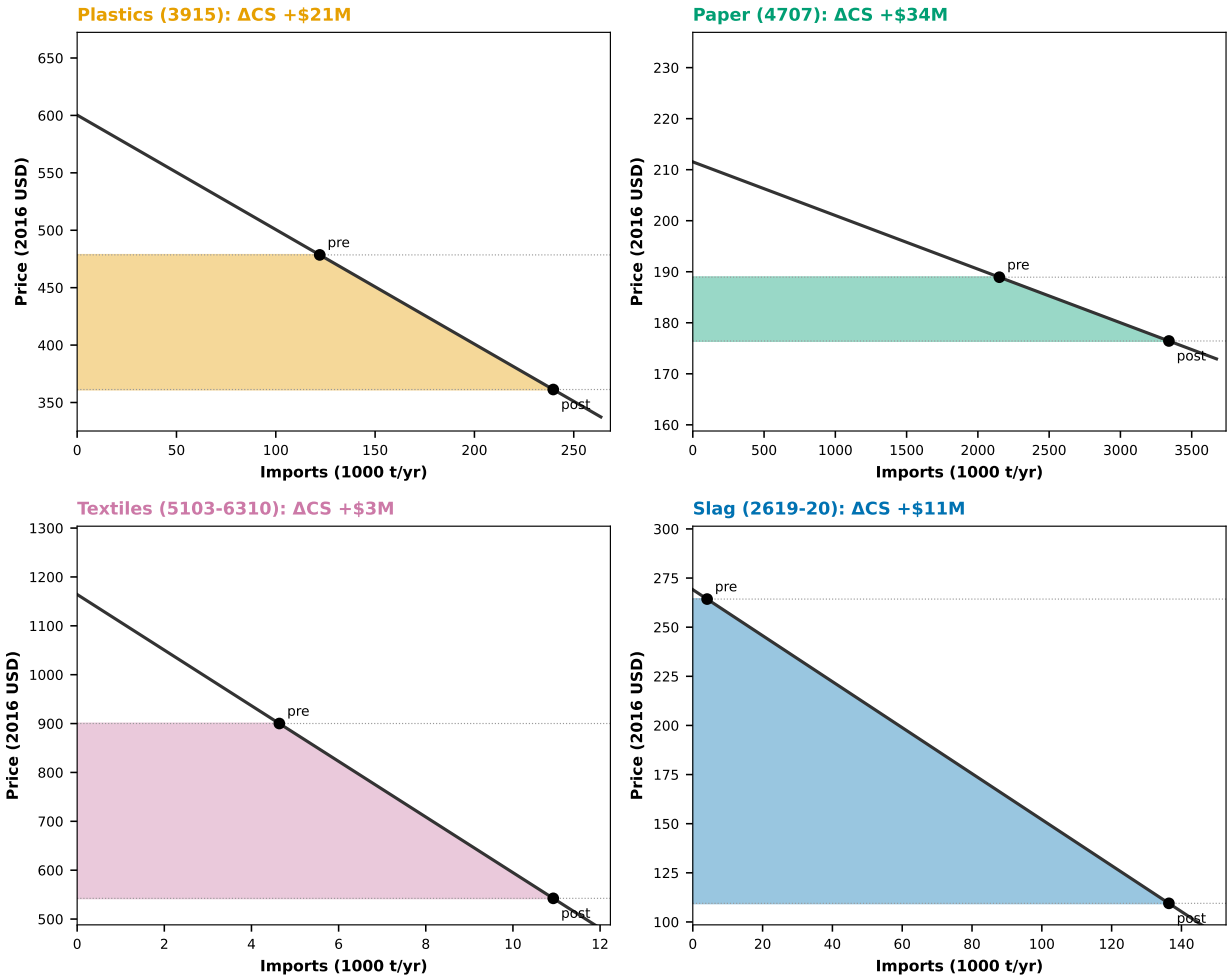
Notes: Time-pooled version of the fire event studies in Figure A18. Equation 11 is re-estimated with the ring-by-quarter interactions replaced by ring-by-announcement-period (July–December 2017) and ring-by-enforcement-period (January 2018–December 2023) interactions. The outcome is an indicator for at least one VIIRS active-fire detection in the cell-month. Each point is the enforcement-period coefficient for that ring: the change in the monthly probability of a fire detection (percentage points) relative to the pre-announcement period (January 2012–June 2017), net of the same change in the omitted 30–40 km ring (marked ×). Points sit at ring midpoints. Bars give 95% confidence intervals with standard errors clustered by district.

Figure A21: Indonesia: Net Imports of Plastic Waste Around the China Ban



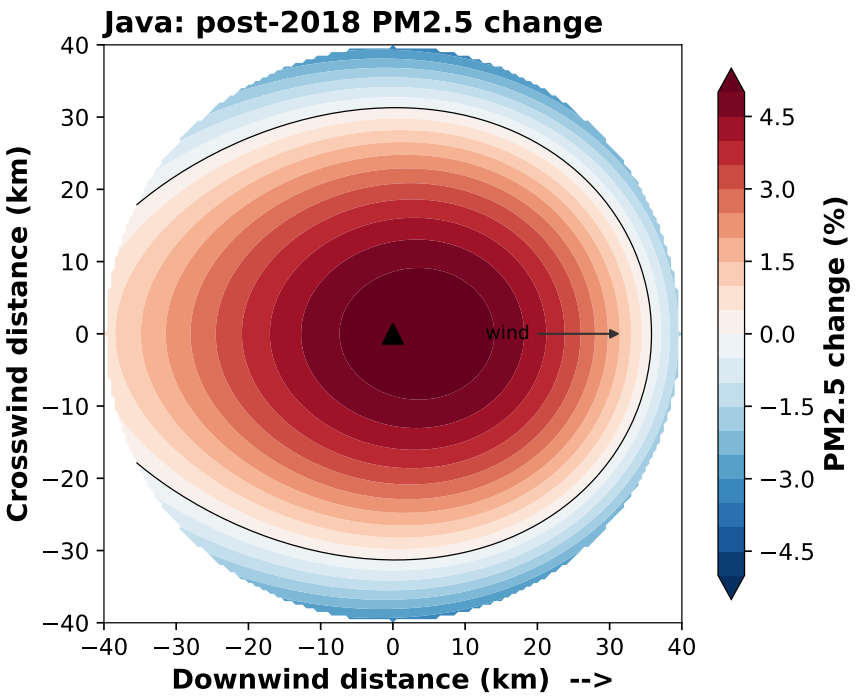
Notes: Annual net imports (imports minus exports, tons) of plastic waste and scrap (HS 3915) for Indonesia, 2012–2023, from CEPII-BACI (Gaulier and Zignago, 2010). Vertical lines mark China's WTO notification of the National Sword Policy (July 2017) and the ban taking effect (January 2018).

Figure A22: Indonesia Consumer-Surplus Gain



Notes: Figure shows Indonesia consumer-surplus gain from the supply shock induced by the 2018 China waste-import ban, by category of banned waste. Each panel reads the surplus change off the observed pre (2012–16) → post (2018–23) price and quantity changes. Points mark pre/post (quantity, price); the solid line is the revealed linear demand; the shaded area is the realized surplus change. Constant 2016 prices deflated using Indonesian CPI converted to 2016 USD using market exchange rates.

Figure A23: Wind-Relative $PM_{2.5}$ Plumes Around Dumps



Notes: Enforcement-period surface $f(d_i, w_{it})$ from Equation 12, plotted as the percent change in $PM_{2.5}$ during the enforcement period (post 2018), relative to the upwind far-field (30–40 km) mean. The horizontal axis is downwind distance from the nearest dump given the month's ERA5 wind direction; the vertical axis is crosswind distance.

D Additional Tables

Table A3: Model Input Data

Source	Features
Sentinel-1	VV and VH backscatter (interferometric wide swath mode, averaged over ascending and descending orbits), plus the three radar indices in Table A4, Panel A.
Sentinel-2	All 13 top-of-atmosphere bands (B1–B12, including B8A), plus the 19 spectral indices in Table A4, Panel B, and the median annual Cloud Score+ value.
Focal statistics	Mean and standard deviation of each Sentinel-1 and Sentinel-2 band and index in a 3-pixel window.
Ports	Distance to the nearest port.
Population	Population density for 2010, 2015, and 2020, at the pixel and averaged over a 9 km $\times$ 9 km neighborhood (Earth Science Data Systems, NASA, 2024).
Dynamic World	Probability of complete coverage by water, trees, grass, flooded vegetation, crops, shrub and scrub, built, bare, and snow and ice land uses.

Table A4: Sentinel-1 and Sentinel-2 Derived Indices

Index	Formula	Description
Panel A: Sentinel-1		
DpRVIVV	$\frac{4\times \mathrm{VH}}{\mathrm{VV}+\mathrm{VH}}$	Dual-polarized radar vegetation index; vegetation density.
VHVVR	$\frac{\mathrm{VH}}{\mathrm{VV}}$	Cross- to co-polarized ratio; vegetation structure and volume scattering.
VVVHR	$\frac{\mathrm{VV}}{\mathrm{VH}}$	Inverse of VHVVR; complementary surface information.
Panel B: Sentinel-2		
NDVI	$\frac{\mathrm{B8}-\mathrm{B4}}{\mathrm{B8}+\mathrm{B4}}$	Normalized difference vegetation index; vegetation greenness.
BSI	$\frac{(\mathrm{B11}+\mathrm{B4})-(\mathrm{B8}+\mathrm{B2})}{(\mathrm{B11}+\mathrm{B4})+(\mathrm{B8}+\mathrm{B2})}$	Bare soil index; bare soil vs. vegetation and water.
NDWI	$\frac{\mathrm{B3}-\mathrm{B8}}{\mathrm{B3}+\mathrm{B8}}$	Normalized difference water index; open water.
UI	$\frac{\mathrm{B12}-\mathrm{B8}}{\mathrm{B12}+\mathrm{B8}}$	Urban index; built-up areas.
MNDVI	$\frac{\mathrm{B8}-\mathrm{B12}}{\mathrm{B8}+\mathrm{B12}}$	Modified NDVI (SWIR2 in place of red).
MNDWI	$\frac{\mathrm{B3}-\mathrm{B11}}{\mathrm{B3}+\mathrm{B11}}$	Modified NDWI; water, suppressing built-up noise.
CSI	$\frac{\mathrm{B8}}{\mathrm{B12}}$	Char soil index; charred or burned soils.
DBSI	$\frac{\mathrm{B11}-\mathrm{B3}}{\mathrm{B11}+\mathrm{B3}} - \frac{\mathrm{B8}-\mathrm{B4}}{\mathrm{B8}+\mathrm{B4}}$	Dry bareness index; dry barren areas.
NDSWIR	$\frac{\mathrm{B8}-\mathrm{B11}}{\mathrm{B8}+\mathrm{B11}}$	Normalized difference NIR–SWIR1; vegetation moisture.
NormNIR	$\frac{\mathrm{B8}}{\mathrm{B8}+\mathrm{B3}+\mathrm{B4}}$	NIR share of visible–NIR reflectance.
NormG	$\frac{\mathrm{B3}}{\mathrm{B8}+\mathrm{B3}+\mathrm{B4}}$	Green share of visible–NIR reflectance.
NormR	$\frac{\mathrm{B4}}{\mathrm{B8}+\mathrm{B3}+\mathrm{B4}}$	Red share of visible–NIR reflectance.
SWIR1SWIR2	$\frac{\mathrm{B11}}{\mathrm{B12}}$	SWIR ratio; mineral and soil composition.
NIRNIR2	$\frac{\mathrm{B8}}{\mathrm{B8A}}$	NIR ratio; vegetation structure.
RE1RE2	$\frac{\mathrm{B5}}{\mathrm{B6}}$	Red-edge ratio; chlorophyll content and vegetation stress.
RE2RE3	$\frac{\mathrm{B6}}{\mathrm{B7}}$	Red-edge ratio; vegetation health.
BG	$\frac{\mathrm{B2}}{\mathrm{B3}}$	Blue/green ratio; water quality and algal blooms.
BR	$\frac{\mathrm{B2}}{\mathrm{B4}}$	Blue/red ratio; atmospheric scattering and water properties.
GR	$\frac{\mathrm{B3}}{\mathrm{B4}}$	Green/red ratio; chlorophyll absorption.

Notes: Sentinel-2 band designations: B2 blue, B3 green, B4 red, B5–B7 red edge 1–3, B8 near-infrared (NIR), B8A narrow NIR, B11 shortwave infrared 1 (SWIR1), B12 SWIR2. Sentinel-1 VV and VH denote co-polarized and cross-polarized backscatter.

Table A5: Classifier Hyperparameters: Search Space and Selected Values

Hyperparameter	Values considered	Selected
num_trees	50, 100, 150, 200	100
max_depth	10, 15, 20, 25, 30	25
min_examples	3, 6, 27	27
shrinkage	0.02, 0.05	0.02
l1_regularization	3.0, 5.0, 7.0	5.0
l2_regularization	3.0, 5.0, 7.0	3.0
num_candidate_attributes_ratio	0.1, 0.2, 0.3	0.1
split_axis	AXIS_ALIGNED, SPARSE_OBLIQUE	AXIS_ALIGNED
growing_strategy	LOCAL, BEST_FIRST_GLOBAL	LOCAL
forest_extraction	DART	DART

Notes: Hyperparameters of the gradient-boosted-trees classifier (TensorFlow Decision Forests), selected by random search (30 trials per fold) within each of three group k-fold cross-validation splits, with training sites grouped by cluster and validation AUC as the objective. Conditional parameters attached to non-selected options (`max_num_nodes` under BEST_FIRST_GLOBAL; the `sparse_oblique_*` settings under SPARSE_OBLIQUE) are omitted. `forest_extraction` was fixed at DART rather than tuned. Remaining parameters were left at library defaults.

Table A6: Correlates of Model Prediction Bias

	Dependent variable: ν_i (× 100)			
	Year	World region	Log GDP	Log pop. density
	(1)	(2)	(3)	(4)
Year 2018	−0.0012*			
	(0.0006)			
Year 2019	−0.0011*			
	(0.0006)			
Year 2020	−0.0009			
	(0.0006)			
Year 2021	−0.0007			
	(0.0006)			
Europe & Central Asia		0.0039**		
		(0.0019)		
Latin America & Caribbean		0.0017		
		(0.0029)		
Middle East & North Africa		−0.0031		
		(0.0072)		
North America		0.0000		
		(0.0041)		
South Asia		−0.0040		
		(0.0056)		
Sub-Saharan Africa		0.0043**		
		(0.0019)		
Log GDP per capita			−0.0045	
			(0.0075)	
Log GDP per capita^2			0.0002	
			(0.0004)	
Log population density				−0.0027***
				(0.0008)
Observations	68,855	68,855	68,855	68,855
R^2	0.0000	0.0000	0.0000	0.0001

Notes: Each column shows a separate regression of prediction errors $(p(k_i) - D_i)\times 100$ on the indicated covariates. Observations are pixel × year, 2017–2021. Regressions are weighted using inverse probability sampling weights. Omitted categories are 2017 in Column 1 and East Asia & Pacific in Column 2. Standard errors are clustered by tile. $^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A7: Share of Open Dumps on Fire

	Share of dump area on fire (%)	
	Verification only	AIPW
	(1)	(2)
All countries	11.3	11.3
	[3.4, 19.8]	[3.3, 19.2]
By income, excluding China		
High income	3.6	3.6
	[1.0, 13.0]	[1.1, 13.1]
Low/middle income	16.7	16.7
	[4.3, 27.0]	[4.3, 26.4]
By population density, excluding China		
Below 80th percentile	13.4	13.4
	[3.3, 23.3]	[3.1, 23.0]
Above 80th percentile	14.6	14.6
	[5.2, 31.6]	[5.0, 31.5]
China	2.4	2.4
	[0.3, 7.1]	[0.5, 7.5]

Notes: Share of dump pixels with at least one VIIRS active-fire detection within 375 m during the year, in percent, pooled over 2012–2023. Column (1) uses a Horvitz–Thompson estimator on the verified labels alone. Column (2) is the bias-corrected (AIPW) estimate: in 2017–2021, the years the model is deployed, it subtracts the prediction error measured on the verification sample from the model's predicted dump and dump-fire areas; in the remaining years it uses the verification labels. High income is GDP per capita above the World Bank threshold of $13,935. High density is defined as the 80th percentile of population density and above (top two deciles in Figure 5). China is excluded from the income and density groups and shown separately. Brackets give 95% confidence intervals from a stratified tile-cluster bootstrap (1,000 replicates)

Table A8: Effects of Change in Net Imports on Open Dump Area (Announcement Year 2017 Shown Separately)

	Dependent variable: Dump area (km^2)	
	OLS	IV
	(1)	(2)
Panel A: Country-Level		
Δ Net Imports (kt)	0.012***	0.034
× 2017 (announced)	(0.004)	(0.028)
	[0.007]	[0.228]
Δ Net Imports (kt)	0.054***	0.064**
× Post-2017	(0.020)	(0.029)
	[0.008]	[0.030]
Implied elasticity, 2017	0.311	0.908
Implied elasticity, post	1.445	1.722
First stage		0.473*** (0.141)
First-stage F		11.3 / 11.3
Observations	1,495	1,495
Countries	115	115
Panel B: Pixel-Level		
Δ Net Imports (kt)	0.013***	0.021*
× 2017 (announced)	(0.003)	(0.012)
	[0.000]	[0.088]
Δ Net Imports (kt)	0.078***	0.076***
× Post-2017	(0.007)	(0.014)
	[0.000]	[0.000]
Implied elasticity, 2017	0.304	0.483
Implied elasticity, post	1.811	1.751
First stage		0.789* (0.407)
First-stage F		5.9 / 5.9
Observations	158,496	158,496
Countries	115	115

Notes: Specification, units, instrument and the elasticity definition are exactly as in Table 1; see its notes. This table differs only in keeping 2017 in the sample with its own interaction, reported here, where Table 1 excludes it from the estimation sample. China notified the WTO in July 2017 and enforced the ban from January 2018, so 2017 is neither a clean pre- nor a clean post-ban year; it carries its own interaction and so enters neither the pre-ban baseline nor the post estimate. The two first-stage Fs are the announcement and enforcement terms, each identified by its own instrument. SEs clustered by country in parentheses, p-values in brackets; China excluded. *p< 0.1; **p< 0.05; ***p< 0.01.

Table A9: Reduced-Form and First-Stage Estimates (Pre/Post Ban)

Outcome	Reduced Form Dump area (km^2) (1)	First Stage Δ Net Imp. (kt) $\times$ Post (2)
Panel A: Country-Level		
Predicted Δ (kt)	0.030**	0.473***
$\times$ Post-2017	(0.014)	(0.140)
	[0.027]	[0.001]
First-stage F		11.3
Observations	1,380	1,380
Countries	115	115
Panel B: Pixel-Level		
Predicted Δ (kt)	0.060*	0.789*
$\times$ Post-2017	(0.035)	(0.407)
	[0.088]	[0.052]
First-stage F		3.8
Observations	146,304	146,304
Countries	115	115

Notes: Table shows the two components of the IV in Table 1. Reduced form (column 1): dump area (km^2) on predicted diversion $\times$ post — coefficients are m^2 per kg of predicted diversion (numerically identical to km^2 per kt). First stage (column 2): realized change in net imports $\times$ post (kt) on predicted diversion $\times$ post; F is the cluster-robust first-stage F. Predicted diversion = pre-ban bilateral trade shares $\times$ each exporter's pre-ban exports to China (per-HS-code shift-share); post = year $>$ 2017. China notified the WTO in July 2017 and enforced the ban from January 2018, so 2017 is neither a clean pre- nor a clean post-ban year; it is excluded from the estimation sample, so the pre-ban baseline is the years through 2016. Panel A: country $\times$ year panel, outcome = verified dump area, country + year FE, unweighted. Panel B: verified-pixel $\times$ year panel, outcome = pixel dump status $\times$ country land area, tile + year FE, weighted by inverse sampling probabilities. SEs clustered by country in parentheses, p-values in brackets; China excluded. *p< 0.1; **p< 0.05; ***p< 0.01.

Table A10: Shift-Share Exposure and Pre-Ban Country Characteristics

	Standardized exposure	
	Bivariate	Joint
	(1)	(2)
Pre-ban dump-area trend (km^2/yr)	0.098	-0.420*
	(0.153)	(0.249)
Pre-ban dump area (km^2)	0.346	0.586***
	(0.302)	(0.209)
Pre-ban net imports (kt/yr)	-0.021	0.085
	(0.171)	(0.136)
Log GDP per capita	0.226***	0.270***
	(0.082)	(0.088)
Log population density	0.216*	0.416***
	(0.115)	(0.122)
Log land area	0.200**	0.405***
	(0.097)	(0.131)
Joint Wald p		0.000
Countries	115	115

Notes: Cross-country regressions of the shift-share exposure (predicted diversion, standardized) on pre-ban country characteristics (standardized): one at a time (column 1, the correlation) and jointly (column 2). Pre-ban = 2011–2016. Dump-area trend = per-country OLS slope of verified dump area on year over the pre-ban years. Correlation with pre-ban levels is absorbed by the country fixed effects in the main specifications; the identifying assumption concerns the trend row. Heteroskedasticity-robust SEs in parentheses. *p< 0.1; **p< 0.05; ***p< 0.01.

Table A11: Label Density and Trade Exposure

	Country level		Pixel level	
	Bivariate	Joint	Bivariate	Joint
	(1)	(2)	(3)	(4)
Δ net imports (kt)	0.106	-0.094**	0.198***	0.038
	(0.071)	(0.045)	(0.074)	(0.076)
Shift-share exposure (kt)	0.202***	0.022	0.311***	-0.021
	(0.045)	(0.057)	(0.092)	(0.133)
Pre-ban dump-area trend (km^2/yr)	0.036	-0.032	0.078	-0.262**
	(0.040)	(0.056)	(0.049)	(0.123)
Pre-ban dump area (km^2)	0.079	0.074	0.265*	0.451***
	(0.061)	(0.061)	(0.141)	(0.168)
Log GDP per capita	0.301***	0.300***	0.066	0.151***
	(0.094)	(0.092)	(0.090)	(0.057)
Log population density	0.426***	0.502***	0.285***	0.205***
	(0.081)	(0.102)	(0.084)	(0.072)
Log land area	-0.169*	0.171	-0.053	-0.091
	(0.100)	(0.119)	(0.077)	(0.074)
Observations	115	115	12,129	12,129
Countries	115	115	115	115

Notes: Label density is the share of a pixel's 26 annotator-year cells (two annotators × 13 years) carrying a recorded value, measured before any of the filling described in Appendix Table A2. It is regressed on the trade exposure and on pre-ban country characteristics, one at a time (columns 1 and 3, the correlation) and jointly (columns 2 and 4); all variables are standardized within each level, so coefficients are standard deviations of density per standard deviation of the covariate. Pre-ban = 2011–2016. Columns 1–2 collapse to one observation per country, unweighted, with heteroskedasticity-robust standard errors; columns 3–4 use the pixel-level sample the dump-prevalence regressions run on, weighted by the design weight and clustered by country. Tile-level design stand-ins, which no annotator was shown, are excluded. *p< 0.1; **p< 0.05; ***p< 0.01.

Table A12: Effects of Change in Net Imports on Open Dump Area, by Trade Definition

	All, Incl. Slag	All, Excl. Slag	Main	Unsorted Paper
	(1)	(2)	(3)	(4)
Panel A: Country-Level				
OLS	0.014	0.053***	0.054***	0.062
	(0.013)	(0.020)	(0.020)	(0.044)
	[0.277]	[0.010]	[0.008]	[0.165]
IV (2SLS)	0.058**	0.062**	0.064**	0.104**
	(0.027)	(0.028)	(0.029)	(0.042)
	[0.036]	[0.030]	[0.030]	[0.013]
AR p-value	[0.028]	[0.025]	[0.025]	[0.037]
First stage	0.501	0.487	0.473	0.800
First-stage F	10.4	12.1	11.3	40.3
Exposure SD (kt)	1318	657	648	339
Observations	1,380	1,380	1,380	1,380
Countries	115	115	115	115
Panel B: Pixel-Level				
OLS	0.032	0.078***	0.078***	0.176**
	(0.022)	(0.007)	(0.007)	(0.078)
	[0.144]	[0.000]	[0.000]	[0.025]
IV (2SLS)	0.063***	0.073***	0.076***	0.157**
	(0.017)	(0.014)	(0.014)	(0.072)
	[0.000]	[0.000]	[0.000]	[0.030]
AR p-value	[0.090]	[0.089]	[0.088]	[0.147]
First stage	0.931	0.806	0.789	1.032
First-stage F	3.6	4.1	3.8	18.6
Exposure SD (kt)	1318	657	648	339
Observations	146,304	146,304	146,304	146,304
Countries	115	115	115	115

Notes: The specification of Table 1 under four definitions of the banned trade basket, widest first: (1) every heading China banned including vanadium slag (HS 2619–2620); (2) every banned heading except slag; (3) the main measure, plastic waste and scrap (HS 3915) plus recovered paper (HS 4707); (4) plastics plus only unsorted recovered paper (HS 4707.90), the one paper subheading named in the ban. Each definition rescales the treatment, so coefficients are NOT comparable across columns in levels — the exposure standard deviation is reported for that reason. The AR p-value inverts the Anderson–Rubin test. SEs clustered by country in parentheses, p-values in brackets; China excluded. *p< 0.1; **p< 0.05; ***p< 0.01.

Table A13: Rotemberg Weights: Which Origins Drive the Shift-Share IV

Origin	HS code	$\hat{g}_k$ (kt/yr)	$\hat{\alpha}_k$	$\hat{\beta}_k$
USA	4707	13059.9	0.297	0.129
CAN	4707	1465.9	0.217	0.086
JPN	4707	2987.8	0.091	-0.009
GBR	4707	3703.7	0.082	0.027
HKG	4707	840.7	0.039	-0.007
AUS	4707	911.7	0.035	0.010
HKG	3915	1636.2	0.028	-0.009
NLD	4707	1351.6	0.027	-0.006
MEX	3915	156.2	0.019	0.086
ITA	4707	1039.0	0.018	-0.007
Cells (origin × code)				199
Sum of positive weights				1.001
Sum of negative weights (share of cells)				-0.001 (0.131)
Top-5 share of positive weights				0.726

Notes: Rotemberg-weight decomposition of the country-level IV (Goldsmith-Pinkham, Sorkin and Swift 2020): $\hat{\beta}_{IV} = \sum_k \hat{\alpha}_k \hat{\beta}_k$ over instrument cells k = (origin × HS code), where $\hat{g}_k$ is the cell's pre-ban exports to China, $\hat{\alpha}_k$ its weight, and $\hat{\beta}_k$ the just-identified IV using only that cell's destination shares. Weights sum to one; the 10 largest in absolute value shown. The pixel-level IV uses the same country-level instrument.

Table A14: Case Study: Near–Far DiD Across Specifications—Java, Indonesia

Ring (km)	Main	Monthly, cell FE	Annual	Border-trimmed	Drop 2020
0-2	+6.09	+6.15	+5.97	+6.26	+4.36
	[+3.28, +8.99]	[+3.48, +8.88]	[+3.19, +8.84]	[+3.33, +9.27]	[+2.13, +6.63]
2-5	+5.10	+5.15	+5.03	+5.27	+3.61
	[+2.49, +7.78]	[+2.68, +7.68]	[+2.41, +7.72]	[+2.54, +8.07]	[+1.53, +5.73]
5-10	+4.13	+4.20	+4.13	+4.31	+2.86
	[+1.67, +6.66]	[+1.87, +6.57]	[+1.61, +6.70]	[+1.72, +6.96]	[+0.89, +4.87]
10-20	+2.49	+2.61	+2.49	+2.65	+1.48
	[+0.06, +4.98]	[+0.31, +4.96]	[+0.03, +5.01]	[+0.09, +5.28]	[-0.49, +3.50]
20-30	+0.52	+0.61	+0.60	+0.63	-0.33
	[-1.41, +2.48]	[-1.20, +2.45]	[-1.33, +2.57]	[-1.43, +2.73]	[-2.04, +1.41]
Time Aggregation	monthly	monthly	annual	monthly	monthly
Cell FE	cell × moy	cell	cell	cell × moy	cell × moy
Cells	106,633	106,633	106,633	104,597	106,633
Clusters (adm-2)	125	125	125	124	125

Notes: Difference-in-differences estimates of the effect of China's waste import ban on $PM_{2.5}$ by distance ring. Equation 11 is re-estimated with the ring-by-quarter interactions replaced by ring-by-announcement-period (July–December 2017) and ring-by-enforcement-period (January 2018–December 2023) interactions. Enforcement period estimates are shown across five specifications. Monthly specifications include cell by month-of-year and district-by-month fixed effects; the second column replaces the cell-by-month-of-year fixed effect by a simple cell fixed effect. The annual specification collapses the panel to cell-by-year with cell and district-by-year fixed effects. Border-trimmed drops cells that are close to region boundaries that might be contaminated by out-of-region dumps. Drop 2020 excludes calendar year 2020 from the post-enforcement period. Effects are in percent, $100(e^{\beta} - 1)$, and relative to the omitted 30–40 km ring, with 95% confidence intervals in brackets (standard errors clustered by district).

Table A15: Case Study: Near–Far DiD Across Specifications—California

Ring (km)	Main	Monthly, cell FE	Annual	Border-trimmed	Drop 2020
0-2	-3.07	-3.14	-4.75	-3.30	-2.09
	[-4.55, -1.58]	[-4.58, -1.68]	[-7.31, -2.11]	[-4.81, -1.77]	[-3.38, -0.78]
2-5	-3.42	-3.49	-5.05	-3.65	-2.51
	[-4.83, -1.99]	[-4.87, -2.09]	[-7.56, -2.47]	[-5.08, -2.20]	[-3.77, -1.24]
5-10	-3.29	-3.35	-4.90	-3.53	-2.45
	[-4.73, -1.82]	[-4.76, -1.91]	[-7.49, -2.23]	[-5.00, -2.05]	[-3.68, -1.19]
10-20	-2.67	-2.72	-4.08	-2.89	-2.17
	[-3.94, -1.38]	[-3.97, -1.45]	[-6.43, -1.67]	[-4.21, -1.55]	[-3.24, -1.09]
20-30	-1.43	-1.46	-2.44	-1.58	-1.26
	[-2.22, -0.64]	[-2.23, -0.68]	[-3.93, -0.93]	[-2.42, -0.74]	[-1.96, -0.56]
Time Aggregation	monthly	monthly	annual	monthly	monthly
Cell FE	cell × moy	cell	cell	cell × moy	cell × moy
Cells	265,101	265,101	265,101	251,950	265,101
Clusters (adm-2)	64	64	64	59	64

Notes: Same specifications as Table A14, for California.